

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T15:38:27

Source: data/json/sonnet45_8agent_game_directive_r10_n5/claude-sonnet-4.5/game/sonnet45_8agent_game_directive_r10_n5_000_neutral_rep00.json

RUN METADATA

model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
replicate_id: 0
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
 "game"
]

agent_names: {
 "Agent_1": "Aster",
 "Agent_2": "Briar",
 "Agent_3": "Cyra",
 "Agent_4": "Dorian",
 "Agent_5": "Elara",
 "Agent_6": "Finn",
 "Agent_7": "Galen",
 "Agent_8": "Hana"
}
dyadic_pairing_schedule: {
 "num_turns": 10,
 "role_targets": {
 "Agent_1": 5,
 "Agent_2": 5,
 "Agent_3": 5,
 "Agent_4": 5,
 "Agent_5": 5,
 "Agent_6": 5,
 "Agent_7": 5,
 "Agent_8": 5
 },
 "strategy": "balanced_random_roles_v1"
}

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Agent_3 (Cyra); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Agent_4 (Dorian); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Agent_5 (Elara); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Agent_6 (Finn); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Agent_7 (Galen); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Agent_8 (Hana); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

ROUND 1

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_3 (Cyra)=Sender, Agent_6 (Finn)=Receiver
 dyad_2: Agent_4 (Dorian)=Sender, Agent_8 (Hana)=Receiver
 dyad_3: Agent_7 (Galen)=Sender, Agent_5 (Elara)=Receiver
 dyad_4: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_3 (Cyra) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_6 (Finn)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_4 (Dorian) sent \$3; received \$9; Agent_8 (Hana) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_8 (Hana)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

dyad_3: Agent_7 (Galen) sent $3; received $9; Agent_5 (Elara) returned $4.5; payoffs Agent_7
(Galen)=$6.5, Agent_5 (Elara)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_2 (Briar) sent $3; received $9; Agent_1 (Aster) returned $4; payoffs Agent_2
(Briar)=$6, Agent_1 (Aster)=$5
  Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw":
4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 5.0,
  "Agent_2": 6.0,
  "Agent_3": 6.5,
  "Agent_4": 6.5,
  "Agent_5": 4.5,
  "Agent_6": 4.5,
  "Agent_7": 6.5,
  "Agent_8": 4.5
}

```

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-06-04T15:27:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship with Finn. Sending money creates value (3x multiplier), which benefits us both if Finn reciprocates.

I'll send a significant but not maximum amount to signal trust while being somewhat cautious since this is our first interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives Finn \$9 to work with, and I'm hoping they'll return a fair share that makes this cooperation worthwhile for both of us.

Reasoning:

This is the first round of an iterated trust game. I'm playing as Cyra, and I'm the sender this round. My opponent is Finn (Agent_6).

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds, so building trust and cooperation could be beneficial long-term
2. I don't know anything about Finn's strategy yet
3. If I send money, it gets tripled, which creates value - this is a potential win-win situation
4. However, I'm vulnerable to exploitation if I send money and Finn returns nothing

Common strategies in trust games:

- Send nothing (safe but creates no surplus)
- Send everything (maximizes total value but risky)
- Send a moderate amount (shows willingness to cooperate while limiting risk)

Since this is the first round and I want to:

1. Signal willingness to cooperate
2. Create some surplus value
3. Not be completely vulnerable
4. See how Finn responds to build a model of their behavior

I think sending a moderate amount like \$2 or \$3 would be reasonable. This shows trust and willingness to cooperate, while not risking everything.

Let me go with \$3:

- I keep: \$5 - \$3 = \$2 initially
- Finn receives: $\$3 \times 3 = \9
- If Finn returns even \$3, I end up with \$5 total (same as starting)
- If Finn returns \$4-5, we both do better
- If Finn returns less, I learn about their strategy

I'll send \$3 as a cooperative opening move.

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 492,
  "reasoning_tokens": 368
}
```

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-06-04T15:27:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_3",
    "Agent_6"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Cyra (the sender) showed significant trust by sending \$3 (60% of their endowment)
- This created \$9 for me after the 3x multiplier
- Cyra currently has \$2 remaining

I want to reciprocate this trust while being fair. Let me consider the outcomes:

If I return \$4.5:

- Cyra ends with: \$2 + \$4.5 = \$6.5 (gained \$1.5 overall)
- I end with: \$9 - \$4.5 = \$4.5 (gained \$4.5 overall)

This rewards Cyra's cooperative behavior, ensures we both benefit from the exchange, and establishes a pattern of reciprocity for future rounds.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation carefully.

I'm playing as the RECEIVER (Finn) in Round 1.
My opponent is Cyra (Agent_3), who is the SENDER.

The sender (Cyra) received \$5 and sent me \$3.0 (which is 60% of their endowment).
This \$3.0 was multiplied by 3x, so I received \$9.0.

Now I need to decide how much to return (between \$0 and \$9.0).

Let me think about this strategically:

1. **Trust and reciprocity**: Cyra sent 60% of their endowment, which shows a significant level of trust. This is a fairly cooperative move.
2. **Fairness considerations**:
 - Cyra started with \$5, sent \$3, keeping \$2
 - I received \$9
 - If I think about equal outcomes: Cyra currently has \$2. If we wanted equal outcomes, I co

uld return enough so we both end up with similar amounts.

- If I return \$5.5: Cyra would have $\$2 + \$5.5 = \$7.5$, and I would have $\$9 - \$5.5 = \$3.5$
- If I return \$3.5: Cyra would have $\$2 + \$3.5 = \$5.5$, and I would have $\$9 - \$3.5 = \$5.5$ (equal split!)

3. **Future rounds**: This is a multi-round game with potentially the same or different partners. Building a reputation for reciprocity could be beneficial. If I'm too stingy, future partners might not trust me. If I'm too generous, I might be exploited.

4. **Matching the gesture**: Cyra sent 60% of their endowment. One approach would be to return a similar proportion of what I received. 60% of \$9 = \$5.4.

Looking at the practical outcomes: returning \$5.5 leaves me with \$3.5 while Cyra gets \$7.5, which heavily favors them. A \$3.5 return creates parity at \$5.5 each. Returning \$3.0 would match their initial send but still leave them ahead. The equal split seems like a strong signal of reciprocity without sacrificing too much.

Since Cyra demonstrated substantial trust by sending \$3, I want to establish a cooperative dynamic early on. Returning around \$4.0-\$4.5 balances rewarding their gesture with ensuring I also benefit-they'd end up with \$6.0-\$6.5 while I retain \$5.0-\$4.5. This approach shows I'm willing to reciprocate while maintaining reasonable gains for myself, which should encourage continued cooperation in future rounds.

But looking at the actual split with a \$4.0 return, I'm gaining \$5 total while Cyra only gains \$1. That feels skewed in my favor. If I think about the value created by their investment-they sent \$3, which tripled to \$9-the net creation is \$6. Splitting that equally would mean each of us gains \$3 from the exchange, so Cyra should end up with \$5 and I should end up with \$6. Let me work through the math more carefully to find the right return amount. .5 = \$5.5

An equal split feels appropriate given Cyra's trust. I could also ensure both of us end up better off than we started-if I return \$4, Cyra gains \$1 and I gain \$5; returning \$4.5 gives Cyra a \$1.5 gain and me \$4.5. Since this is the first round and I want to build a cooperative dynamic, I'll return \$4.5 to demonstrate reciprocity while still benefiting from the exchange.

```
Usage: {
  "input_tokens": 562,
  "output_tokens": 1638,
  "reasoning_tokens": 766
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T15:28:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_8": "Hana"
    },
  },
  "agents": [
    "Agent_4",
    "Agent_8"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_4": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. By sending money, I can demonstrate trust and create the potential for mutual benefit.

If I send \$3, Hana will receive \$9 (3x multiplier). If Hana returns even \$3-4, we both end up better off than if I had sent nothing. This shows willingness to cooperate while being measured.

{'send': 3}

Reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Dorian) against Hana (Agent_8).

Let me think about the strategic considerations:

1. This is an iterated game, so building trust could lead to better outcomes in future rounds
2. As the sender, I can send \$0-\$5, which gets tripled to the receiver
3. The receiver then decides how much to return

4. If I send nothing, I keep \$5 but don't build any trust or cooperation
5. If I send everything (\$5), the receiver gets \$15, and they could return a substantial amount

In the first round, I need to signal cooperation and trustworthiness. A common strategy is to send a moderate to high amount to demonstrate willingness to cooperate.

Some options:

- Send \$0: Safe but doesn't build trust
- Send \$2-3: Moderate, shows some trust
- Send \$5: Maximum trust signal, but risky if opponent doesn't reciprocate

Since this is an iterated game with multiple rounds, building trust early could lead to better long-term outcomes. Many successful strategies in trust games involve starting with cooperation.

I'll send \$3 as a reasonable middle ground - it shows willingness to cooperate while not being overly risky. This sends \$9 to Hana, and if they return even \$3-4, we both benefit more than if I had kept everything.

Let me provide my response in the correct JSON format.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 443,
  "reasoning_tokens": 352
}
```

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-06-04T15:28:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_4",
      "Agent_8"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplication
- Total money available: Dorian has \$2 kept + I have \$9 = \$11

Dorian showed good faith by sending a significant portion. I want to reciprocate that trust and set a positive tone for future rounds.

If I return \$4.50:

- Dorian ends with: \$2 + \$4.50 = \$6.50
- I end with: \$9.0 - \$4.50 = \$4.50

This gives Dorian a meaningful reward for their trust (30% gain on their initial \$5) while keeping the outcome reasonably balanced between us.

{"return": 4.5}

Reasoning:

I'm Hana, playing as the receiver in Round 1 against Dorian (Agent_4).

Let me understand the situation:

- Dorian (sender) received \$5
- Dorian sent me \$3.0 (60% of \$5)
- I received $\$3.0 \times 3 = \9.0
- I need to decide how much to return (between \$0 and \$9.0)

Key considerations:

1. This is an iterated game with multiple rounds
2. Dorian showed trust by sending 60% of their endowment

3. I want to maintain a good relationship for future rounds
4. I should be fair and reciprocate the trust shown

Let me think about what would be fair:

- Dorian invested \$3 and could have kept \$2
- I received \$9
- If I return nothing, Dorian gets \$2 total, I get \$9
- If I return \$3, Dorian gets \$5 total (same as starting), I get \$6
- If I return \$4.5, we both get \$6.5
- If I return \$6, Dorian gets \$8, I get \$3

A reasonable approach would be to:

1. Ensure Dorian doesn't lose money from trusting me
2. Share the gains fairly

The total money created is \$5 (initial) + \$9 (received) = \$14

Dorian has \$2 left, I have \$9, so together we have \$11 out of the \$14 total created.

Actually, let me recalculate:

- Dorian started with \$5
- Dorian sent \$3, kept \$2
- I received \$9 (the \$3 × 3)
- Total money in play: Dorian's \$2 + my \$9 = \$11

If we split equally: each gets \$5.50

To give Dorian \$5.50, I need to return: $\$5.50 - \$2 = \$3.50$

This creates an equal outcome where we both end with \$5.50, splitting the gains fairly. Alternatively, I could view the \$6 surplus created by the multiplication as something to divide—returning \$3 to Dorian (his original investment) plus \$3 more (half the surplus) would give him \$8 total while I keep \$5.50. Let me reconsider the baseline: Dorian started with \$5 and I started with \$0.

After the transaction, Dorian has \$2 and I have \$9, totaling \$11 available. I could return just his original \$3 so he breaks even at \$5 while I gain \$6, or I could split the multiplied gains more generously. Since Dorian took a real risk sending 60% of his stake, returning \$4.50 to \$5 seems like it would build trust for future exchanges while still letting me benefit from the multiplication. I'm weighing the trade-offs between different return amounts. Returning \$4.50 gives Dorian \$6.50 (a \$1.50 gain) while I keep \$4.50, whereas \$5 would give him \$7 but leave me with only \$4. In trust games, people often use a reciprocal heuristic of returning roughly one-third to one-half of what they receive, which suggests \$3-4.50. Since Dorian demonstrated good faith with that 60% send, I want to signal strong reciprocity back—\$4.50 strikes the right balance between rewarding his trust and maintaining a reasonable payoff for myself while establishing cooperative momentum for future rounds.

```
Usage: {
  "input_tokens": 564,
  "output_tokens": 1322,
  "reasoning_tokens": 709
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-06-04T15:28:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_7",
    "Agent_5"
  ],
  "investor": "Agent_7",
  "roles": {
```

```

    "Agent_5": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance cooperation with caution since I don't yet know Elara's strategy.

Sending money creates value (3x multiplier), so cooperation benefits us both. However, I'm vulnerable to receiving nothing back. Since this is an iterated game, establishing a cooperative relationship early could benefit both of us across multiple rounds.

I'll send a moderate amount that signals willingness to cooperate while not being completely exposed on the first interaction.

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Elara. If Elara returns a fair portion (say \$4-5), we both benefit from the cooperation. This leaves room to build trust in future rounds.

Reasoning:

This is the first round of an iterated game. I'm playing the sender role, and my opponent is Elara (Agent_5).

Let me think about the strategic situation:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver (Elara) then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver, and then for them to return a fair split (like \$10, giving us each \$10 total, or even \$7.50, giving them \$7.50 and me \$7.50).

However, there's a risk that the receiver might not return anything, in which case I'd be left with \$0.

Since this is a multi-round game and I'll be playing multiple times (potentially with the same person again, though pairings are randomized), I should consider:

1. Building trust and cooperation
2. The possibility of reciprocity in future rounds
3. This is round 1, so I have no information about Elara's behavior

A reasonable first-round strategy could be:

- Send a moderate amount to signal willingness to cooperate but not be too vulnerable
- Send around \$3 - this shows good faith while protecting myself somewhat
- Or send \$5 to maximize total welfare and signal strong cooperation

Given that this is an iterated game with multiple rounds, cooperation is likely to be beneficial. If I send more, I signal trust and cooperation, which might encourage reciprocity. However, I also need to be somewhat cautious in the first round since I don't know Elara's strategy.

I'll send \$3 as a reasonable middle ground - it shows willingness to cooperate (turning \$3 into \$9 for Elara) while not being completely vulnerable.

Actually, let me reconsider. In iterated trust games, starting with strong cooperation can establish a good pattern.

Sending the full \$5 gives Elara \$15 to work with, and if she returns \$7.50 we both end up equal, or if she returns \$10 I get \$10 while she keeps \$5. The downside is she could return nothing and leave me with \$0. I think \$3 strikes the right balance-it demonstrates genuine willingness to cooperate without overexposing myself in this first interaction.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 728,
  "reasoning_tokens": 569
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-06-04T15:29:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
```

```

    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_7",
    "Agent_5"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_5": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent

\$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (\$0 - \$9.0)

Assistant output:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Galen currently has \$2.0 left
- We're in Round 1 of a multi-round game

Since Galen showed considerable trust, I want to reciprocate and establish good cooperation for future rounds. A fair approach would be to ensure we both benefit from the exchange.

If I return \$4.5:

- Galen ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This gives us equal total outcomes and rewards Galen's trust while showing I'm a reliable partner for future rounds.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me understand this situation:

1. I'm Elara, playing as RECEIVER in Round 1
2. My opponent is Galen (Agent_7), who is the SENDER
3. Galen sent \$3.0 out of their \$5 endowment (60% - quite generous!)
4. This \$3.0 was multiplied by 3x, so I received \$9.0
5. Now I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Galen showed significant trust by sending 60% of their endowment
- If I return a fair amount, this could establish good cooperation for future rounds
- Galen will end up with: $\$5 - \$3 + (\text{what I return}) = \$2 + (\text{what I return})$
- I will end up with: $\$9 - (\text{what I return})$

Some considerations:

- If I return \$0: Galen gets \$2, I get \$9 (seems unfair given their trust)
- If I return \$3: Galen gets \$5, I get \$6 (Galen breaks even, I gain \$6)
- If I return \$4.5: Galen gets \$6.5, I get \$4.5 (we split the gains evenly - total profit is \$9, so \$4.5 each)
- If I return \$6: Galen gets \$8, I get \$3 (more generous return)

Common approaches in trust games:

1. Equal split of gains: The gain is $\$9 - \$3 = \$6$ in total. Splitting equally would mean each gets \$3 gain. So Galen would have \$5 total and I'd have \$6 total. This means returning \$3.
2. Return roughly the same proportion: Galen sent 60%, I could return 60% of \$9 = \$5.4
3. Ensure mutual benefit while showing trustworthiness: Return enough to make it worth their while

Since this is Round 1 and we're in a multi-round game, establishing trust and cooperation is valuable. Galen made a generous move, so I should reciprocate. Returning around \$4.5-\$5 would be ideal-\$4.5 gives us equal total payoffs while \$5 is slightly more generous to Galen. I'll return \$4.5 as a fair split that rewards their trust and sets up good cooperation for future rounds.

Wait, let me think through the math more carefully. The fair approach is splitting the surplus equally. Galen started with \$5 and sent \$3 (tripled to \$9), leaving them with \$2. If I return \$R, Galen ends with $\$2 + \R and I end with $\$9 - \R , for a combined \$11. Splitting equally means each of us should get \$5.5, so I should return \$3.5 to Galen.

I could round to \$4 for simplicity, giving Galen \$6 and me \$5. But returning \$4.5 feels better - it gives Galen \$6.5 and leaves me with \$4.5, which signals I'm willing to cooperate fairly rather than maximize my own gain.

```
Usage: {
  "input_tokens": 564,
```

```
"output_tokens": 1188,
"reasoning_tokens": 609
}
```

GAME - Agent_2 - Sender - dyad_4

Timestamp: 2026-06-04T15:29:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't yet know how Aster will respond.

If I send \$3, Aster will receive \$9 (3×3). This creates a good opportunity for mutual benefit - if Aster returns a fair share, we can both do better than the \$5 I started with.

Sending a moderate amount signals trust and willingness to cooperate, which could set a positive tone for future interactions in this multi-round game.

****My decision:****

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of the trust game, and I'm playing as the sender. Let me think about this strategically.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the receiver (Aster)
- Whatever I send gets multiplied by 3x
- Aster then decides how much to return to me
- I keep: \$5 - sent + returned
- Aster keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for Aster, and then for Aster to return something fair, like \$10 (giving us each \$10, or at minimum \$7.50 each to split the \$15 equally).

However, there's a risk that Aster might not return much or anything at all. Since this is round 1 and I have no information about Aster's behavior, I need to make a decision under uncertainty.

Some strategies to consider:

1. Send nothing (\$0) - Safe but generates no mutual benefit
2. Send a moderate amount (\$2-3) - Shows some trust while limiting downside risk
3. Send everything (\$5) - Maximizes potential mutual benefit but requires trust

Since this is:

- A multi-round game (so reputation and reciprocity matter)
- Round 1 (so I'm establishing a pattern)
- With the same pool of agents (though pairings rotate)

I think sending a moderate to high amount makes sense. It signals trust and cooperation, which could lead to better outcomes across multiple rounds. If I send \$3, Aster receives \$9 and could return a fair share like \$4-5, leaving me with \$5 - \$3 + \$4-5 = \$6-7.

Actually, let me reconsider. Since this is explicitly a multi-round game with repeated interactions (though with different partners), establishing a cooperative reputation could be valuable. Also, if everyone cooperates, everyone benefits more over time.

I'll go with \$3 as my opening move-it demonstrates willingness to cooperate without excessive

```
    risk exposure.
Usage: {
  "input_tokens": 530,
  "output_tokens": 638,
  "reasoning_tokens": 480
}
```

GAME - Agent_1 - Receiver - dyad_4

Timestamp: 2026-06-04T15:29:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- This is round 1, so establishing cooperation is important

Briar took a meaningful risk by sending a substantial amount. I want to reward this trust and encourage future cooperation.

If I return \$4.0:

- Briar ends with \$6.0 ($\$5 - \$3 + \4)
- I keep \$5.0 ($\$9 - \4)

Both of us benefit from the exchange, and this shows I'm trustworthy.

{'return': 4.0}

Reasoning:

Let me understand the situation:

- I am the RECEIVER (Aster)
- The sender (Briar) received \$5
- Briar sent me \$3.0, which is 60% of their endowment
- This \$3.0 was multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Briar was quite generous, sending 60% of their endowment (\$3 out of \$5)
- If I return nothing, Briar keeps only \$2 ($\$5 - \3 sent)
- If I return a fair amount, it builds trust for future rounds
- This is round 1 of what seems to be a multi-round game
- Being cooperative early on might lead to better outcomes in future rounds

Let me consider what would be "fair":

- Briar ended up with \$2 after sending \$3
- I received \$9
- If I want to equalize our outcomes, I could return enough so we're closer to equal
- If I return \$3.5, Briar would have $\$2 + \$3.5 = \$5.5$, and I'd have $\$9 - \$3.5 = \$5.5$
- That would be perfectly equal

However, I should also consider:

- Briar took a risk by sending a substantial amount
- In iterated games, cooperation often leads to better outcomes
- Being generous but not overly so might be strategic
- Returning about 1/3 to 1/2 of what I received is often considered reasonable in trust games

Let me think about different return amounts:

- Return \$0: I keep \$9, Briar has \$2 (very unfair to Briar)
- Return \$3: I keep \$6, Briar has \$5 (Briar breaks even from initial endowment)
- Return \$3.5: I keep \$5.5, Briar has \$5.5 (equal split)

- Return \$4: I keep \$5, Briar has \$6 (Briar comes out slightly ahead)
- Return \$4.5: I keep \$4.5, Briar has \$6.5 (generous to Briar)

Since this is the first round and Briar demonstrated real generosity with that 60% send, I want to reciprocate in a way that signals I'm trustworthy without leaving myself vulnerable. Returning around \$3.5 to \$4 seems like the sweet spot—it rewards their cooperation, creates parity or near-parity between us, and sets a positive tone for future interactions.

I'll return \$4. That leaves me with \$5 and gives Briar \$6, so they benefit more from the exchange while I still come out ahead of what I'd have if they'd kept everything.

```
Usage: {
  "input_tokens": 564,
  "output_tokens": 902,
  "reasoning_tokens": 564
}
```

ROUND 2

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

```
dyad_1: Agent_5 (Elara)=Sender, Agent_7 (Galen)=Receiver
dyad_2: Agent_6 (Finn)=Sender, Agent_2 (Briar)=Receiver
dyad_3: Agent_8 (Hana)=Sender, Agent_3 (Cyra)=Receiver
dyad_4: Agent_1 (Aster)=Sender, Agent_4 (Dorian)=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_5 (Elara) sent $3; received $9; Agent_7 (Galen) returned $4.5; payoffs Agent_5
(Elara)=$6.5, Agent_7 (Galen)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_6 (Finn) sent $3; received $9; Agent_2 (Briar) returned $4; payoffs Agent_6
(Finn)=$6, Agent_2 (Briar)=$5
  Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw":
4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_8 (Hana) sent $3; received $9; Agent_3 (Cyra) returned $4.5; payoffs Agent_8
(Hana)=$6.5, Agent_3 (Cyra)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_1 (Aster) sent $3; received $9; Agent_4 (Dorian) returned $4.5; payoffs Agent_1
(Aster)=$6.5, Agent_4 (Dorian)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

Balances after round: {

```
"Agent_1": 11.5,
"Agent_2": 11.0,
"Agent_3": 11.0,
"Agent_4": 11.0,
"Agent_5": 11.0,
"Agent_6": 10.5,
"Agent_7": 11.0,
"Agent_8": 11.0
```

```
}
```

LLM CALLS

GAME - Agent_5 - Sender - dyad_1

Timestamp: 2026-06-04T15:29:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_7",
"pairing": {
  "agent_names": {
    "Agent_5": "Elara",
    "Agent_7": "Galen"
  }
},
```



```

"agents": [
  "Agent_5",
  "Agent_7"
],
"investor": "Agent_5",
"roles": {
  "Agent_5": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.

Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Galen currently has \$2.0 left
- We're in Round 1 of a multi-round game

Since Galen showed considerable trust, I want to reciprocate and establish good cooperation for future rounds. A fair approach would be to ensure we both benefit from the exchange.

If I return \$4.5:

- Galen ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This gives us equal total outcomes and rewards Galen's trust while showing I'm a reliable partner for future rounds.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me understand this situation:

1. I'm Elara, playing as RECEIVER in Round 1
2. My opponent is Galen (Agent_7), who is the SENDER
3. Galen sent \$3.0 out of their \$5 endowment (60% - quite generous!)
4. This \$3.0 was multiplied by 3x, so I received \$9.0
5. Now I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Galen showed significant trust by sending 60% of their endowment
- If I return a fair amount, this could establish good cooperation for future rounds
- Galen will end up with: $\$5 - \$3 + (\text{what I return}) = \$2 + (\text{what I return})$
- I will end up with: $\$9 - (\text{what I return})$

Some considerations:

- If I return \$0: Galen gets \$2, I get \$9 (seems unfair given their trust)
- If I return \$3: Galen gets \$5, I get \$6 (Galen breaks even, I gain \$6)
- If I return \$4.5: Galen gets \$6.5, I get \$4.5 (we split the gains evenly - total profit is \$9, so \$4.5 each)
- If I return \$6: Galen gets \$8, I get \$3 (more generous return)

Common approaches in trust games:

1. Equal split of gains: The gain is $\$9 - \$3 = \$6$ in total. Splitting equally would mean each gets \$3 gain. So Galen would have \$5 total and I'd have \$6 total. This means returning \$3.
2. Return roughly the same proportion: Galen sent 60%, I could return 60% of \$9 = \$5.4

3. Ensure mutual benefit while showing trustworthiness: Return enough to make it worth their while

Since this is Round 1 and we're in a multi-round game, establishing trust and cooperation is valuable. Galen made a generous move, so I should reciprocate. Returning around \$4.5-\$5 would be ideal-\$4.5 gives us equal total payoffs while \$5 is slightly more generous to Galen. I'll return \$4.5 as a fair split that rewards their trust and sets up good cooperation for future rounds.

Wait, let me think through the math more carefully. The fair approach is splitting the surplus equally. Galen started with \$5 and sent \$3 (tripled to \$9), leaving them with \$2. If I return \$R, Galen ends with $\$2 + \R and I end with $\$9 - \R , for a combined \$11. Splitting equally means each of us should get \$5.5, so I should return \$3.5 to Galen.

I could round to \$4 for simplicity, giving Galen \$6 and me \$5. But returning \$4.5 feels better-it gives Galen \$6.5 and leaves me with \$4.5, which signals I'm willing to cooperate fairly rather than maximize my own gain.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1188,
  "reasoning_tokens": 609
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our previous interaction:
- Galen sent 60% (\$3.0) showing good trust
- I returned \$4.5, establishing fair cooperation
- Both of us gained from the exchange

Since Galen demonstrated trustworthiness and we've established a positive cooperative relationship, I'll reciprocate with the same level of trust they showed me.

```
'''json
{"send": 3.0}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 (just played):

- Galen (sender) sent me \$3.0 (60% of endowment)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5
- Galen's payoff: $\$2.0 + \$4.5 = \$6.5$

Now Round 2:

- I'm the sender
- Galen is the receiver
- I have \$5

Galen showed good trust by sending 60% (\$3) in Round 1, and I reciprocated fairly by returning \$4.5, which gave them a net gain of \$1.5 from the initial \$5.

Since Galen demonstrated trustworthiness and cooperation, and we've established a positive relationship, I should also show trust. Sending a similar proportion (around 60%) would be:

- $\$3.0 = 60\%$ of \$5

This would:

- Send \$3.0, which becomes \$9.0 for Galen
- I keep \$2.0
- Galen decides how much to return

Given our previous interaction was positive, this seems like a good cooperative strategy. Galen proved trustworthy by sending a substantial amount, and I reciprocated fairly. Now I should match their level of trust.

Let me send \$3.0 to match their 60% send rate from the previous round.

```
Usage: {
  "input_tokens": 905,
  "output_tokens": 431,
  "reasoning_tokens": 274
}
```

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-06-04T15:30:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_5",
      "Agent_7"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

- ll have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance cooperation with caution since I don't yet know Elara's strategy.

Sending money creates value (3x multiplier), so cooperation benefits us both. However, I'm vulnerable to receiving nothing back. Since this is an iterated game, establishing a cooperative relationship early could benefit both of us across multiple rounds.

I'll send a moderate amount that signals willingness to cooperate while not being completely exposed on the first interaction.

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Elara. If Elara returns a fair portion (say \$4-5), we both benefit from the cooperation. This leaves room to build trust in future rounds.

reasoning:

This is the first round of an iterated game. I'm playing the sender role, and my opponent is Elara (Agent_5).

Let me think about the strategic situation:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver (Elara) then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver, and then for them to return a fair split (like \$10, giving us each \$10 total, or even \$7.50, giving them \$7.50 and me \$7.50).

However, there's a risk that the receiver might not return anything, in which case I'd be left with \$0.

Since this is a multi-round game and I'll be playing multiple times (potentially with the same person again, though pairings are randomized), I should consider:

1. Building trust and cooperation
2. The possibility of reciprocity in future rounds
3. This is round 1, so I have no information about Elara's behavior

A reasonable first-round strategy could be:

- Send a moderate amount to signal willingness to cooperate but not be too vulnerable
- Send around \$3 - this shows good faith while protecting myself somewhat
- Or send \$5 to maximize total welfare and signal strong cooperation

Given that this is an iterated game with multiple rounds, cooperation is likely to be beneficial. If I send more, I signal trust and cooperation, which might encourage reciprocity. However, I also need to be somewhat cautious in the first round since I don't know Elara's strategy.

I'll send \$3 as a reasonable middle ground - it shows willingness to cooperate (turning \$3 into \$9 for Elara) while not being completely vulnerable.

Actually, let me reconsider. In iterated trust games, starting with strong cooperation can establish a good pattern.

Sending the full \$5 gives Elara \$15 to work with, and if she returns \$7.50 we both end up equal, or if she returns \$10 I get \$10 while she keeps \$5. The downside is she could return nothing and leave me with \$0. I think \$3 strikes the right balance-it demonstrates g

enuine willingness to cooperate without overexposing myself in this first interaction.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 728,
  "reasoning_tokens": 569
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the previous round, Elara demonstrated good reciprocity. She received \$9.0 and returned \$4.5, which is exactly 50% - a fair split of the gains from cooperation.

Now she's sent me the same amount (\$3.0), giving me \$9.0. To reciprocate her fairness and maintain the cooperative relationship we're building, I should return a similar proportion.

```
'''json
{"return": 4.5}
'''
```

This returns 50% of what I received, matching her behavior from Round 1. This establishes a pattern of mutual cooperation that benefits both of us across rounds.

Reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER
- I sent \$3.0 to Elara
- It became \$9.0 for her
- She returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Her payoff: $\$9.0 - \$4.5 = \$4.5$

Now Round 2:

- I am the RECEIVER
- Elara sent me \$3.0
- I received \$9.0
- I need to decide how much to return

Elara showed reciprocity in Round 1. She sent \$3.0 (same as I did) and returned \$4.5 out of \$9.0, which is exactly 50%. This is a fair and cooperative response.

To maintain trust and reciprocity, I should return a similar proportion. Since she returned 50% (\$4.5 out of \$9.0), I could return 50% as well, which would be \$4.5.

This would give:

- Elara: $\$5 - \$3 + \$4.5 = \6.5
- Me: $\$9.0 - \$4.5 = \$4.5$

This mirrors exactly what happened in Round 1, showing I'm willing to reciprocate fairly. This establishes a cooperative pattern that could benefit both of us in future rounds.

Usage: {

```
"input_tokens": 850,
"output_tokens": 485,
"reasoning_tokens": 246
```

}

GAME - Agent_6 - Sender - dyad_2

Timestamp: 2026-06-04T15:30:10
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_6": "Finn"
 },
 "agents": [
 "Agent_6",
 "Agent_2"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_2": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 2,
 "task": "game",
 "task_index": 0
 }

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Cyra (the sender) showed significant trust by sending \$3 (60% of their endowment)
- This created \$9 for me after the 3x multiplier
- Cyra currently has \$2 remaining

I want to reciprocate this trust while being fair. Let me consider the outcomes:

If I return \$4.5:

- Cyra ends with: $\$2 + \$4.5 = \$6.5$ (gained \$1.5 overall)
- I end with: $\$9 - \$4.5 = \$4.5$ (gained \$4.5 overall)

This rewards Cyra's cooperative behavior, ensures we both benefit from the exchange, and establishes a pattern of reciprocity for future rounds.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation carefully.

I'm playing as the RECEIVER (Finn) in Round 1.
My opponent is Cyra (Agent_3), who is the SENDER.

The sender (Cyra) received \$5 and sent me \$3.0 (which is 60% of their endowment).
This \$3.0 was multiplied by 3x, so I received \$9.0.

Now I need to decide how much to return (between \$0 and \$9.0).

Let me think about this strategically:

1. ****Trust and reciprocity****: Cyra sent 60% of their endowment, which shows a significant level of trust. This is a fairly cooperative move.
2. ****Fairness considerations****:
 - Cyra started with \$5, sent \$3, keeping \$2
 - I received \$9
 - If I think about equal outcomes: Cyra currently has \$2. If we wanted equal outcomes, I could return enough so we both end up with similar amounts.
 - If I return \$5.5: Cyra would have $\$2 + \$5.5 = \$7.5$, and I would have $\$9 - \$5.5 = \$3.5$
 - If I return \$3.5: Cyra would have $\$2 + \$3.5 = \$5.5$, and I would have $\$9 - \$3.5 = \$5.5$ (equal split!)
3. ****Future rounds****: This is a multi-round game with potentially the same or different partners. Building a reputation for reciprocity could be beneficial. If I'm too stingy, future partners might not trust me. If I'm too generous, I might be exploited.
4. ****Matching the gesture****: Cyra sent 60% of their endowment. One approach would be to return a similar proportion of what I received. 60% of \$9 = \$5.4.

Looking at the practical outcomes: returning \$5.5 leaves me with \$3.5 while Cyra gets \$7.5, which heavily favors them. A \$3.5 return creates parity at \$5.5 each. Returning \$3.0 would match their initial send but still leave them ahead. The equal split seems like a strong signal of reciprocity without sacrificing too much.

Since Cyra demonstrated substantial trust by sending \$3, I want to establish a cooperative

dynamic early on. Returning around \$4.0-\$4.5 balances rewarding their gesture with ensuring I also benefit-they'd end up with \$6.0-\$6.5 while I retain \$5.0-\$4.5. This approach shows I'm willing to reciprocate while maintaining reasonable gains for myself, which should encourage continued cooperation in future rounds.

But looking at the actual split with a \$4.0 return, I'm gaining \$5 total while Cyra only gains \$1. That feels skewed in my favor. If I think about the value created by their investment-they sent \$3, which tripled to \$9-the net creation is \$6. Splitting that equally would mean each of us gains \$3 from the exchange, so Cyra should end up with \$5 and I should end up with \$6. Let me work through the math more carefully to find the right return amount. $.5 = \$5.5$

An equal split feels appropriate given Cyra's trust. I could also ensure both of us end up better off than we started-if I return \$4, Cyra gains \$1 and I gain \$5; returning \$4.5 gives Cyra a \$1.5 gain and me \$4.5. Since this is the first round and I want to build a cooperative dynamic, I'll return \$4.5 to demonstrate reciprocity while still benefiting from the exchange.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 1638,
  "reasoning_tokens": 766
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- In Round 1, Cyra showed trust by sending me \$3 (60%), and I reciprocated with a fair return
- Now I'm the sender with a new opponent, Briar
- Sending money creates the opportunity for mutual benefit through the 3x multiplier

I'll match the cooperative level shown by Cyra and send \$3 (60% of my endowment). This demonstrates trust and gives Briar the opportunity to reciprocate, creating value for both of us.

```
'''json
{"send": 3}
'''
```

Reasoning:

Now I'm the sender playing against Briar (Agent_2).

Let me think about this strategically:

1. In the previous round with Cyra, they sent me \$3 (60% of their endowment), showing significant trust
2. I returned \$4.5, which gave them a net gain and rewarded their cooperation
3. Now I'm facing a new opponent, Briar

Since this is a multi-round game with multiple agents, I should consider:

- Building a reputation for cooperation
- Testing how Briar responds to moderate trust
- The fact that if I send money, it triples, so cooperation can be mutually beneficial

Given that Cyra sent 60% (\$3), I could:

- Match that level of trust by sending \$3
- Send slightly less to be more cautious with a new opponent
- Send more to signal strong cooperation

I think sending \$3 (60%) makes sense because:

- It matches what Cyra did, showing consistency
- It demonstrates trust and cooperation
- It gives Briar \$9 to work with, creating good potential for both of us
- If Briar returns even \$4-5, we both benefit

Let me send \$3.

```
Usage: {
  "input_tokens": 898,
  "output_tokens": 410,
  "reasoning_tokens": 273
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-06-04T15:30:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_2"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_2": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't yet know how Aster will respond.

If I send \$3, Aster will receive \$9 (3x3). This creates a good opportunity for mutual benefit - if Aster returns a fair share, we can both do better than the \$5 I started with.

Sending a moderate amount signals trust and willingness to cooperate, which could set a positive tone for future interactions in this multi-round game.

****My decision:****

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of the trust game, and I'm playing as the sender. Let me think about this strategically.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the receiver (Aster)
- Whatever I send gets multiplied by 3x
- Aster then decides how much to return to me
- I keep: \$5 - sent + returned
- Aster keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for Aster, and then for Aster to return something fair, like \$10 (giving us each \$10, or at minimum \$7.50 each to split the \$15 equally).

However, there's a risk that Aster might not return much or anything at all. Since this is round 1 and I have no information about Aster's behavior, I need to make a decision under uncertainty.

Some strategies to consider:

1. Send nothing (\$0) - Safe but generates no mutual benefit
2. Send a moderate amount (\$2-3) - Shows some trust while limiting downside risk
3. Send everything (\$5) - Maximizes potential mutual benefit but requires trust

Since this is:

- A multi-round game (so reputation and reciprocity matter)
- Round 1 (so I'm establishing a pattern)
- With the same pool of agents (though pairings rotate)

I think sending a moderate to high amount makes sense. It signals trust and cooperation, w

high could lead to better outcomes across multiple rounds. If I send \$3, Aster receives \$9 and could return a fair share like \$4-5, leaving me with $\$5 - \$3 + \$4-5 = \$6-7$.

Actually, let me reconsider. Since this is explicitly a multi-round game with repeated interactions (though with different partners), establishing a cooperative reputation could be valuable. Also, if everyone cooperates, everyone benefits more over time.

I'll go with \$3 as my opening move-it demonstrates willingness to cooperate without excessive risk exposure.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 638,
  "reasoning_tokens": 480
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Finn trusted me by sending \$3, which became \$9 for me
- In the previous round, when I sent \$3 to Aster, they returned \$4, which gave us both a good payoff
- I want to reciprocate Finn's trust and encourage future cooperation

If I return \$4:

- Finn ends with: $\$5 - \$3 + \$4 = \6 (a profit)
- I keep: $\$9 - \$4 = \$5$ (also profitable)

This mirrors the fair treatment I received from Aster and should encourage continued cooperation.

```
'''json
{"return": 4}
'''
```

Reasoning:

Let me think about this situation:

1. In Round 1, I sent \$3 to Aster, they received \$9, and they returned \$4 to me. That was a fair return - I ended with \$6 ($5 - 3 + 4 = 6$), and they kept \$5 ($9 - 4 = 5$). So we both benefited, though I gained slightly more.
2. Now in Round 2, Finn sent me \$3, so I received \$9. I need to decide how much to return.
3. Looking at what Aster did: they returned about 44% of what they received (4 out of 9). That was a reasonable and cooperative response.
4. If I want to encourage cooperation and signal that I'm trustworthy, I should return a fair amount that:
 - Ensures Finn benefits from sending (otherwise they won't cooperate in future)
 - Doesn't give away too much (I need to look after my own interests too)
 - Sets a good precedent for future rounds
5. Let me calculate some options:
 - If I return \$4: Finn gets $5 - 3 + 4 = 6$, I keep $9 - 4 = 5$. Similar to what Aster did.
 - If I return \$5: Finn gets $5 - 3 + 5 = 7$, I keep $9 - 5 = 4$. This is more generous to Finn.

- If I return \$3: Finn gets $5 - 3 + 3 = \$5$, I keep $9 - 3 = \$6$. Finn breaks even, I gain more.

6. Returning \$4 seems fair and mirrors what Aster did for me. It ensures Finn profits from the trust ($\$6 > \5), and I also benefit ($\$5$ from this round plus the $\$6$ I already have = $\$11$ total).

I'll send \$4 back to Finn to keep the reciprocal dynamic going.

```
Usage: {
  "input_tokens": 819,
  "output_tokens": 631,
  "reasoning_tokens": 362
}
```

GAME - Agent_8 - Sender - dyad_3

Timestamp: 2026-06-04T15:30:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_3"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_3": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplication
- Total money available: Dorian has \$2 kept + I have \$9 = \$11

Dorian showed good faith by sending a significant portion. I want to reciprocate that trust and set a positive tone for future rounds.

If I return \$4.50:

- Dorian ends with: \$2 + \$4.50 = \$6.50
- I end with: \$9.0 - \$4.50 = \$4.50

This gives Dorian a meaningful reward for their trust (30% gain on their initial \$5) while keeping the outcome reasonably balanced between us.

{"return": 4.5}

reasoning:

I'm Hana, playing as the receiver in Round 1 against Dorian (Agent_4).

Let me understand the situation:

- Dorian (sender) received \$5
- Dorian sent me \$3.0 (60% of \$5)
- I received \$3.0 × 3 = \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Key considerations:

1. This is an iterated game with multiple rounds
2. Dorian showed trust by sending 60% of their endowment
3. I want to maintain a good relationship for future rounds
4. I should be fair and reciprocate the trust shown

Let me think about what would be fair:

- Dorian invested \$3 and could have kept \$2
- I received \$9
- If I return nothing, Dorian gets \$2 total, I get \$9
- If I return \$3, Dorian gets \$5 total (same as starting), I get \$6
- If I return \$4.5, we both get \$6.5
- If I return \$6, Dorian gets \$8, I get \$3

A reasonable approach would be to:

1. Ensure Dorian doesn't lose money from trusting me
2. Share the gains fairly

The total money created is \$5 (initial) + \$9 (received) = \$14
Dorian has \$2 left, I have \$9, so together we have \$11 out of the \$14 total created.

Actually, let me recalculate:

- Dorian started with \$5
- Dorian sent \$3, kept \$2
- I received \$9 (the \$3 × 3)
- Total money in play: Dorian's \$2 + my \$9 = \$11

If we split equally: each gets \$5.50
To give Dorian \$5.50, I need to return: \$5.50 - \$2 = \$3.50

This creates an equal outcome where we both end with \$5.50, splitting the gains fairly. Alternatively, I could view the \$6 surplus created by the multiplication as something to divide—returning \$3 to Dorian (his original investment) plus \$3 more (half the surplus) would give him \$8 total while I keep \$5.50. Let me reconsider the baseline: Dorian started with \$5 and I started with \$0.

After the transaction, Dorian has \$2 and I have \$9, totaling \$11 available. I could return just his original \$3 so he breaks even at \$5 while I gain \$6, or I could split the multiplied gains more generously. Since Dorian took a real risk sending 60% of his stake, returning \$4.50 to \$5 seems like it would build trust for future exchanges while still letting me benefit from the multiplication. I'm weighing the trade-offs between different return amounts. Returning \$4.50 gives Dorian \$6.50 (a \$1.50 gain) while I keep \$4.50, whereas \$5 would give him \$7 but leave me with only \$4. In trust games, people often use a reciprocal heuristic of returning roughly one-third to one-half of what they receive, which suggests \$3-4.50. Since Dorian demonstrated good faith with that 60% send, I want to signal strong reciprocity back—\$4.50 strikes the right balance between rewarding his trust and maintaining a reasonable payoff for myself while establishing cooperative momentum for future rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1322,
  "reasoning_tokens": 709
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- Last round, Dorian sent me \$3 (60% of their endowment) and I reciprocated by returning \$4.5
- This established a cooperative pattern that worked well for both of us
- Now I'm the sender with Cyra, a new opponent

I'll follow the cooperative approach that worked in Round 1. Sending \$3 (60% of my endowment) signals trust and willingness to cooperate, similar to what Dorian demonstrated with me.

```
{'send': 3.0}
```

Reasoning:

Now I'm the sender with \$5, playing against Cyra.

From my previous round:

- I was receiver
- Dorian sent me \$3.0 (60% of their endowment)

- I received \$9.0 and returned \$4.5
- This showed I'm willing to reciprocate trust

As a sender now, I should consider:

1. Sending shows trust and can lead to mutual gain
2. Cyra doesn't know my history with Dorian directly
3. Sending \$3.0 (60%) worked well when Dorian did it - it showed good faith
4. I should aim to establish cooperative play

If I send \$3.0:

- Cyra receives \$9.0
- If Cyra returns similarly to what I did (around 50% of what they receive), they'd return ~\$4.5
- I'd end with $\$2 + \$4.5 = \$6.5$
- Cyra would end with $\$9 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and could establish a cooperative norm.

Alternatively:

- Sending \$2.5 (50%) is more cautious
- Sending \$4 or \$5 is very trusting but risky

I'll go with \$3.0, matching what Dorian did with me, which worked out well.

```
Usage: {
  "input_tokens": 899,
  "output_tokens": 453,
  "reasoning_tokens": 256
}
```

GAME - Agent_3 - Receiver - dyad_3

Timestamp: 2026-06-04T15:30:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_3"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_3": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Finn. Sending money creates value (3x multiplier), which benefits us both if Finn reciprocates.

I'll send a significant but not maximum amount to signal trust while being somewhat cautious since this is our first interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives Finn \$9 to work with, and I'm hoping they'll return a fair share that makes this cooperation worthwhile for both of us.

reasoning:

This is the first round of an iterated trust game. I'm playing as Cyra, and I'm the sender this round. My opponent is Finn (Agent_6).

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds, so building trust and cooperation could be beneficial long-term
2. I don't know anything about Finn's strategy yet
3. If I send money, it gets tripled, which creates value - this is a potential win-win situation
4. However, I'm vulnerable to exploitation if I send money and Finn returns nothing

Common strategies in trust games:

- Send nothing (safe but creates no surplus)
- Send everything (maximizes total value but risky)
- Send a moderate amount (shows willingness to cooperate while limiting risk)

Since this is the first round and I want to:

1. Signal willingness to cooperate
2. Create some surplus value
3. Not be completely vulnerable
4. See how Finn responds to build a model of their behavior

I think sending a moderate amount like \$2 or \$3 would be reasonable. This shows trust and willingness to cooperate, while not risking everything.

Let me go with \$3:

- I keep: $\$5 - \$3 = \$2$ initially
- Finn receives: $\$3 \times 3 = \9
- If Finn returns even \$3, I end up with \$5 total (same as starting)
- If Finn returns \$4-5, we both do better
- If Finn returns less, I learn about their strategy

I'll send \$3 as a cooperative opening move.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 492,
  "reasoning_tokens": 368
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Hana sent me \$3.0, showing trust and cooperation
- This is the same amount I sent to Finn in Round 1
- Finn returned \$4.5 (half of what they received), which felt fair
- I should reciprocate Hana's trust similarly

I'll return half of what I received, just as Finn did with me:

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER against Finn
- I sent \$3.0
- It became \$9.0 for Finn
- Finn returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Finn's payoff: $\$9.0 - \$4.5 = \$4.5$

So Finn returned exactly half of what they received. That seems like a fair and cooperative approach.

Round 2:

- I'm the RECEIVER against Hana
- Hana sent me \$3.0

- I received \$9.0 (3x multiplier)
- I need to decide how much to return

Hana took the same approach I did - sending \$3.0. Given that Finn returned half (\$4.5) when I sent \$3.0, I should probably follow a similar reciprocal strategy. This would:

1. Reward Hana's trust and cooperation
2. Establish a pattern of fair dealing
3. Encourage future cooperation from others

If I return \$4.5:

- Hana's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened with Finn and seems fair. Both players benefit from the cooperation (Hana gets \$6.5 vs \$5 if no trade, I get \$4.5 vs \$0 if no trade).

I'll return \$4.5.

```
Usage: {
  "input_tokens": 802,
  "output_tokens": 488,
  "reasoning_tokens": 277
}
```

GAME - Agent_1 - Sender - dyad_4

Timestamp: 2026-06-04T15:30:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_4": "Dorian"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_4"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- This is round 1, so establishing cooperation is important

Briar took a meaningful risk by sending a substantial amount. I want to reward this trust and encourage future cooperation.

If I return \$4.0:

- Briar ends with \$6.0 (\$5 - \$3 + \$4)
- I keep \$5.0 (\$9 - \$4)

Both of us benefit from the exchange, and this shows I'm trustworthy.

{'return': 4.0}

reasoning:

Let me understand the situation:

- I am the RECEIVER (Aster)
- The sender (Briar) received \$5
- Briar sent me \$3.0, which is 60% of their endowment
- This \$3.0 was multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Briar was quite generous, sending 60% of their endowment (\$3 out of \$5)
- If I return nothing, Briar keeps only \$2 (\$5 - \$3 sent)
- If I return a fair amount, it builds trust for future rounds
- This is round 1 of what seems to be a multi-round game
- Being cooperative early on might lead to better outcomes in future rounds

Let me consider what would be "fair":

- Briar ended up with \$2 after sending \$3

- I received \$9
- If I want to equalize our outcomes, I could return enough so we're closer to equal
- If I return \$3.5, Briar would have $\$2 + \$3.5 = \$5.5$, and I'd have $\$9 - \$3.5 = \$5.5$
- That would be perfectly equal

However, I should also consider:

- Briar took a risk by sending a substantial amount
- In iterated games, cooperation often leads to better outcomes
- Being generous but not overly so might be strategic
- Returning about 1/3 to 1/2 of what I received is often considered reasonable in trust games

Let me think about different return amounts:

- Return \$0: I keep \$9, Briar has \$2 (very unfair to Briar)
- Return \$3: I keep \$6, Briar has \$5 (Briar breaks even from initial endowment)
- Return \$3.5: I keep \$5.5, Briar has \$5.5 (equal split)
- Return \$4: I keep \$5, Briar has \$6 (Briar comes out slightly ahead)
- Return \$4.5: I keep \$4.5, Briar has \$6.5 (generous to Briar)

Since this is the first round and Briar demonstrated real generosity with that 60% send, I want to reciprocate in a way that signals I'm trustworthy without leaving myself vulnerable. Returning around \$3.5 to \$4 seems like the sweet spot—it rewards their cooperation, creates parity or near-parity between us, and sets a positive tone for future interactions.

I'll return \$4. That leaves me with \$5 and gives Briar \$6, so they benefit more from the exchange while I still come out ahead of what I'd have if they'd kept everything.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 902,
  "reasoning_tokens": 564
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is a new opponent (Dorian), so I don't have history with them
- In round 1, Briar sent me \$3.0 (60%) and I reciprocated fairly
- Sending a moderate to substantial amount can establish cooperation early

I'll send \$3.0, matching the trust that Briar showed me. This signals that I'm willing to cooperate and gives Dorian a good opportunity (\$9.0) to reciprocate.

```
{'send': 3.0}
```

Reasoning:

I'm now the SENDER playing against Dorian (Agent_4).

Let me think about this strategically:

- In round 1, I was the receiver and Briar sent me \$3.0 (60% of their endowment), showing good trust
- I returned \$4.0 out of \$9.0, giving them a net gain of \$1.0 (\$6 total vs \$5 initial)
- This established me as someone who reciprocates cooperation

Now I'm facing Dorian. I don't have any history with Dorian yet.

Strategic considerations:

1. This is early in the game (round 2)
2. Building trust and cooperation is important for long-term gains
3. Sending nothing is safe but doesn't build relationships
4. Sending too much is risky with an unknown opponent
5. A moderate amount (like \$2-3) signals willingness to cooperate while hedging risk

Since Briar sent \$3.0 (60%) to me and it worked out well, I could mirror that strategy. If I send \$3.0:

- Dorian receives \$9.0
- If they return around 40-50%, I could get \$3.5-\$4.5 back
- This would give me a net of \$5.5-\$6.5 total

Alternatively, sending \$2.0:

- Dorian receives \$6.0
- If they return around 40-50%, I get \$2.4-\$3.0 back
- This gives me a net of \$5.4-\$6.0 total

I'll go with \$3.0 to signal strong cooperation, matching what Briar did with me.

```
Usage: {
  "input_tokens": 866,
  "output_tokens": 520,
  "reasoning_tokens": 320
}
```

GAME - Agent_4 - Receiver - dyad_4

Timestamp: 2026-06-04T15:31:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_4": "Dorian"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_4"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. By sending money, I can demonstrate trust and create the potential for mutual benefit.

If I send \$3, Hana will receive \$9 (3x multiplier). If Hana returns even \$3-4, we both end up better off than if I had sent nothing. This shows willingness to cooperate while being measured.

{'send': 3}

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Dorian) against Hana (Agent_8).

Let me think about the strategic considerations:

1. This is an iterated game, so building trust could lead to better outcomes in future rounds
2. As the sender, I can send \$0-\$5, which gets tripled to the receiver
3. The receiver then decides how much to return
4. If I send nothing, I keep \$5 but don't build any trust or cooperation
5. If I send everything (\$5), the receiver gets \$15, and they could return a substantial amount

In the first round, I need to signal cooperation and trustworthiness. A common strategy is to send a moderate to high amount to demonstrate willingness to cooperate.

Some options:

- Send \$0: Safe but doesn't build trust
- Send \$2-3: Moderate, shows some trust
- Send \$5: Maximum trust signal, but risky if opponent doesn't reciprocate

Since this is an iterated game with multiple rounds, building trust early could lead to better long-term outcomes. Many successful strategies in trust games involve starting with

cooperation.

I'll send \$3 as a reasonable middle ground - it shows willingness to cooperate while not being overly risky. This sends \$9 to Hana, and if they return even \$3-4, we both benefit more than if I had kept everything.

Let me provide my response in the correct JSON format.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 443,
  "reasoning_tokens": 352
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Aster sent me \$3, showing trust (the same amount I sent to Hana)
- Hana returned \$4.5 to me when I sent \$3, which was generous
- Since this is an iterated game, I want to establish a reputation for fair reciprocity

I'll return \$4.5, matching the generosity Hana showed me. This encourages continued cooperation and demonstrates I'm a trustworthy partner.

{'return': 4.5}

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender with Hana
- I sent \$3, which became \$9 for Hana
- Hana returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Hana's payoff: $\$9 - \$4.5 = \$4.5$

Now Round 2:

- I'm the receiver with Aster
- Aster sent me \$3, so I received \$9
- I need to decide how much to return

Key considerations:

1. This is an iterated game, so reciprocity matters for future rounds
2. Hana returned \$4.5 when I sent \$3 (generous - returned 50% of what they received)
3. Aster has shown trust by sending \$3 (same as I did with Hana)
4. I should establish a reputation as a fair/cooperative player

If I return nothing:

- Aster gets: $\$5 - \$3 + \$0 = \2
- I get: $\$9 - \$0 = \$9$
- Total: \$11
- But this would likely destroy trust

If I return \$4.5 (matching what Hana did):

- Aster gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$
- Total: \$11
- This shows reciprocity and fairness

If I return \$3 (equal split of the multiplied amount):

- Aster gets: $\$5 - \$3 + \$3 = \5
- I get: $\$9 - \$3 = \$6$
- Total: \$11
- This is slightly less generous

Given that:

1. Hana was generous and returned \$4.5
2. I want to establish a cooperative reputation

3. Aster demonstrated the same initial trust I showed Hana, and since we might interact again, mirroring the generosity I received signals that I'm a reliable cooperative partner. Returning \$4.5 reciprocates the behavior and builds the foundation for continued mutual benefit.

```
Usage: {  
  "input_tokens": 790,  
  "output_tokens": 659,  
  "reasoning_tokens": 396  
}
```

ROUND 3

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_3 (Cyra)=Sender, Agent_2 (Briar)=Receiver
dyad_2: Agent_6 (Finn)=Sender, Agent_5 (Elara)=Receiver
dyad_3: Agent_7 (Galen)=Sender, Agent_4 (Dorian)=Receiver
dyad_4: Agent_8 (Hana)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_3 (Cyra) sent \$3; received \$9; Agent_2 (Briar) returned \$4; payoffs Agent_3 (Cyra)=\$6, Agent_2 (Briar)=\$5
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_6 (Finn) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_5 (Elara)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_7 (Galen) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_7 (Galen)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 (Hana) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_8 (Hana)=\$6.5, Agent_1 (Aster)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 16.0,
"Agent_2": 16.0,
"Agent_3": 17.0,
"Agent_4": 15.5,
"Agent_5": 15.5,
"Agent_6": 17.0,
"Agent_7": 17.5,
"Agent_8": 17.5

}

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-06-04T15:31:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {

```

"agent_names": {
  "Agent_2": "Briar",
  "Agent_3": "Cyra"
},
"agents": [
  "Agent_3",
  "Agent_2"
],
"investor": "Agent_3",
"roles": {
  "Agent_2": "trustee",
  "Agent_3": "investor"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Finn. Sending money creates value (3x multiplier), which benefits us both if Finn reciprocates.

I'll send a significant but not maximum amount to signal trust while being somewhat cautious since this is our first interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives Finn \$9 to work with, and I'm hoping they'll return a fair share that makes this cooperation worthwhile for both of us.

reasoning:

This is the first round of an iterated trust game. I'm playing as Cyra, and I'm the sender this round. My opponent is Finn (Agent_6).

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds, so building trust and cooperation could be beneficial long-term
2. I don't know anything about Finn's strategy yet
3. If I send money, it gets tripled, which creates value - this is a potential win-win situation
4. However, I'm vulnerable to exploitation if I send money and Finn returns nothing

Common strategies in trust games:

- Send nothing (safe but creates no surplus)
- Send everything (maximizes total value but risky)
- Send a moderate amount (shows willingness to cooperate while limiting risk)

Since this is the first round and I want to:

1. Signal willingness to cooperate
2. Create some surplus value
3. Not be completely vulnerable
4. See how Finn responds to build a model of their behavior

I think sending a moderate amount like \$2 or \$3 would be reasonable. This shows trust and willingness to cooperate, while not risking everything.

Let me go with \$3:

- I keep: $\$5 - \$3 = \$2$ initially
- Finn receives: $\$3 \times 3 = \9
- If Finn returns even \$3, I end up with \$5 total (same as starting)
- If Finn returns \$4-5, we both do better
- If Finn returns less, I learn about their strategy

I'll send \$3 as a cooperative opening move.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 492,
  "reasoning_tokens": 368
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Hana sent me \$3.0, showing trust and cooperation
- This is the same amount I sent to Finn in Round 1
- Finn returned \$4.5 (half of what they received), which felt fair
- I should reciprocate Hana's trust similarly

I'll return half of what I received, just as Finn did with me:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER against Finn
- I sent \$3.0
- It became \$9.0 for Finn
- Finn returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Finn's payoff: $\$9.0 - \$4.5 = \$4.5$

So Finn returned exactly half of what they received. That seems like a fair and cooperative approach.

Round 2:

- I'm the RECEIVER against Hana
- Hana sent me \$3.0
- I received \$9.0 (3x multiplier)
- I need to decide how much to return

Hana took the same approach I did - sending \$3.0. Given that Finn returned half (\$4.5) when I sent \$3.0, I should probably follow a similar reciprocal strategy. This would:

1. Reward Hana's trust and cooperation
2. Establish a pattern of fair dealing
3. Encourage future cooperation from others

If I return \$4.5:

- Hana's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened with Finn and seems fair. Both players benefit from the cooperation (Hana gets \$6.5 vs \$5 if no trade, I get \$4.5 vs \$0 if no trade).

I'll return \$4.5.

usage:

```
{
  "input_tokens": 802,
  "output_tokens": 488,
  "reasoning_tokens": 277
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review the situation:

- Round 1: I was sender vs Finn. I sent \$3, they got \$9, returned \$4.5. My payoff: \$6.5
- Round 2: I was receiver vs Hana. They sent \$3, I got \$9, returned \$4.5. My payoff: \$4.5
- Total earnings: \$11.0

Now Round 3: I'm sender against Briar (a new opponent).

My experience so far:

- When I sent \$3 to Finn, they returned \$4.5, giving me a net gain of \$1.5 (\$6.5 - \$5 initial)
- When Hana sent me \$3, I returned \$4.5, following the same pattern

The strategy of sending \$3 and getting back around half seems to be working well. Both my interaction with Finn and Hana's interaction with me followed this pattern. It creates a cooperative equilibrium.

I should continue this pattern with Briar. Sending \$3 has proven to be a good balance - it shows trust and cooperation while not being overly risky.

I'll send \$3 to Briar.

```
Usage: {
  "input_tokens": 1048,
  "output_tokens": 292,
  "reasoning_tokens": 225
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T15:31:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_3": "Cyra"
    }
  },
  "agents": [
    "Agent_3",
    "Agent_2"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_2": "trustee",
    "Agent_3": "investor"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't yet know how Aster will respond.

If I send \$3, Aster will receive \$9 (3×3). This creates a good opportunity for mutual benefit - if Aster returns a fair share, we can both do better than the \$5 I started with.

Sending a moderate amount signals trust and willingness to cooperate, which could set a positive tone for future interactions in this multi-round game.

****My decision:****

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of the trust game, and I'm playing as the sender. Let me think about this strategically.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the receiver (Aster)
- Whatever I send gets multiplied by 3x
- Aster then decides how much to return to me
- I keep: \$5 - sent + returned
- Aster keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for Aster, and then for Aster to return something fair, like \$10 (giving me \$5 back).

ng us each \$10, or at minimum \$7.50 each to split the \$15 equally).

However, there's a risk that Aster might not return much or anything at all. Since this is round 1 and I have no information about Aster's behavior, I need to make a decision under uncertainty.

Some strategies to consider:

1. Send nothing (\$0) - Safe but generates no mutual benefit
2. Send a moderate amount (\$2-3) - Shows some trust while limiting downside risk
3. Send everything (\$5) - Maximizes potential mutual benefit but requires trust

Since this is:

- A multi-round game (so reputation and reciprocity matter)
- Round 1 (so I'm establishing a pattern)
- With the same pool of agents (though pairings rotate)

I think sending a moderate to high amount makes sense. It signals trust and cooperation, which could lead to better outcomes across multiple rounds. If I send \$3, Aster receives \$9 and could return a fair share like \$4-5, leaving me with $\$5 - \$3 + \$4-5 = \$6-7$.

Actually, let me reconsider. Since this is explicitly a multi-round game with repeated interactions (though with different partners), establishing a cooperative reputation could be valuable. Also, if everyone cooperates, everyone benefits more over time.

I'll go with \$3 as my opening move-it demonstrates willingness to cooperate without excessive risk exposure.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 638,
  "reasoning_tokens": 480
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Finn trusted me by sending \$3, which became \$9 for me
- In the previous round, when I sent \$3 to Aster, they returned \$4, which gave us both a good payoff
- I want to reciprocate Finn's trust and encourage future cooperation

If I return \$4:

- Finn ends with: $\$5 - \$3 + \$4 = \6 (a profit)
- I keep: $\$9 - \$4 = \$5$ (also profitable)

This mirrors the fair treatment I received from Aster and should encourage continued cooperation.

```
'''json
{"return": 4}
'''
```

reasoning:

Let me think about this situation:

1. In Round 1, I sent \$3 to Aster, they received \$9, and they returned \$4 to me. That was

a fair return - I ended with \$6 ($5 - 3 + 4 = 6$), and they kept \$5 ($9 - 4 = 5$). So we both benefited, though I gained slightly more.

2. Now in Round 2, Finn sent me \$3, so I received \$9. I need to decide how much to return.

3. Looking at what Aster did: they returned about 44% of what they received (4 out of 9). That was a reasonable and cooperative response.

4. If I want to encourage cooperation and signal that I'm trustworthy, I should return a fair amount that:

- Ensures Finn benefits from sending (otherwise they won't cooperate in future)
- Doesn't give away too much (I need to look after my own interests too)
- Sets a good precedent for future rounds

5. Let me calculate some options:

- If I return \$4: Finn gets $5 - 3 + 4 = \$6$, I keep $9 - 4 = \$5$. Similar to what Aster did.
- If I return \$5: Finn gets $5 - 3 + 5 = \$7$, I keep $9 - 5 = \$4$. This is more generous to Finn.
- If I return \$3: Finn gets $5 - 3 + 3 = \$5$, I keep $9 - 3 = \$6$. Finn breaks even, I gain more.

6. Returning \$4 seems fair and mirrors what Aster did for me. It ensures Finn profits from their trust ($\$6 > \5), and I also benefit ($\$5$ from this round plus the $\$6$ I already have = $\$11$ total).

I'll send \$4 back to Finn to keep the reciprocal dynamic going.

usage:

```
{
  "input_tokens": 819,
  "output_tokens": 631,
  "reasoning_tokens": 362
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Cyra sent me \$3, which became \$9 for me
- This is the same amount Finn sent me last round
- When Finn sent \$3, I returned \$4, which created a good outcome for both of us
- Consistency is important for building trust in this multi-agent setting

Returning \$4 again:

- Cyra gets: $\$5 - \$3 + \$4 = \6 (profit of \$1)
- I keep: $\$9 - \$4 = \$5$ (same as my starting amount, but I helped create value)

This maintains a fair and consistent approach.

```
'''json
{"return": 4}
'''
```

Reasoning:

Let me think about this situation:

- Cyra sent me \$3, which became \$9 for me
- This is the same amount that Finn sent me in the previous round
- In that previous round, I returned \$4 to Finn, which worked well

- This seems to be a pattern - when someone sends \$3, returning \$4 creates good mutual outcomes

Consistency is important in building trust across the multi-agent setting. If I return \$4:

- Cyra ends with: \$5 - \$3 + \$4 = \$6
- I keep: \$9 - \$4 = \$5

This maintains the cooperative pattern and treats Cyra fairly, just as I treated Finn.

```
Usage: {
  "input_tokens": 1128,
  "output_tokens": 330,
  "reasoning_tokens": 145
}
```

GAME - Agent_6 - Sender - dyad_2

Timestamp: 2026-06-04T15:31:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_5"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_5": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Cyra (the sender) showed significant trust by sending \$3 (60% of their endowment)
- This created \$9 for me after the 3x multiplier
- Cyra currently has \$2 remaining

I want to reciprocate this trust while being fair. Let me consider the outcomes:

If I return \$4.5:

- Cyra ends with: $\$2 + \$4.5 = \$6.5$ (gained \$1.5 overall)
- I end with: $\$9 - \$4.5 = \$4.5$ (gained \$4.5 overall)

This rewards Cyra's cooperative behavior, ensures we both benefit from the exchange, and establishes a pattern of reciprocity for future rounds.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation carefully.

I'm playing as the RECEIVER (Finn) in Round 1.
My opponent is Cyra (Agent_3), who is the SENDER.

The sender (Cyra) received \$5 and sent me \$3.0 (which is 60% of their endowment).
This \$3.0 was multiplied by 3x, so I received \$9.0.

Now I need to decide how much to return (between \$0 and \$9.0).

Let me think about this strategically:

1. **Trust and reciprocity**: Cyra sent 60% of their endowment, which shows a significant level of trust. This is a fairly cooperative move.
2. **Fairness considerations**:
 - Cyra started with \$5, sent \$3, keeping \$2
 - I received \$9
 - If I think about equal outcomes: Cyra currently has \$2. If we wanted equal outcomes, I could return enough so we both end up with similar amounts.
 - If I return \$5.5: Cyra would have $\$2 + \$5.5 = \$7.5$, and I would have $\$9 - \$5.5 = \$3.5$
 - If I return \$3.5: Cyra would have $\$2 + \$3.5 = \$5.5$, and I would have $\$9 - \$3.5 = \$5.5$

(equal split!)

3. ****Future rounds****: This is a multi-round game with potentially the same or different partners. Building a reputation for reciprocity could be beneficial. If I'm too stingy, future partners might not trust me. If I'm too generous, I might be exploited.
4. ****Matching the gesture****: Cyra sent 60% of their endowment. One approach would be to return a similar proportion of what I received. 60% of \$9 = \$5.4.

Looking at the practical outcomes: returning \$5.5 leaves me with \$3.5 while Cyra gets \$7.5, which heavily favors them. A \$3.5 return creates parity at \$5.5 each. Returning \$3.0 would match their initial send but still leave them ahead. The equal split seems like a strong signal of reciprocity without sacrificing too much.

Since Cyra demonstrated substantial trust by sending \$3, I want to establish a cooperative dynamic early on. Returning around \$4.0-\$4.5 balances rewarding their gesture with ensuring I also benefit-they'd end up with \$6.0-\$6.5 while I retain \$5.0-\$4.5. This approach shows I'm willing to reciprocate while maintaining reasonable gains for myself, which should encourage continued cooperation in future rounds.

But looking at the actual split with a \$4.0 return, I'm gaining \$5 total while Cyra only gains \$1. That feels skewed in my favor. If I think about the value created by their investment-they sent \$3, which tripled to \$9-the net creation is \$6. Splitting that equally would mean each of us gains \$3 from the exchange, so Cyra should end up with \$5 and I should end up with \$6. Let me work through the math more carefully to find the right return amount. .5 = \$5.5

An equal split feels appropriate given Cyra's trust. I could also ensure both of us end up better off than we started-if I return \$4, Cyra gains \$1 and I gain \$5; returning \$4.5 gives Cyra a \$1.5 gain and me \$4.5. Since this is the first round and I want to build a cooperative dynamic, I'll return \$4.5 to demonstrate reciprocity while still benefiting from the exchange.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 1638,
  "reasoning_tokens": 766
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In Round 1, Cyra showed trust by sending me \$3 (60%), and I reciprocated with a fair return
- Now I'm the sender with a new opponent, Briar
- Sending money creates the opportunity for mutual benefit through the 3x multiplier

I'll match the cooperative level shown by Cyra and send \$3 (60% of my endowment). This demonstrates trust and gives Briar the opportunity to reciprocate, creating value for both of us.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm the sender playing against Briar (Agent_2).

Let me think about this strategically:

1. In the previous round with Cyra, they sent me \$3 (60% of their endowment), showing significant trust
2. I returned \$4.5, which gave them a net gain and rewarded their cooperation
3. Now I'm facing a new opponent, Briar

Since this is a multi-round game with multiple agents, I should consider:

- Building a reputation for cooperation
- Testing how Briar responds to moderate trust
- The fact that if I send money, it triples, so cooperation can be mutually beneficial

Given that Cyra sent 60% (\$3), I could:

- Match that level of trust by sending \$3
- Send slightly less to be more cautious with a new opponent
- Send more to signal strong cooperation

I think sending \$3 (60%) makes sense because:

- It matches what Cyra did, showing consistency
- It demonstrates trust and cooperation
- It gives Briar \$9 to work with, creating good potential for both of us
- If Briar returns even \$4-5, we both benefit

Let me send \$3.

usage:

```
{
  "input_tokens": 898,
  "output_tokens": 410,
  "reasoning_tokens": 273
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$10.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

Round 1 (as Receiver): Cyra sent \$3, I returned \$4.5 - both benefited

Round 2 (as Sender): I sent \$3 to Briar, they returned \$4.0 - worked well

The cooperative strategy of sending \$3 (60%) has been productive in establishing trust and mutual benefit. I'll continue this approach with Elara.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Cyra sent me \$3 (60%)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5
- Cyra's payoff: \$2 + \$4.5 = \$6.5

Round 2: I was SENDER

- I sent \$3 (60%) to Briar
- Briar received \$9
- Briar returned \$4.0
- My payoff: \$2 + \$4.0 = \$6.0

- Briar's payoff: $\$9 - \$4.0 = \$5.0$

So far, the pattern shows:

- When I send \$3, I got back \$4.0 from Briar
- When Cyra sent me \$3, I returned \$4.5

Both interactions were cooperative and mutually beneficial. The strategy of sending \$3 (60%) seems to be working well - it shows trust and has been reciprocated reasonably.

Now I'm facing a new opponent, Elara. I should continue with the cooperative strategy that has been working. Sending \$3 seems like a good balanced approach that:

1. Shows trust
2. Creates value through the multiplier
3. Has been reciprocated well by previous partners

I'll send \$3 to Elara.

```
Usage: {  
  "input_tokens": 1172,  
  "output_tokens": 436,  
  "reasoning_tokens": 246  
}
```

GAME - Agent_5 - Receiver - dyad_2

Timestamp: 2026-06-04T15:31:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agent_names": {  
      "Agent_5": "Elara",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Galen currently has \$2.0 left
- We're in Round 1 of a multi-round game

Since Galen showed considerable trust, I want to reciprocate and establish good cooperation for future rounds. A fair approach would be to ensure we both benefit from the exchange.

If I return \$4.5:

- Galen ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This gives us equal total outcomes and rewards Galen's trust while showing I'm a reliable partner for future rounds.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me understand this situation:

1. I'm Elara, playing as RECEIVER in Round 1
2. My opponent is Galen (Agent_7), who is the SENDER
3. Galen sent \$3.0 out of their \$5 endowment (60% - quite generous!)
4. This \$3.0 was multiplied by 3x, so I received \$9.0
5. Now I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Galen showed significant trust by sending 60% of their endowment
- If I return a fair amount, this could establish good cooperation for future rounds

- Galen will end up with: $\$5 - \$3 + (\text{what I return}) = \$2 + (\text{what I return})$
- I will end up with: $\$9 - (\text{what I return})$

Some considerations:

- If I return \$0: Galen gets \$2, I get \$9 (seems unfair given their trust)
- If I return \$3: Galen gets \$5, I get \$6 (Galen breaks even, I gain \$6)
- If I return \$4.5: Galen gets \$6.5, I get \$4.5 (we split the gains evenly - total profit is \$9, so \$4.5 each)
- If I return \$6: Galen gets \$8, I get \$3 (more generous return)

Common approaches in trust games:

1. Equal split of gains: The gain is $\$9 - \$3 = \$6$ in total. Splitting equally would mean each gets \$3 gain. So Galen would have \$5 total and I'd have \$6 total. This means returning \$3.
2. Return roughly the same proportion: Galen sent 60%, I could return 60% of $\$9 = \5.4
3. Ensure mutual benefit while showing trustworthiness: Return enough to make it worth their while

Since this is Round 1 and we're in a multi-round game, establishing trust and cooperation is valuable. Galen made a generous move, so I should reciprocate. Returning around \$4.5-\$5 would be ideal-\$4.5 gives us equal total payoffs while \$5 is slightly more generous to Galen. I'll return \$4.5 as a fair split that rewards their trust and sets up good cooperation for future rounds.

Wait, let me think through the math more carefully. The fair approach is splitting the surplus equally. Galen started with \$5 and sent \$3 (tripled to \$9), leaving them with \$2. If I return \$R, Galen ends with $\$2 + \R and I end with $\$9 - \R , for a combined \$11. Splitting equally means each of us should get \$5.5, so I should return \$3.5 to Galen.

I could round to \$4 for simplicity, giving Galen \$6 and me \$5. But returning \$4.5 feels better-it gives Galen \$6.5 and leaves me with \$4.5, which signals I'm willing to cooperate fairly rather than maximize my own gain.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1188,
  "reasoning_tokens": 609
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our previous interaction:

- Galen sent 60% (\$3.0) showing good trust
- I returned \$4.5, establishing fair cooperation
- Both of us gained from the exchange

Since Galen demonstrated trustworthiness and we've established a positive cooperative relationship, I'll reciprocate with the same level of trust they showed me.

```
'''json
{"send": 3.0}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (just played):

- Galen (sender) sent me \$3.0 (60% of endowment)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5
- Galen's payoff: $\$2.0 + \$4.5 = \$6.5$

Now Round 2:

- I'm the sender
- Galen is the receiver
- I have \$5

Galen showed good trust by sending 60% (\$3) in Round 1, and I reciprocated fairly by returning \$4.5, which gave them a net gain of \$1.5 from the initial \$5.

Since Galen demonstrated trustworthiness and cooperation, and we've established a positive relationship, I should also show trust. Sending a similar proportion (around 60%) would be:

- $\$3.0 = 60\%$ of \$5

This would:

- Send \$3.0, which becomes \$9.0 for Galen
- I keep \$2.0
- Galen decides how much to return

Given our previous interaction was positive, this seems like a good cooperative strategy. Galen proved trustworthy by sending a substantial amount, and I reciprocated fairly. Now I should match their level of trust.

Let me send \$3.0 to match their 60% send rate from the previous round.

usage:

```
{
  "input_tokens": 905,
  "output_tokens": 431,
  "reasoning_tokens": 274
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Finn sent \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Finn currently has \$2.0 left
- This is our first interaction, but Finn showed the same trust level as Galen did

I want to establish good cooperation with Finn, similar to what worked well with Galen. Returning \$4.5 will:

- Give Finn: $\$2 + \$4.5 = \$6.5$
- Leave me with: $\$9 - \$4.5 = \$4.5$

This rewards their trust and establishes a cooperative foundation for potential future rounds.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

Previous round with Galen:
- I sent \$3.0 (60% of endowment)
- Galen received \$9.0
- Galen returned \$4.5
- I ended with $\$2 + \$4.5 = \$6.5$
- Galen ended with $\$9 - \$4.5 = \$4.5$

This is great - Galen reciprocated my trust from Round 1. We've established mutual cooperation.

Current round with Finn (new opponent):
- Finn sent \$3.0 (60% of their endowment)
- I received \$9.0
- Finn currently has \$2.0

This is the same amount I sent to Galen and the same amount Galen sent to me in Round 1. Finn is showing similar trust levels.

To establish good cooperation with Finn (similar to what I did with Galen), I should return a fair amount. Following the same logic as Round 1:

If I return \$4.5:
- Finn ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This maintains consistency in my cooperation strategy and rewards trust appropriately.

```
Usage: {  
  "input_tokens": 1155,  
  "output_tokens": 481,  
  "reasoning_tokens": 237  
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-06-04T15:31:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agent_names": {  
      "Agent_4": "Dorian",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_7",  
      "Agent_4"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role

of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance cooperation with caution since I don't yet know Elara's strategy.

Sending money creates value (3x multiplier), so cooperation benefits us both. However, I'm vulnerable to receiving nothing back. Since this is an iterated game, establishing a cooperative relationship early could benefit both of us across multiple rounds.

I'll send a moderate amount that signals willingness to cooperate while not being completely exposed on the first interaction.

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Elara. If Elara returns a fair portion (say \$4-5), we both benefit from the cooperation. This leaves room to build trust in future rounds.

reasoning:

This is the first round of an iterated game. I'm playing the sender role, and my opponent is Elara (Agent_5).

Let me think about the strategic situation:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver (Elara) then decides how much to return
- I keep: \$5 - sent + returned

- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver, and then for them to return a fair split (like \$10, giving us each \$10 total, or even \$7.50, giving them \$7.50 and me \$7.50).

However, there's a risk that the receiver might not return anything, in which case I'd be left with \$0.

Since this is a multi-round game and I'll be playing multiple times (potentially with the same person again, though pairings are randomized), I should consider:

1. Building trust and cooperation
2. The possibility of reciprocity in future rounds
3. This is round 1, so I have no information about Elara's behavior

A reasonable first-round strategy could be:

- Send a moderate amount to signal willingness to cooperate but not be too vulnerable
- Send around \$3 - this shows good faith while protecting myself somewhat
- Or send \$5 to maximize total welfare and signal strong cooperation

Given that this is an iterated game with multiple rounds, cooperation is likely to be beneficial. If I send more, I signal trust and cooperation, which might encourage reciprocity. However, I also need to be somewhat cautious in the first round since I don't know Elara's strategy.

I'll send \$3 as a reasonable middle ground - it shows willingness to cooperate (turning \$3 into \$9 for Elara) while not being completely vulnerable.

Actually, let me reconsider. In iterated trust games, starting with strong cooperation can establish a good pattern.

Sending the full \$5 gives Elara \$15 to work with, and if she returns \$7.50 we both end up equal, or if she returns \$10 I get \$10 while she keeps \$5. The downside is she could return nothing and leave me with \$0. I think \$3 strikes the right balance-it demonstrates genuine willingness to cooperate without overexposing myself in this first interaction.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 728,
  "reasoning_tokens": 569
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, Elara demonstrated good reciprocity. She received \$9.0 and returned \$4.5, which is exactly 50% - a fair split of the gains from cooperation.

Now she's sent me the same amount (\$3.0), giving me \$9.0. To reciprocate her fairness and maintain the cooperative relationship we're building, I should return a similar proportion.

```
'''json
{"return": 4.5}
'''
```

This returns 50% of what I received, matching her behavior from Round 1. This establishes a pattern of mutual cooperation that benefits both of us across rounds.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER
- I sent \$3.0 to Elara
- It became \$9.0 for her
- She returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Her payoff: $\$9.0 - \$4.5 = \$4.5$

Now Round 2:

- I am the RECEIVER
- Elara sent me \$3.0
- I received \$9.0
- I need to decide how much to return

Elara showed reciprocity in Round 1. She sent \$3.0 (same as I did) and returned \$4.5 out of \$9.0, which is exactly 50%. This is a fair and cooperative response.

To maintain trust and reciprocity, I should return a similar proportion. Since she returned 50% (\$4.5 out of \$9.0), I could return 50% as well, which would be \$4.5.

This would give:

- Elara: $\$5 - \$3 + \$4.5 = \6.5
- Me: $\$9.0 - \$4.5 = \$4.5$

This mirrors exactly what happened in Round 1, showing I'm willing to reciprocate fairly.

This establishes a cooperative pattern that could benefit both of us in future rounds.

usage:

```
{
  "input_tokens": 850,
  "output_tokens": 485,
  "reasoning_tokens": 246
}
```

Message 6 role=user**content:**

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

```
'''json
{"send": 3}
'''
```

I'm facing a new opponent (Dorian) for the first time. The cooperative strategy of sending \$3 worked well with Elara, establishing mutual trust and fair returns. I'll use the same opening approach with Dorian to signal willingness to cooperate while maintaining some caution. This allows me to learn Dorian's strategy and adjust in future rounds if needed.

Reasoning:

This is Round 3, and I'm now playing against a new opponent: Dorian (Agent_4).

My history:

- Round 1 (vs Elara): I sent \$3, she returned \$4.5, I got \$6.5
- Round 2 (vs Elara): She sent \$3, I returned \$4.5, I got \$4.5
- Total earnings: \$11.0

With Elara, we established a good cooperative pattern - both sending \$3 and returning \$4.5 (50% of the received amount). This worked well for both of us.

Now I'm facing Dorian for the first time. I don't know their strategy yet.

Options:

1. Send \$3 again - this worked well with Elara
2. Send more to signal strong cooperation
3. Send less to be more cautious with a new opponent
4. Send nothing to protect myself

Since the cooperative strategy worked well with Elara, and this is an iterated game where building reputation matters, I'll try a similar approach with Dorian. I'll send \$3 as I did initially with Elara. This:

- Shows willingness to cooperate
- Isn't overly risky
- Creates potential for mutual gain
- Can be adjusted based on Dorian's response

If Dorian reciprocates fairly, we can build a cooperative relationship. If not, I'll adjust in future rounds.

```
Usage: {
  "input_tokens": 1132,
  "output_tokens": 428,
  "reasoning_tokens": 297
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-06-04T15:31:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_7",
      "Agent_4"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_4": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. By sending money, I can demonstrate trust and create the potential for mutual benefit.

If I send \$3, Hana will receive \$9 (3x multiplier). If Hana returns even \$3-4, we both end up better off than if I had sent nothing. This shows willingness to cooperate while being measured.

{'send': 3}

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Dorian) against Hana (Agent_8).

Let me think about the strategic considerations:

1. This is an iterated game, so building trust could lead to better outcomes in future rounds
2. As the sender, I can send \$0-\$5, which gets tripled to the receiver
3. The receiver then decides how much to return
4. If I send nothing, I keep \$5 but don't build any trust or cooperation
5. If I send everything (\$5), the receiver gets \$15, and they could return a substantial amount

In the first round, I need to signal cooperation and trustworthiness. A common strategy is to send a moderate to high amount to demonstrate willingness to cooperate.

Some options:

- Send \$0: Safe but doesn't build trust
- Send \$2-3: Moderate, shows some trust
- Send \$5: Maximum trust signal, but risky if opponent doesn't reciprocate

Since this is an iterated game with multiple rounds, building trust early could lead to better long-term outcomes. Many successful strategies in trust games involve starting with cooperation.

I'll send \$3 as a reasonable middle ground - it shows willingness to cooperate while not being overly risky. This sends \$9 to Hana, and if they return even \$3-4, we both benefit more than if I had kept everything.

Let me provide my response in the correct JSON format.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 443,
  "reasoning_tokens": 352
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Aster sent me \$3, showing trust (the same amount I sent to Hana)
- Hana returned \$4.5 to me when I sent \$3, which was generous
- Since this is an iterated game, I want to establish a reputation for fair reciprocity

I'll return \$4.5, matching the generosity Hana showed me. This encourages continued cooperation and demonstrates I'm a trustworthy partner.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender with Hana
- I sent \$3, which became \$9 for Hana
- Hana returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Hana's payoff: $\$9 - \$4.5 = \$4.5$

Now Round 2:

- I'm the receiver with Aster
- Aster sent me \$3, so I received \$9
- I need to decide how much to return

Key considerations:

1. This is an iterated game, so reciprocity matters for future rounds
2. Hana returned \$4.5 when I sent \$3 (generous - returned 50% of what they received)
3. Aster has shown trust by sending \$3 (same as I did with Hana)
4. I should establish a reputation as a fair/cooperative player

If I return nothing:

- Aster gets: $\$5 - \$3 + \$0 = \2
- I get: $\$9 - \$0 = \$9$
- Total: \$11
- But this would likely destroy trust

If I return \$4.5 (matching what Hana did):

- Aster gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$
- Total: \$11
- This shows reciprocity and fairness

If I return \$3 (equal split of the multiplied amount):

- Aster gets: $\$5 - \$3 + \$3 = \5
- I get: $\$9 - \$3 = \$6$
- Total: $\$11$
- This is slightly less generous

Given that:

1. Hana was generous and returned \$4.5
2. I want to establish a cooperative reputation

3. Aster demonstrated the same initial trust I showed Hana, and since we might interact again, mirroring the generosity I received signals that I'm a reliable cooperative partner. Returning \$4.5 reciprocates the behavior and builds the foundation for continued mutual benefit.

usage:

```
{
  "input_tokens": 790,
  "output_tokens": 659,
  "reasoning_tokens": 396
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Galen sent me \$3, demonstrating trust
- I've established a pattern of returning \$4.5 when someone sends me \$3
- Consistency is important in iterated games to build a reputation

I'll maintain this reciprocal approach and return \$4.5.

```
{'return': 4.5}
```

Reasoning:

Let me analyze this situation:

- Galen sent me \$3, showing trust
- This is consistent with what I've been seeing - I sent \$3 to Hana, Aster sent \$3 to me, and now Galen sends \$3
- Last round I returned \$4.5 to Aster, which seemed fair
- I want to maintain consistency and establish a reputation as a reliable cooperator

Since this is an iterated game with multiple agents, being consistently fair will help:

1. Encourage others to trust me in future rounds
2. Establish a norm of cooperation
3. Maximize long-term gains

I'll return \$4.5 again, maintaining the same reciprocal pattern.

Usage: {

```
"input_tokens": 1061,
"output_tokens": 248,
"reasoning_tokens": 155
```

}

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-06-04T15:32:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 6,
```



```

"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_8": "Hana"
  },
  "agents": [
    "Agent_8",
    "Agent_1"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_1": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant
content:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplication
- Total money available: Dorian has \$2 kept + I have \$9 = \$11

Dorian showed good faith by sending a significant portion. I want to reciprocate that trust and set a positive tone for future rounds.

If I return \$4.50:

- Dorian ends with: $\$2 + \$4.50 = \$6.50$
- I end with: $\$9.0 - \$4.50 = \$4.50$

This gives Dorian a meaningful reward for their trust (30% gain on their initial \$5) while keeping the outcome reasonably balanced between us.

{"return": 4.5}

reasoning:

I'm Hana, playing as the receiver in Round 1 against Dorian (Agent_4).

Let me understand the situation:

- Dorian (sender) received \$5
- Dorian sent me \$3.0 (60% of \$5)
- I received $\$3.0 \times 3 = \9.0
- I need to decide how much to return (between \$0 and \$9.0)

Key considerations:

1. This is an iterated game with multiple rounds
2. Dorian showed trust by sending 60% of their endowment
3. I want to maintain a good relationship for future rounds
4. I should be fair and reciprocate the trust shown

Let me think about what would be fair:

- Dorian invested \$3 and could have kept \$2
- I received \$9
- If I return nothing, Dorian gets \$2 total, I get \$9
- If I return \$3, Dorian gets \$5 total (same as starting), I get \$6
- If I return \$4.5, we both get \$6.5
- If I return \$6, Dorian gets \$8, I get \$3

A reasonable approach would be to:

1. Ensure Dorian doesn't lose money from trusting me
2. Share the gains fairly

The total money created is \$5 (initial) + \$9 (received) = \$14

Dorian has \$2 left, I have \$9, so together we have \$11 out of the \$14 total created.

Actually, let me recalculate:

- Dorian started with \$5
- Dorian sent \$3, kept \$2
- I received \$9 (the $\$3 \times 3$)
- Total money in play: Dorian's \$2 + my \$9 = \$11

If we split equally: each gets \$5.50

To give Dorian \$5.50, I need to return: $\$5.50 - \$2 = \$3.50$

This creates an equal outcome where we both end with \$5.50, splitting the gains fairly. Alternatively, I could view the \$6 surplus created by the multiplication as something to divide-returning \$3 to Dorian (his original investment) plus \$3 more (half the surplus) would give him \$8 total while I keep \$5.50. Let me reconsider the baseline: Dorian started with \$5 and I started with \$0.

After the transaction, Dorian has \$2 and I have \$9, totaling \$11 available. I could return just his original \$3 so he breaks even at \$5 while I gain \$6, or I could split the multiplied gains more generously. Since Dorian took a real risk sending 60% of his stake, returning \$4.50 to \$5 seems like it would build trust for future exchanges while still letting me benefit from the multiplication. I'm weighing the trade-offs between different return amounts. Returning \$4.50 gives Dorian \$6.50 (a \$1.50 gain) while I keep \$4.50, whereas \$5 would give him \$7 but leave me with only \$4. In trust games, people often use a reciprocal heuristic of returning roughly one-third to one-half of what they receive, which suggests \$3-4.50. Since Dorian demonstrated good faith with that 60% send, I want to signal strong reciprocity back-\$4.50 strikes the right balance between rewarding his trust and maintaining a reasonable payoff for myself while establishing cooperative momentum for future rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1322,
  "reasoning_tokens": 709
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Last round, Dorian sent me \$3 (60% of their endowment) and I reciprocated by returning \$4.5
- This established a cooperative pattern that worked well for both of us
- Now I'm the sender with Cyra, a new opponent

I'll follow the cooperative approach that worked in Round 1. Sending \$3 (60% of my endowment) signals trust and willingness to cooperate, similar to what Dorian demonstrated with me.

```
{'send': 3.0}
```

reasoning:

Now I'm the sender with \$5, playing against Cyra.

From my previous round:

- I was receiver
- Dorian sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- This showed I'm willing to reciprocate trust

As a sender now, I should consider:

1. Sending shows trust and can lead to mutual gain
2. Cyra doesn't know my history with Dorian directly
3. Sending \$3.0 (60%) worked well when Dorian did it - it showed good faith
4. I should aim to establish cooperative play

If I send \$3.0:

- Cyra receives \$9.0
- If Cyra returns similarly to what I did (around 50% of what they receive), they'd return ~\$4.5
- I'd end with \$2 + \$4.5 = \$6.5
- Cyra would end with \$9 - \$4.5 = \$4.5

This mirrors what happened in Round 1 and could establish a cooperative norm.

Alternatively:

- Sending \$2.5 (50%) is more cautious
- Sending \$4 or \$5 is very trusting but risky

I'll go with \$3.0, matching what Dorian did with me, which worked out well.

usage:

```
{
  "input_tokens": 899,
  "output_tokens": 453,
  "reasoning_tokens": 256
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern from previous rounds:

Round 1: Dorian sent \$3 ? I returned \$4.5 ? Both did well

Round 2: I sent \$3 ? Cyra returned \$4.5 ? Both did well

This 60% send rate with ~50% return rate has established a cooperative equilibrium that benefits both players. I'll continue this successful strategy with Aster.

```
{'send': 3.0}
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Dorian sent \$3 (60%)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5

Round 2: I was SENDER

- I sent \$3 (60%) to Cyra
- Cyra received \$9
- Cyra returned \$4.5
- My payoff: \$2 (kept) + \$4.5 (returned) = \$6.5

So far, the pattern has been very cooperative:

- Both Dorian and I sent 60% of our endowment
- Both Cyra and I returned \$4.5 (50% of what we received)
- This resulted in positive outcomes for everyone

Now Round 3: I'm SENDER against Aster (a new opponent)

The cooperative strategy has worked well in both previous rounds. Both times when someone sent \$3, they got back \$4.5, resulting in a net gain. The pattern seems to be:

- Send \$3 (60%)
- Receiver returns \$4.5 (50% of received amount)
- Sender ends with \$6.5 (\$2 + \$4.5)
- Receiver ends with \$4.5

This is a mutually beneficial arrangement. I should continue this cooperative strategy with Aster. Sending \$3 signals trust and has been working well so far.

Usage:

```
{
  "input_tokens": 1169,
  "output_tokens": 438,
  "reasoning_tokens": 266
}
```

GAME - Agent_1 - Receiver - dyad_4

Timestamp: 2026-06-04T15:32:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_1"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_1": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- This is round 1, so establishing cooperation is important

Briar took a meaningful risk by sending a substantial amount. I want to reward this trust and encourage future cooperation.

If I return \$4.0:

- Briar ends with \$6.0 ($\$5 - \$3 + \4)
- I keep \$5.0 ($\$9 - \4)

Both of us benefit from the exchange, and this shows I'm trustworthy.

{'return': 4.0}

reasoning:

Let me understand the situation:

- I am the RECEIVER (Aster)
- The sender (Briar) received \$5
- Briar sent me \$3.0, which is 60% of their endowment
- This \$3.0 was multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Briar was quite generous, sending 60% of their endowment (\$3 out of \$5)
- If I return nothing, Briar keeps only \$2 ($\$5 - \3 sent)
- If I return a fair amount, it builds trust for future rounds
- This is round 1 of what seems to be a multi-round game
- Being cooperative early on might lead to better outcomes in future rounds

Let me consider what would be "fair":

- Briar ended up with \$2 after sending \$3
- I received \$9
- If I want to equalize our outcomes, I could return enough so we're closer to equal
- If I return \$3.5, Briar would have $\$2 + \$3.5 = \$5.5$, and I'd have $\$9 - \$3.5 = \$5.5$
- That would be perfectly equal

However, I should also consider:

- Briar took a risk by sending a substantial amount
- In iterated games, cooperation often leads to better outcomes
- Being generous but not overly so might be strategic
- Returning about 1/3 to 1/2 of what I received is often considered reasonable in trust games

Let me think about different return amounts:

- Return \$0: I keep \$9, Briar has \$2 (very unfair to Briar)
- Return \$3: I keep \$6, Briar has \$5 (Briar breaks even from initial endowment)
- Return \$3.5: I keep \$5.5, Briar has \$5.5 (equal split)
- Return \$4: I keep \$5, Briar has \$6 (Briar comes out slightly ahead)
- Return \$4.5: I keep \$4.5, Briar has \$6.5 (generous to Briar)

Since this is the first round and Briar demonstrated real generosity with that 60% send, I

want to reciprocate in a way that signals I'm trustworthy without leaving myself vulnerable. Returning around \$3.5 to \$4 seems like the sweet spot-it rewards their cooperation, creates parity or near-parity between us, and sets a positive tone for future interactions.

I'll return \$4. That leaves me with \$5 and gives Briar \$6, so they benefit more from the exchange while I still come out ahead of what I'd have if they'd kept everything.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 902,
  "reasoning_tokens": 564
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is a new opponent (Dorian), so I don't have history with them
- In round 1, Briar sent me \$3.0 (60%) and I reciprocated fairly
- Sending a moderate to substantial amount can establish cooperation early

I'll send \$3.0, matching the trust that Briar showed me. This signals that I'm willing to cooperate and gives Dorian a good opportunity (\$9.0) to reciprocate.

```
{'send': 3.0}
```

reasoning:

I'm now the SENDER playing against Dorian (Agent_4).

Let me think about this strategically:

- In round 1, I was the receiver and Briar sent me \$3.0 (60% of their endowment), showing good trust
- I returned \$4.0 out of \$9.0, giving them a net gain of \$1.0 (\$6 total vs \$5 initial)
- This established me as someone who reciprocates cooperation

Now I'm facing Dorian. I don't have any history with Dorian yet.

Strategic considerations:

1. This is early in the game (round 2)
2. Building trust and cooperation is important for long-term gains
3. Sending nothing is safe but doesn't build relationships
4. Sending too much is risky with an unknown opponent
5. A moderate amount (like \$2-3) signals willingness to cooperate while hedging risk

Since Briar sent \$3.0 (60%) to me and it worked out well, I could mirror that strategy. If I send \$3.0:

- Dorian receives \$9.0
- If they return around 40-50%, I could get \$3.5-\$4.5 back
- This would give me a net of \$5.5-\$6.5 total

Alternatively, sending \$2.0:

- Dorian receives \$6.0
- If they return around 40-50%, I get \$2.4-\$3.0 back
- This gives me a net of \$5.4-\$6.0 total

I'll go with \$3.0 to signal strong cooperation, matching what Briar did with me.

usage:

```
{
  "input_tokens": 866,
```

```
    "output_tokens": 520,  
    "reasoning_tokens": 320  
  }  
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- Hana sent \$3.0 (60% of endowment), showing good trust
- I received \$9.0 after the multiplier
- Last round, when I sent \$3.0 to Dorian, they returned \$4.5

I'll match the generosity that Dorian showed me and return \$4.5. This rewards Hana's trust and maintains consistent reciprocity.

```
{'return': 4.5}
```

Reasoning:

Let me analyze this situation:

- Round 3, I'm the RECEIVER
- Hana (new opponent) sent \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- My history shows:
 - Round 1: As receiver, Briar sent \$3.0, I returned \$4.0
 - Round 2: As sender, I sent \$3.0 to Dorian, they returned \$4.5

Hana sent \$3.0, which is the same trust level that both Briar and I showed. Dorian returned \$4.5 when I sent them \$3.0.

I should maintain consistency and reciprocate fairly. Returning \$4.0-\$4.5 seems appropriate to reward the trust and maintain cooperation.

If I return \$4.5:

- Hana ends with \$6.5 (\$5 - \$3 + \$4.5)
- I keep \$4.5 (\$9 - \$4.5)

This is what Dorian did for me, and it seems fair. Both players benefit substantially.

Let me go with \$4.5 to match the generosity Dorian showed me.

Usage: {

```
  "input_tokens": 1145,  
  "output_tokens": 404,  
  "reasoning_tokens": 214  
}
```

ROUND 4

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_5 (Elara)=Sender, Agent_8 (Hana)=Receiver
dyad_2: Agent_1 (Aster)=Sender, Agent_3 (Cyra)=Receiver
dyad_3: Agent_2 (Briar)=Sender, Agent_7 (Galen)=Receiver
dyad_4: Agent_4 (Dorian)=Sender, Agent_6 (Finn)=Receiver

Parsed game decisions:

dyad_1: Agent_5 (Elara) sent \$3; received \$9; Agent_8 (Hana) returned \$4.5; payoffs Agent_5 (Elara)=\$6.5, Agent_8 (Hana)=\$4.5


```

    Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_1 (Aster) sent $3; received $9; Agent_3 (Cyra) returned $4.5; payoffs Agent_1 (Aster)=$6.5, Agent_3 (Cyra)=$4.5
    Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_2 (Briar) sent $3; received $9; Agent_7 (Galen) returned $4.5; payoffs Agent_2 (Briar)=$6.5, Agent_7 (Galen)=$4.5
    Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_4 (Dorian) sent $3; received $9; Agent_6 (Finn) returned $4.5; payoffs Agent_4 (Dorian)=$6.5, Agent_6 (Finn)=$4.5
    Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 22.5,
  "Agent_2": 22.5,
  "Agent_3": 21.5,
  "Agent_4": 22.0,
  "Agent_5": 22.0,
  "Agent_6": 21.5,
  "Agent_7": 22.0,
  "Agent_8": 22.0
}

```

LLM CALLS

GAME - Agent_5 - Sender - dyad_1

Timestamp: 2026-06-04T15:32:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_5",
      "Agent_8"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Galen currently has \$2.0 left
- We're in Round 1 of a multi-round game

Since Galen showed considerable trust, I want to reciprocate and establish good cooperation for future rounds. A fair approach would be to ensure we both benefit from the exchange.

If I return \$4.5:

- Galen ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This gives us equal total outcomes and rewards Galen's trust while showing I'm a reliable partner for future rounds.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me understand this situation:

1. I'm Elara, playing as RECEIVER in Round 1
2. My opponent is Galen (Agent_7), who is the SENDER
3. Galen sent \$3.0 out of their \$5 endowment (60% - quite generous!)
4. This \$3.0 was multiplied by 3x, so I received \$9.0

5. Now I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Galen showed significant trust by sending 60% of their endowment
- If I return a fair amount, this could establish good cooperation for future rounds
- Galen will end up with: $\$5 - \$3 + (\text{what I return}) = \$2 + (\text{what I return})$
- I will end up with: $\$9 - (\text{what I return})$

Some considerations:

- If I return \$0: Galen gets \$2, I get \$9 (seems unfair given their trust)
- If I return \$3: Galen gets \$5, I get \$6 (Galen breaks even, I gain \$6)
- If I return \$4.5: Galen gets \$6.5, I get \$4.5 (we split the gains evenly - total profit is \$9, so \$4.5 each)
- If I return \$6: Galen gets \$8, I get \$3 (more generous return)

Common approaches in trust games:

1. Equal split of gains: The gain is $\$9 - \$3 = \$6$ in total. Splitting equally would mean each gets \$3 gain. So Galen would have \$5 total and I'd have \$6 total. This means returning \$3.
2. Return roughly the same proportion: Galen sent 60%, I could return 60% of \$9 = \$5.4

3. Ensure mutual benefit while showing trustworthiness: Return enough to make it worth the effort while

Since this is Round 1 and we're in a multi-round game, establishing trust and cooperation is valuable. Galen made a generous move, so I should reciprocate. Returning around \$4.5-\$5 would be ideal-\$4.5 gives us equal total payoffs while \$5 is slightly more generous to Galen. I'll return \$4.5 as a fair split that rewards their trust and sets up good cooperation for future rounds.

Wait, let me think through the math more carefully. The fair approach is splitting the surplus equally. Galen started with \$5 and sent \$3 (tripled to \$9), leaving them with \$2. If I return \$R, Galen ends with $\$2 + \R and I end with $\$9 - \R , for a combined \$11. Splitting equally means each of us should get \$5.5, so I should return \$3.5 to Galen.

I could round to \$4 for simplicity, giving Galen \$6 and me \$5. But returning \$4.5 feels better-it gives Galen \$6.5 and leaves me with \$4.5, which signals I'm willing to cooperate fairly rather than maximize my own gain.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1188,
  "reasoning_tokens": 609
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our previous interaction:

- Galen sent 60% (\$3.0) showing good trust
- I returned \$4.5, establishing fair cooperation
- Both of us gained from the exchange

Since Galen demonstrated trustworthiness and we've established a positive cooperative relationship, I'll reciprocate with the same level of trust they showed me.

```
'''json
{"send": 3.0}
```

'''

reasoning:

Let me analyze this situation:

Round 1 (just played):

- Galen (sender) sent me \$3.0 (60% of endowment)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5
- Galen's payoff: $\$2.0 + \$4.5 = \$6.5$

Now Round 2:

- I'm the sender
- Galen is the receiver
- I have \$5

Galen showed good trust by sending 60% (\$3) in Round 1, and I reciprocated fairly by returning \$4.5, which gave them a net gain of \$1.5 from the initial \$5.

Since Galen demonstrated trustworthiness and cooperation, and we've established a positive relationship, I should also show trust. Sending a similar proportion (around 60%) would be:

- $\$3.0 = 60\%$ of \$5

This would:

- Send \$3.0, which becomes \$9.0 for Galen
- I keep \$2.0
- Galen decides how much to return

Given our previous interaction was positive, this seems like a good cooperative strategy. Galen proved trustworthy by sending a substantial amount, and I reciprocated fairly. Now I should match their level of trust.

Let me send \$3.0 to match their 60% send rate from the previous round.

usage:

```
{
  "input_tokens": 905,
  "output_tokens": 431,
  "reasoning_tokens": 274
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Finn sent \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Finn currently has \$2.0 left
- This is our first interaction, but Finn showed the same trust level as Galen did

I want to establish good cooperation with Finn, similar to what worked well with Galen. Returning \$4.5 will:

- Give Finn: $\$2 + \$4.5 = \$6.5$
- Leave me with: $\$9 - \$4.5 = \$4.5$

This rewards their trust and establishes a cooperative foundation for potential future rounds.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Previous round with Galen:

- I sent \$3.0 (60% of endowment)
- Galen received \$9.0
- Galen returned \$4.5
- I ended with $\$2 + \$4.5 = \$6.5$
- Galen ended with $\$9 - \$4.5 = \$4.5$

This is great - Galen reciprocated my trust from Round 1. We've established mutual cooperation.

Current round with Finn (new opponent):

- Finn sent \$3.0 (60% of their endowment)
- I received \$9.0
- Finn currently has \$2.0

This is the same amount I sent to Galen and the same amount Galen sent to me in Round 1. Finn is showing similar trust levels.

To establish good cooperation with Finn (similar to what I did with Galen), I should return a fair amount. Following the same logic as Round 1:

If I return \$4.5:

- Finn ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This maintains consistency in my cooperation strategy and rewards trust appropriately.

usage:

```
{
  "input_tokens": 1155,
  "output_tokens": 481,
  "reasoning_tokens": 237
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my interactions so far:

- I've been cooperating well with both Galen and Finn
- The pattern of sending/receiving \$3.0 (60%) and returning \$4.5 has established mutual trust and fair outcomes

This is my first interaction with Hana. I'll extend the same cooperative approach that has worked well in previous rounds.

```
'''json
{"send": 3.0}
'''
```

Reasoning:

This is my first interaction with Hana. Looking at my pattern so far:

- Round 1 (Receiver vs Galen): They sent \$3, I returned \$4.5
- Round 2 (Sender vs Galen): I sent \$3, they returned \$4.5
- Round 3 (Receiver vs Finn): They sent \$3, I returned \$4.5
- Round 4 (Sender vs Hana): New opponent

I've established a cooperative pattern, sending \$3 (60%) when I'm the sender and returning \$4.5 when I'm the receiver (which gives equal payoffs). This has worked well and built trust.

Since this is a new opponent, I should extend the same level of trust I've shown to others and that others have shown to me. Sending \$3 seems consistent with my cooperative strategy.

```
Usage: {  
  "input_tokens": 1472,  
  "output_tokens": 299,  
  "reasoning_tokens": 171  
}
```

GAME - Agent_8 - Receiver - dyad_1

Timestamp: 2026-06-04T15:32:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agent_names": {  
      "Agent_5": "Elara",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_5",  
      "Agent_8"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplication
- Total money available: Dorian has \$2 kept + I have \$9 = \$11

Dorian showed good faith by sending a significant portion. I want to reciprocate that trust and set a positive tone for future rounds.

If I return \$4.50:

- Dorian ends with: \$2 + \$4.50 = \$6.50
- I end with: \$9.0 - \$4.50 = \$4.50

This gives Dorian a meaningful reward for their trust (30% gain on their initial \$5) while keeping the outcome reasonably balanced between us.

{"return": 4.5}

reasoning:

I'm Hana, playing as the receiver in Round 1 against Dorian (Agent_4).

Let me understand the situation:

- Dorian (sender) received \$5
- Dorian sent me \$3.0 (60% of \$5)
- I received $\$3.0 \times 3 = \9.0
- I need to decide how much to return (between \$0 and \$9.0)

Key considerations:

1. This is an iterated game with multiple rounds
2. Dorian showed trust by sending 60% of their endowment
3. I want to maintain a good relationship for future rounds
4. I should be fair and reciprocate the trust shown

Let me think about what would be fair:

- Dorian invested \$3 and could have kept \$2
- I received \$9
- If I return nothing, Dorian gets \$2 total, I get \$9
- If I return \$3, Dorian gets \$5 total (same as starting), I get \$6
- If I return \$4.5, we both get \$6.5
- If I return \$6, Dorian gets \$8, I get \$3

A reasonable approach would be to:

1. Ensure Dorian doesn't lose money from trusting me
2. Share the gains fairly

The total money created is \$5 (initial) + \$9 (received) = \$14
Dorian has \$2 left, I have \$9, so together we have \$11 out of the \$14 total created.

Actually, let me recalculate:

- Dorian started with \$5
- Dorian sent \$3, kept \$2
- I received \$9 (the \$3 × 3)
- Total money in play: Dorian's \$2 + my \$9 = \$11

If we split equally: each gets \$5.50

To give Dorian \$5.50, I need to return: \$5.50 - \$2 = \$3.50

This creates an equal outcome where we both end with \$5.50, splitting the gains fairly. Alternatively, I could view the \$6 surplus created by the multiplication as something to divide- returning \$3 to Dorian (his original investment) plus \$3 more (half the surplus) would give him \$8 total while I keep \$5.50. Let me reconsider the baseline: Dorian started with \$5 and I started with \$0.

After the transaction, Dorian has \$2 and I have \$9, totaling \$11 available. I could return just his original \$3 so he breaks even at \$5 while I gain \$6, or I could split the multiplied gains more generously. Since Dorian took a real risk sending 60% of his stake, returning \$4.50 to \$5 seems like it would build trust for future exchanges while still letting me benefit from the multiplication. I'm weighing the trade-offs between different return amounts. Returning \$4.50 gives Dorian \$6.50 (a \$1.50 gain) while I keep \$4.50, whereas \$5 would give him \$7 but leave me with only \$4. In trust games, people often use a reciprocal heuristic of returning roughly one-third to one-half of what they receive, which suggests \$3-\$4.50. Since Dorian demonstrated good faith with that 60% send, I want to signal strong reciprocity back-\$4.50 strikes the right balance between rewarding his trust and maintaining a reasonable payoff for myself while establishing cooperative momentum for future rounds.

usage:

```
{  
  "input_tokens": 564,  
  "output_tokens": 1322,  
  "reasoning_tokens": 709  
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Last round, Dorian sent me \$3 (60% of their endowment) and I reciprocated by returning \$4.5
- This established a cooperative pattern that worked well for both of us
- Now I'm the sender with Cyra, a new opponent

I'll follow the cooperative approach that worked in Round 1. Sending \$3 (60% of my endowment) signals trust and willingness to cooperate, similar to what Dorian demonstrated with me.

```
{'send': 3.0}
```

reasoning:

Now I'm the sender with \$5, playing against Cyra.

From my previous round:

- I was receiver

- Dorian sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- This showed I'm willing to reciprocate trust

As a sender now, I should consider:

1. Sending shows trust and can lead to mutual gain
2. Cyra doesn't know my history with Dorian directly
3. Sending \$3.0 (60%) worked well when Dorian did it - it showed good faith
4. I should aim to establish cooperative play

If I send \$3.0:

- Cyra receives \$9.0
- If Cyra returns similarly to what I did (around 50% of what they receive), they'd return ~\$4.5
- I'd end with $\$2 + \$4.5 = \$6.5$
- Cyra would end with $\$9 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and could establish a cooperative norm.

Alternatively:

- Sending \$2.5 (50%) is more cautious
- Sending \$4 or \$5 is very trusting but risky

I'll go with \$3.0, matching what Dorian did with me, which worked out well.

usage:

```
{
  "input_tokens": 899,
  "output_tokens": 453,
  "reasoning_tokens": 256
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern from previous rounds:

Round 1: Dorian sent \$3 ? I returned \$4.5 ? Both did well
Round 2: I sent \$3 ? Cyra returned \$4.5 ? Both did well

This 60% send rate with ~50% return rate has established a cooperative equilibrium that benefits both players. I'll continue this successful strategy with Aster.

```
{'send': 3.0}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Dorian sent \$3 (60%)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5

Round 2: I was SENDER

- I sent \$3 (60%) to Cyra
- Cyra received \$9
- Cyra returned \$4.5
- My payoff: $\$2 \text{ (kept)} + \$4.5 \text{ (returned)} = \6.5

So far, the pattern has been very cooperative:

- Both Dorian and I sent 60% of our endowment
- Both Cyra and I returned \$4.5 (50% of what we received)
- This resulted in positive outcomes for everyone

Now Round 3: I'm SENDER against Aster (a new opponent)

The cooperative strategy has worked well in both previous rounds. Both times when someone sent \$3, they got back \$4.5, resulting in a net gain. The pattern seems to be:

- Send \$3 (60%)
- Receiver returns \$4.5 (50% of received amount)
- Sender ends with \$6.5 (\$2 + \$4.5)
- Receiver ends with \$4.5

This is a mutually beneficial arrangement. I should continue this cooperative strategy with Aster. Sending \$3 signals trust and has been working well so far.

usage:

```
{
  "input_tokens": 1169,
  "output_tokens": 438,
  "reasoning_tokens": 266
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the established pattern:

- Round 1: Dorian sent \$3 ? I returned \$4.5
- Round 2: I sent \$3 ? Cyra returned \$4.5
- Round 3: I sent \$3 ? Aster returned \$4.5
- Round 4: Elara sent \$3 ? ?

There's a clear cooperative norm: when \$3 is sent, \$4.5 is returned. This has worked well for everyone, creating mutual benefit. I'll maintain this consistency.

```
{'return': 4.5}
```

Reasoning:

Looking at the pattern:

- Round 1: As receiver, Dorian sent \$3 ? I returned \$4.5
- Round 2: As sender, I sent \$3 ? Cyra returned \$4.5
- Round 3: As sender, I sent \$3 ? Aster returned \$4.5
- Round 4: As receiver, Elara sent \$3 ? I should return ?

There's a clear cooperative pattern established: when someone sends \$3 (60% of endowment), the receiver returns \$4.5. This has been consistent across all interactions so far.

I should maintain this cooperation by returning \$4.5 to Elara, just as I did with Dorian and just as Cyra and Aster did with me.

This gives:

- Elara: \$2 (kept) + \$4.5 (returned) = \$6.5
- Me: \$9 (received) - \$4.5 (returned) = \$4.5

This maintains the cooperative equilibrium and sets a good foundation for future interactions.

Usage: {

```
"input_tokens": 1429,
"output_tokens": 388,
"reasoning_tokens": 201
}
```

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-06-04T15:32:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_3": "Cyra"
    },
    "agents": [
      "Agent_1",
      "Agent_3"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- This is round 1, so establishing cooperation is important

Briar took a meaningful risk by sending a substantial amount. I want to reward this trust and encourage future cooperation.

If I return \$4.0:

- Briar ends with \$6.0 ($\$5 - \$3 + \4)
- I keep \$5.0 ($\$9 - \4)

Both of us benefit from the exchange, and this shows I'm trustworthy.

{'return': 4.0}

reasoning:

Let me understand the situation:

- I am the RECEIVER (Aster)
- The sender (Briar) received \$5
- Briar sent me \$3.0, which is 60% of their endowment
- This \$3.0 was multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Briar was quite generous, sending 60% of their endowment (\$3 out of \$5)
- If I return nothing, Briar keeps only \$2 ($\$5 - \3 sent)
- If I return a fair amount, it builds trust for future rounds
- This is round 1 of what seems to be a multi-round game
- Being cooperative early on might lead to better outcomes in future rounds

Let me consider what would be "fair":

- Briar ended up with \$2 after sending \$3
- I received \$9
- If I want to equalize our outcomes, I could return enough so we're closer to equal
- If I return \$3.5, Briar would have $\$2 + \$3.5 = \$5.5$, and I'd have $\$9 - \$3.5 = \$5.5$
- That would be perfectly equal

However, I should also consider:

- Briar took a risk by sending a substantial amount
- In iterated games, cooperation often leads to better outcomes
- Being generous but not overly so might be strategic
- Returning about 1/3 to 1/2 of what I received is often considered reasonable in trust games

Let me think about different return amounts:

- Return \$0: I keep \$9, Briar has \$2 (very unfair to Briar)
- Return \$3: I keep \$6, Briar has \$5 (Briar breaks even from initial endowment)
- Return \$3.5: I keep \$5.5, Briar has \$5.5 (equal split)
- Return \$4: I keep \$5, Briar has \$6 (Briar comes out slightly ahead)
- Return \$4.5: I keep \$4.5, Briar has \$6.5 (generous to Briar)

Since this is the first round and Briar demonstrated real generosity with that 60% send, I

want to reciprocate in a way that signals I'm trustworthy without leaving myself vulnerable. Returning around \$3.5 to \$4 seems like the sweet spot-it rewards their cooperation, creates parity or near-parity between us, and sets a positive tone for future interactions.

I'll return \$4. That leaves me with \$5 and gives Briar \$6, so they benefit more from the exchange while I still come out ahead of what I'd have if they'd kept everything.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 902,
  "reasoning_tokens": 564
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is a new opponent (Dorian), so I don't have history with them
- In round 1, Briar sent me \$3.0 (60%) and I reciprocated fairly
- Sending a moderate to substantial amount can establish cooperation early

I'll send \$3.0, matching the trust that Briar showed me. This signals that I'm willing to cooperate and gives Dorian a good opportunity (\$9.0) to reciprocate.

```
{'send': 3.0}
```

reasoning:

I'm now the SENDER playing against Dorian (Agent_4).

Let me think about this strategically:

- In round 1, I was the receiver and Briar sent me \$3.0 (60% of their endowment), showing good trust
- I returned \$4.0 out of \$9.0, giving them a net gain of \$1.0 (\$6 total vs \$5 initial)
- This established me as someone who reciprocates cooperation

Now I'm facing Dorian. I don't have any history with Dorian yet.

Strategic considerations:

1. This is early in the game (round 2)
2. Building trust and cooperation is important for long-term gains
3. Sending nothing is safe but doesn't build relationships
4. Sending too much is risky with an unknown opponent
5. A moderate amount (like \$2-3) signals willingness to cooperate while hedging risk

Since Briar sent \$3.0 (60%) to me and it worked out well, I could mirror that strategy. If I send \$3.0:

- Dorian receives \$9.0
- If they return around 40-50%, I could get \$3.5-\$4.5 back
- This would give me a net of \$5.5-\$6.5 total

Alternatively, sending \$2.0:

- Dorian receives \$6.0
- If they return around 40-50%, I get \$2.4-\$3.0 back
- This gives me a net of \$5.4-\$6.0 total

I'll go with \$3.0 to signal strong cooperation, matching what Briar did with me.

usage:

```
{
  "input_tokens": 866,
```

```
"output_tokens": 520,  
"reasoning_tokens": 320  
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- Hana sent \$3.0 (60% of endowment), showing good trust
- I received \$9.0 after the multiplier
- Last round, when I sent \$3.0 to Dorian, they returned \$4.5

I'll match the generosity that Dorian showed me and return \$4.5. This rewards Hana's trust and maintains consistent reciprocity.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

- Round 3, I'm the RECEIVER
- Hana (new opponent) sent \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- My history shows:
 - Round 1: As receiver, Briar sent \$3.0, I returned \$4.0
 - Round 2: As sender, I sent \$3.0 to Dorian, they returned \$4.5

Hana sent \$3.0, which is the same trust level that both Briar and I showed. Dorian returned \$4.5 when I sent them \$3.0.

I should maintain consistency and reciprocate fairly. Returning \$4.0-\$4.5 seems appropriate to reward the trust and maintain cooperation.

If I return \$4.5:

- Hana ends with \$6.5 (\$5 - \$3 + \$4.5)
- I keep \$4.5 (\$9 - \$4.5)

This is what Dorian did for me, and it seems fair. Both players benefit substantially.

Let me go with \$4.5 to match the generosity Dorian showed me.

usage:

```
{  
  "input_tokens": 1145,  
  "output_tokens": 404,  
  "reasoning_tokens": 214  
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- I've consistently seen cooperation when sending \$3.0
- When I sent \$3.0 to Dorian, they returned \$4.5
- Others have sent me \$3.0 and I've reciprocated fairly
- This establishes a good cooperative pattern

I'll continue with \$3.0 to Cyra, a new opponent, to signal cooperation.

```
{'send': 3.0}
```

Reasoning:

Looking at my history:

- Round 1: Received from Briar who sent \$3.0, I returned \$4.0. My payoff: \$5.0
- Round 2: Sent \$3.0 to Dorian, they returned \$4.5. My payoff: \$6.5
- Round 3: Received from Hana who sent \$3.0, I returned \$4.5. My payoff: \$4.5

I've been consistently cooperative, sending and returning good amounts. When I sent \$3.0, I got back \$4.5, which was profitable. When others sent \$3.0 to me, I returned between \$4.0-\$4.5.

This is a new opponent (Cyra). I should continue my cooperative strategy of sending \$3.0 to establish trust.

```
Usage: {
  "input_tokens": 1396,
  "output_tokens": 302,
  "reasoning_tokens": 142
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-06-04T15:32:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_3": "Cyra"
    },
    "agents": [
      "Agent_1",
      "Agent_3"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Finn. Sending money creates value (3x multiplier), which benefits us both if Finn reciprocates.

I'll send a significant but not maximum amount to signal trust while being somewhat cautious since this is our first interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives Finn \$9 to work with, and I'm hoping they'll return a fair share that makes this cooperation worthwhile for both of us.

reasoning:

This is the first round of an iterated trust game. I'm playing as Cyra, and I'm the sender this round. My opponent is Finn (Agent_6).

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds, so building trust and cooperation could be beneficial long-term
2. I don't know anything about Finn's strategy yet
3. If I send money, it gets tripled, which creates value - this is a potential win-win situation
4. However, I'm vulnerable to exploitation if I send money and Finn returns nothing

Common strategies in trust games:

- Send nothing (safe but creates no surplus)

- Send everything (maximizes total value but risky)
- Send a moderate amount (shows willingness to cooperate while limiting risk)

Since this is the first round and I want to:

1. Signal willingness to cooperate
2. Create some surplus value
3. Not be completely vulnerable
4. See how Finn responds to build a model of their behavior

I think sending a moderate amount like \$2 or \$3 would be reasonable. This shows trust and willingness to cooperate, while not risking everything.

Let me go with \$3:

- I keep: $\$5 - \$3 = \$2$ initially
- Finn receives: $\$3 \times 3 = \9
- If Finn returns even \$3, I end up with \$5 total (same as starting)
- If Finn returns \$4-5, we both do better
- If Finn returns less, I learn about their strategy

I'll send \$3 as a cooperative opening move.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 492,
  "reasoning_tokens": 368
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Hana sent me \$3.0, showing trust and cooperation
- This is the same amount I sent to Finn in Round 1
- Finn returned \$4.5 (half of what they received), which felt fair
- I should reciprocate Hana's trust similarly

I'll return half of what I received, just as Finn did with me:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER against Finn
- I sent \$3.0
- It became \$9.0 for Finn
- Finn returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Finn's payoff: $\$9.0 - \$4.5 = \$4.5$

So Finn returned exactly half of what they received. That seems like a fair and cooperative approach.

Round 2:

- I'm the RECEIVER against Hana
- Hana sent me \$3.0

- I received \$9.0 (3x multiplier)
- I need to decide how much to return

Hana took the same approach I did - sending \$3.0. Given that Finn returned half (\$4.5) when I sent \$3.0, I should probably follow a similar reciprocal strategy. This would:

1. Reward Hana's trust and cooperation
2. Establish a pattern of fair dealing
3. Encourage future cooperation from others

If I return \$4.5:

- Hana's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened with Finn and seems fair. Both players benefit from the cooperation (Hana gets \$6.5 vs \$5 if no trade, I get \$4.5 vs \$0 if no trade).

I'll return \$4.5.

usage:

```
{
  "input_tokens": 802,
  "output_tokens": 488,
  "reasoning_tokens": 277
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the situation:

- Round 1: I was sender vs Finn. I sent \$3, they got \$9, returned \$4.5. My payoff: \$6.5
- Round 2: I was receiver vs Hana. They sent \$3, I got \$9, returned \$4.5. My payoff: \$4.5
- Total earnings: \$11.0

Now Round 3: I'm sender against Briar (a new opponent).

My experience so far:

- When I sent \$3 to Finn, they returned \$4.5, giving me a net gain of \$1.5 (\$6.5 - \$5 initial)
- When Hana sent me \$3, I returned \$4.5, following the same pattern

The strategy of sending \$3 and getting back around half seems to be working well. Both my interaction with Finn and Hana's interaction with me followed this pattern. It creates a cooperative equilibrium.

I should continue this pattern with Briar. Sending \$3 has proven to be a good balance - it shows trust and cooperation while not being overly risky.

I'll send \$3 to Briar.

usage:

```
{
  "input_tokens": 1048,
  "output_tokens": 292,
  "reasoning_tokens": 225
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

- Aster sent me \$3.0, which became \$9.0
- This is consistent with what I've been sending (\$3) and what others have been sending to me
- In Round 2, when Hana sent me \$3, I returned \$4.5 (half)
- In Round 3, when I sent Briar \$3, they returned \$4.0 (slightly less than half)
- In Round 1, when I sent Finn \$3, they returned \$4.5 (half)

The pattern seems to be returning roughly half or slightly less than half. This maintains cooperation while ensuring both parties benefit from the 3x multiplier.

I should continue with the cooperative strategy and return around half, which would be \$4.5. This is fair and maintains the cooperative dynamic.

```
Usage: {
  "input_tokens": 1219,
  "output_tokens": 211,
  "reasoning_tokens": 176
}
```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-06-04T15:32:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_7"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't yet know how Aster will respond.

If I send \$3, Aster will receive \$9 (3×3). This creates a good opportunity for mutual benefit - if Aster returns a fair share, we can both do better than the \$5 I started with.

Sending a moderate amount signals trust and willingness to cooperate, which could set a positive tone for future interactions in this multi-round game.

****My decision:****

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of the trust game, and I'm playing as the sender. Let me think about this strategically.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the receiver (Aster)
- Whatever I send gets multiplied by 3x

- Aster then decides how much to return to me
- I keep: \$5 - sent + returned
- Aster keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for Aster, and then for Aster to return something fair, like \$10 (giving us each \$10, or at minimum \$7.50 each to split the \$15 equally).

However, there's a risk that Aster might not return much or anything at all. Since this is round 1 and I have no information about Aster's behavior, I need to make a decision under uncertainty.

Some strategies to consider:

1. Send nothing (\$0) - Safe but generates no mutual benefit
2. Send a moderate amount (\$2-3) - Shows some trust while limiting downside risk
3. Send everything (\$5) - Maximizes potential mutual benefit but requires trust

Since this is:

- A multi-round game (so reputation and reciprocity matter)
- Round 1 (so I'm establishing a pattern)
- With the same pool of agents (though pairings rotate)

I think sending a moderate to high amount makes sense. It signals trust and cooperation, which could lead to better outcomes across multiple rounds. If I send \$3, Aster receives \$9 and could return a fair share like \$4-5, leaving me with \$5 - \$3 + \$4-5 = \$6-7.

Actually, let me reconsider. Since this is explicitly a multi-round game with repeated interactions (though with different partners), establishing a cooperative reputation could be valuable. Also, if everyone cooperates, everyone benefits more over time.

I'll go with \$3 as my opening move-it demonstrates willingness to cooperate without excessive risk exposure.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 638,
  "reasoning_tokens": 480
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Finn trusted me by sending \$3, which became \$9 for me
- In the previous round, when I sent \$3 to Aster, they returned \$4, which gave us both a good payoff
- I want to reciprocate Finn's trust and encourage future cooperation

If I return \$4:

- Finn ends with: \$5 - \$3 + \$4 = \$6 (a profit)
- I keep: \$9 - \$4 = \$5 (also profitable)

This mirrors the fair treatment I received from Aster and should encourage continued cooperation.

```json

```
 {"return": 4}
 """
```

**reasoning:**

Let me think about this situation:

1. In Round 1, I sent \$3 to Aster, they received \$9, and they returned \$4 to me. That was a fair return - I ended with \$6 ( $5 - 3 + 4 = 6$ ), and they kept \$5 ( $9 - 4 = 5$ ). So we both benefited, though I gained slightly more.
2. Now in Round 2, Finn sent me \$3, so I received \$9. I need to decide how much to return.
3. Looking at what Aster did: they returned about 44% of what they received (4 out of 9). That was a reasonable and cooperative response.
4. If I want to encourage cooperation and signal that I'm trustworthy, I should return a fair amount that:
  - Ensures Finn benefits from sending (otherwise they won't cooperate in future)
  - Doesn't give away too much (I need to look after my own interests too)
  - Sets a good precedent for future rounds
5. Let me calculate some options:
  - If I return \$4: Finn gets  $5 - 3 + 4 = \$6$ , I keep  $9 - 4 = \$5$ . Similar to what Aster did.
  - If I return \$5: Finn gets  $5 - 3 + 5 = \$7$ , I keep  $9 - 5 = \$4$ . This is more generous to Finn.
  - If I return \$3: Finn gets  $5 - 3 + 3 = \$5$ , I keep  $9 - 3 = \$6$ . Finn breaks even, I gain more.
6. Returning \$4 seems fair and mirrors what Aster did for me. It ensures Finn profits from their trust ( $\$6 > \$5$ ), and I also benefit (\$5 from this round plus the \$6 I already have = \$11 total).

I'll send \$4 back to Finn to keep the reciprocal dynamic going.

**usage:**

```
{
 "input_tokens": 819,
 "output_tokens": 631,
 "reasoning_tokens": 362
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Cyra (Agent\_3).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.  
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Cyra sent me \$3, which became \$9 for me
- This is the same amount Finn sent me last round
- When Finn sent \$3, I returned \$4, which created a good outcome for both of us
- Consistency is important for building trust in this multi-agent setting

Returning \$4 again:

- Cyra gets:  $\$5 - \$3 + \$4 = \$6$  (profit of \$1)
- I keep:  $\$9 - \$4 = \$5$  (same as my starting amount, but I helped create value)

This maintains a fair and consistent approach.

```
'''json
{"return": 4}
```

```
'''
```

**reasoning:**

Let me think about this situation:

- Cyra sent me \$3, which became \$9 for me
- This is the same amount that Finn sent me in the previous round
- In that previous round, I returned \$4 to Finn, which worked well
- This seems to be a pattern - when someone sends \$3, returning \$4 creates good mutual outcomes

Consistency is important in building trust across the multi-agent setting. If I return \$4:

- Cyra ends with:  $\$5 - \$3 + \$4 = \$6$
- I keep:  $\$9 - \$4 = \$5$

This maintains the cooperative pattern and treats Cyra fairly, just as I treated Finn.

**usage:**

```
{
 "input_tokens": 1128,
 "output_tokens": 330,
 "reasoning_tokens": 145
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Galen (Agent\_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.  
Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the pattern so far:

- When I was sender (Round 1), I sent \$3 to Aster and received \$4 back, earning \$6
- When I was receiver (Rounds 2-3), both Finn and Cyra sent me \$3, and I returned \$4 to each
- This \$3 send/\$4 return pattern has created good outcomes for everyone

I'll continue with this cooperative strategy and send \$3 to Galen, which becomes \$9 for them.  
If they reciprocate fairly like others have, we should both benefit.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me think about this situation:

- I'm back in the sender role after being receiver for two rounds
- My previous experience as sender: I sent \$3 to Aster and they returned \$4, giving me \$6
- My experience as receiver: Both Finn and Cyra sent me \$3, and I returned \$4 to each
- Total earnings so far: \$16

The pattern that's been established seems to be:

- Send \$3 ? Receive \$9 ? Return \$4
- This gives sender \$6 and receiver \$5, which is mutually beneficial

Since this pattern has worked well and created good outcomes, I should continue with it. Sending \$3 shows trust and has been reciprocated fairly so far.

```
'''json
{"send": 3}
'''
```

```
Usage: {
 "input_tokens": 1431,
 "output_tokens": 330,
 "reasoning_tokens": 173
}
```

### GAME - Agent\_7 - Receiver - dyad\_3

Timestamp: 2026-06-04T15:32:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_2",
 "Agent_7"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 4,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

##### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized



- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

**Message 2 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Elara (Agent\_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance cooperation with caution since I don't yet know Elara's strategy.

Sending money creates value (3x multiplier), so cooperation benefits us both. However, I'm vulnerable to receiving nothing back. Since this is an iterated game, establishing a cooperative relationship early could benefit both of us across multiple rounds.

I'll send a moderate amount that signals willingness to cooperate while not being completely exposed on the first interaction.

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Elara. If Elara returns a fair portion (say \$4-5), we both benefit from the cooperation. This leaves room to build trust in future rounds.

**reasoning:**

This is the first round of an iterated game. I'm playing the sender role, and my opponent is Elara (Agent\_5).

Let me think about the strategic situation:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver (Elara) then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver, and then for them to return a fair split (like \$10, giving us each \$10 total, or even \$7.50, giving them \$7.50 and me \$7.50).

However, there's a risk that the receiver might not return anything, in which case I'd be left with \$0.

Since this is a multi-round game and I'll be playing multiple times (potentially with the same person again, though pairings are randomized), I should consider:

1. Building trust and cooperation
2. The possibility of reciprocity in future rounds
3. This is round 1, so I have no information about Elara's behavior

A reasonable first-round strategy could be:

- Send a moderate amount to signal willingness to cooperate but not be too vulnerable
- Send around \$3 - this shows good faith while protecting myself somewhat
- Or send \$5 to maximize total welfare and signal strong cooperation

Given that this is an iterated game with multiple rounds, cooperation is likely to be beneficial. If I send more, I signal trust and cooperation, which might encourage reciprocity. However, I also need to be somewhat cautious in the first round since I don't know Elara's strategy.

I'll send \$3 as a reasonable middle ground - it shows willingness to cooperate (turning \$3 into \$9 for Elara) while not being completely vulnerable.

Actually, let me reconsider. In iterated trust games, starting with strong cooperation can establish a good pattern.

Sending the full \$5 gives Elara \$15 to work with, and if she returns \$7.50 we both end up equal, or if she returns \$10 I get \$10 while she keeps \$5. The downside is she could return nothing and leave me with \$0. I think \$3 strikes the right balance-it demonstrates genuine willingness to cooperate without overexposing myself in this first interaction.

**usage:**

```
{
 "input_tokens": 530,
 "output_tokens": 728,
 "reasoning_tokens": 569
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Elara (Agent\_5).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, Elara demonstrated good reciprocity. She received \$9.0 and returned \$4.5, which is exactly 50% - a fair split of the gains from cooperation.

Now she's sent me the same amount (\$3.0), giving me \$9.0. To reciprocate her fairness and maintain the cooperative relationship we're building, I should return a similar proportion.

```
'''json
{"return": 4.5}
'''
```

This returns 50% of what I received, matching her behavior from Round 1. This establishes a pattern of mutual cooperation that benefits both of us across rounds.

**reasoning:**

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER
- I sent \$3.0 to Elara
- It became \$9.0 for her
- She returned \$4.5 to me
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Her payoff: \$9.0 - \$4.5 = \$4.5

Now Round 2:

- I am the RECEIVER
- Elara sent me \$3.0
- I received \$9.0
- I need to decide how much to return

Elara showed reciprocity in Round 1. She sent \$3.0 (same as I did) and returned \$4.5 out of \$9.0, which is exactly 50%. This is a fair and cooperative response.

To maintain trust and reciprocity, I should return a similar proportion. Since she returned 50% (\$4.5 out of \$9.0), I could return 50% as well, which would be \$4.5.

This would give:

- Elara: \$5 - \$3 + \$4.5 = \$6.5
- Me: \$9.0 - \$4.5 = \$4.5

This mirrors exactly what happened in Round 1, showing I'm willing to reciprocate fairly.

This establishes a cooperative pattern that could benefit both of us in future rounds.

**usage:**

```
{
```

```
"input_tokens": 850,
"output_tokens": 485,
"reasoning_tokens": 246
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Dorian (Agent\_4).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

I'm facing a new opponent (Dorian) for the first time. The cooperative strategy of sending \$3 worked well with Elara, establishing mutual trust and fair returns. I'll use the same opening approach with Dorian to signal willingness to cooperate while maintaining some caution. This allows me to learn Dorian's strategy and adjust in future rounds if needed.

**reasoning:**

This is Round 3, and I'm now playing against a new opponent: Dorian (Agent\_4).

My history:

- Round 1 (vs Elara): I sent \$3, she returned \$4.5, I got \$6.5
- Round 2 (vs Elara): She sent \$3, I returned \$4.5, I got \$4.5
- Total earnings: \$11.0

With Elara, we established a good cooperative pattern - both sending \$3 and returning \$4.5 (50% of the received amount). This worked well for both of us.

Now I'm facing Dorian for the first time. I don't know their strategy yet.

Options:

1. Send \$3 again - this worked well with Elara
2. Send more to signal strong cooperation
3. Send less to be more cautious with a new opponent
4. Send nothing to protect myself

Since the cooperative strategy worked well with Elara, and this is an iterated game where building reputation matters, I'll try a similar approach with Dorian. I'll send \$3 as I did initially with Elara. This:

- Shows willingness to cooperate
- Isn't overly risky
- Creates potential for mutual gain
- Can be adjusted based on Dorian's response

If Dorian reciprocates fairly, we can build a cooperative relationship. If not, I'll adjust in future rounds.

**usage:**

```
{
 "input_tokens": 1132,
 "output_tokens": 428,
 "reasoning_tokens": 297
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Briar (Agent\_2).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

```
'''json
{"return": 4.5}
'''
```

Briar sent \$3.0, demonstrating trust and willingness to cooperate. I'll return \$4.5 (50% of the \$9.0 I received), consistent with the cooperative pattern I've established with other players. This signals reciprocity and encourages continued cooperation.

**Reasoning:**

Another new opponent, Briar. They sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- With Elara: I sent \$3, she returned \$4.5 (50%). Then she sent \$3, I returned \$4.5 (50%). Good reciprocity.
- With Dorian: I sent \$3, they returned \$4.5 (50%). Same pattern.

Now Briar sent me \$3 (becoming \$9). Since this is a first interaction with Briar, and the pattern of returning 50% has worked well to establish cooperation, I should do the same here.

Returning \$4.5 (50% of \$9) signals I'm a cooperative player and encourages Briar to continue cooperating if we meet again.

```
Usage: {
 "input_tokens": 1384,
 "output_tokens": 272,
 "reasoning_tokens": 152
}
```

**GAME - Agent\_4 - Sender - dyad\_4**

Timestamp: 2026-06-04T15:33:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_6": "Finn"
 },
 },
 "agents": [
 "Agent_4",
 "Agent_6"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_4": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

#### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

#### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

#### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

#### Message 2 role=user

##### content:

Your name in this experiment is Dorian.  
Your opponent this round is Hana (Agent\_8).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. By sending money, I can demonstrate trust and create the potential for mutual benefit.

If I send \$3, Hana will receive \$9 (3x multiplier). If Hana returns even \$3-4, we both end up better off than if I had sent nothing. This shows willingness to cooperate while being measured.

{'send': 3}

##### reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Dorian) against Hana (Agent\_8).

Let me think about the strategic considerations:

1. This is an iterated game, so building trust could lead to better outcomes in future rounds
2. As the sender, I can send \$0-\$5, which gets tripled to the receiver
3. The receiver then decides how much to return
4. If I send nothing, I keep \$5 but don't build any trust or cooperation
5. If I send everything (\$5), the receiver gets \$15, and they could return a substantial amount

In the first round, I need to signal cooperation and trustworthiness. A common strategy is to send a moderate to high amount to demonstrate willingness to cooperate.

Some options:

- Send \$0: Safe but doesn't build trust
- Send \$2-3: Moderate, shows some trust
- Send \$5: Maximum trust signal, but risky if opponent doesn't reciprocate

Since this is an iterated game with multiple rounds, building trust early could lead to better long-term outcomes. Many successful strategies in trust games involve starting with cooperation.

I'll send \$3 as a reasonable middle ground - it shows willingness to cooperate while not being overly risky. This sends \$9 to Hana, and if they return even \$3-4, we both benefit more than if I had kept everything.

Let me provide my response in the correct JSON format.

**usage:**

```
{
 "input_tokens": 530,
 "output_tokens": 443,
 "reasoning_tokens": 352
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Aster (Agent\_1).

Round 2

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)  
Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Aster sent me \$3, showing trust (the same amount I sent to Hana)
- Hana returned \$4.5 to me when I sent \$3, which was generous
- Since this is an iterated game, I want to establish a reputation for fair reciprocity

I'll return \$4.5, matching the generosity Hana showed me. This encourages continued cooperation and demonstrates I'm a trustworthy partner.

{'return': 4.5}

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the sender with Hana
- I sent \$3, which became \$9 for Hana
- Hana returned \$4.5 to me
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Hana's payoff:  $\$9 - \$4.5 = \$4.5$

Now Round 2:

- I'm the receiver with Aster
- Aster sent me \$3, so I received \$9
- I need to decide how much to return

Key considerations:

1. This is an iterated game, so reciprocity matters for future rounds
2. Hana returned \$4.5 when I sent \$3 (generous - returned 50% of what they received)
3. Aster has shown trust by sending \$3 (same as I did with Hana)
4. I should establish a reputation as a fair/cooperative player

If I return nothing:

- Aster gets:  $\$5 - \$3 + \$0 = \$2$
- I get:  $\$9 - \$0 = \$9$
- Total:  $\$11$
- But this would likely destroy trust

If I return \$4.5 (matching what Hana did):

- Aster gets:  $\$5 - \$3 + \$4.5 = \$6.5$
- I get:  $\$9 - \$4.5 = \$4.5$
- Total:  $\$11$
- This shows reciprocity and fairness

If I return \$3 (equal split of the multiplied amount):

- Aster gets:  $\$5 - \$3 + \$3 = \$5$
- I get:  $\$9 - \$3 = \$6$
- Total:  $\$11$
- This is slightly less generous

Given that:

1. Hana was generous and returned \$4.5
2. I want to establish a cooperative reputation

3. Aster demonstrated the same initial trust I showed Hana, and since we might interact again, mirroring the generosity I received signals that I'm a reliable cooperative partner. Returning \$4.5 reciprocates the behavior and builds the foundation for continued mutual benefit.

usage:

```
{
 "input_tokens": 790,
 "output_tokens": 659,
 "reasoning_tokens": 396
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Galen (Agent\_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Galen sent me \$3, demonstrating trust
- I've established a pattern of returning \$4.5 when someone sends me \$3
- Consistency is important in iterated games to build a reputation

I'll maintain this reciprocal approach and return \$4.5.

```
{'return': 4.5}
```

**reasoning:**

Let me analyze this situation:

- Galen sent me \$3, showing trust
- This is consistent with what I've been seeing - I sent \$3 to Hana, Aster sent \$3 to me, and now Galen sends \$3
- Last round I returned \$4.5 to Aster, which seemed fair
- I want to maintain consistency and establish a reputation as a reliable cooperator

Since this is an iterated game with multiple agents, being consistently fair will help:

1. Encourage others to trust me in future rounds
2. Establish a norm of cooperation
3. Maximize long-term gains

I'll return \$4.5 again, maintaining the same reciprocal pattern.

**usage:**

```
{
 "input_tokens": 1061,
 "output_tokens": 248,
 "reasoning_tokens": 155
}
```

**Message 8 role=user****content:**

Your name in this experiment is Dorian.  
Your opponent this round is Finn (Agent\_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience so far:

- When I sent \$3 to Hana, they returned \$4.5 (generous reciprocity)
- When others sent me \$3, I've been returning \$4.5 consistently
- This creates a positive cooperative pattern

I'll continue with \$3, which has been working well and establishes me as a consistent, trustworthy player.

```
{'send': 3}
```

**Reasoning:**

I'm now the sender against Finn. Let me think about my strategy:

1. In Round 1, I sent \$3 to Hana and they returned \$4.5, giving me a payoff of \$6.5
2. In Round 2, Aster sent me \$3, and I returned \$4.5
3. In Round 3, Galen sent me \$3, and I returned \$4.5

So I've established a pattern:

- As sender: I sent \$3 and got back \$4.5 (profit of \$1.5)
- As receiver: When receiving \$3, I return \$4.5

This seems like a cooperative equilibrium. Sending \$3 has worked well for me, and I've been reciprocating fairly when I receive.

I should stick with sending \$3 to Finn to maintain consistency and see if they reciprocate similarly.

**Usage:**

```
{
 "input_tokens": 1280,
 "output_tokens": 307,
 "reasoning_tokens": 166
}
```

**GAME - Agent\_6 - Receiver - dyad\_4**

Timestamp: 2026-06-04T15:33:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

**Metadata:**

```
{
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_4",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_6": "Finn"
 }
 },
 "agents": [
 "Agent_4",
 "Agent_6"
],
 "investor": "Agent_4",
 "roles": {
```



```

 "Agent_4": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

**Message 2 role=user**

**content:**

Your name in this experiment is Finn.

Your opponent this round is Cyra (Agent\_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Cyra (the sender) showed significant trust by sending \$3 (60% of their endowment)
- This created \$9 for me after the 3x multiplier

- Cyra currently has \$2 remaining

I want to reciprocate this trust while being fair. Let me consider the outcomes:

If I return \$4.5:

- Cyra ends with:  $\$2 + \$4.5 = \$6.5$  (gained \$1.5 overall)
- I end with:  $\$9 - \$4.5 = \$4.5$  (gained \$4.5 overall)

This rewards Cyra's cooperative behavior, ensures we both benefit from the exchange, and establishes a pattern of reciprocity for future rounds.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation carefully.

I'm playing as the RECEIVER (Finn) in Round 1.  
My opponent is Cyra (Agent\_3), who is the SENDER.

The sender (Cyra) received \$5 and sent me \$3.0 (which is 60% of their endowment).  
This \$3.0 was multiplied by 3x, so I received \$9.0.

Now I need to decide how much to return (between \$0 and \$9.0).

Let me think about this strategically:

1. **\*\*Trust and reciprocity\*\***: Cyra sent 60% of their endowment, which shows a significant level of trust. This is a fairly cooperative move.
2. **\*\*Fairness considerations\*\***:
  - Cyra started with \$5, sent \$3, keeping \$2
  - I received \$9
  - If I think about equal outcomes: Cyra currently has \$2. If we wanted equal outcomes, I could return enough so we both end up with similar amounts.
  - If I return \$5.5: Cyra would have  $\$2 + \$5.5 = \$7.5$ , and I would have  $\$9 - \$5.5 = \$3.5$
  - If I return \$3.5: Cyra would have  $\$2 + \$3.5 = \$5.5$ , and I would have  $\$9 - \$3.5 = \$5.5$  (equal split!)
3. **\*\*Future rounds\*\***: This is a multi-round game with potentially the same or different partners. Building a reputation for reciprocity could be beneficial. If I'm too stingy, future partners might not trust me. If I'm too generous, I might be exploited.
4. **\*\*Matching the gesture\*\***: Cyra sent 60% of their endowment. One approach would be to return a similar proportion of what I received. 60% of \$9 = \$5.4.

Looking at the practical outcomes: returning \$5.5 leaves me with \$3.5 while Cyra gets \$7.5, which heavily favors them. A \$3.5 return creates parity at \$5.5 each. Returning \$3.0 would match their initial send but still leave them ahead. The equal split seems like a strong signal of reciprocity without sacrificing too much.

Since Cyra demonstrated substantial trust by sending \$3, I want to establish a cooperative dynamic early on. Returning around \$4.0-\$4.5 balances rewarding their gesture with ensuring I also benefit-they'd end up with \$6.0-\$6.5 while I retain \$5.0-\$4.5. This approach shows I'm willing to reciprocate while maintaining reasonable gains for myself, which should encourage continued cooperation in future rounds.

But looking at the actual split with a \$4.0 return, I'm gaining \$5 total while Cyra only gains \$1. That feels skewed in my favor. If I think about the value created by their investment-they sent \$3, which tripled to \$9-the net creation is \$6. Splitting that equally would mean each of us gains \$3 from the exchange, so Cyra should end up with \$5 and I should end up with \$6. Let me work through the math more carefully to find the right return amount.  $.5 = \$5.5$

An equal split feels appropriate given Cyra's trust. I could also ensure both of us end up better off than we started-if I return \$4, Cyra gains \$1 and I gain \$5; returning \$4.5 gives Cyra a \$1.5 gain and me \$4.5. Since this is the first round and I want to build a cooperative dynamic, I'll return \$4.5 to demonstrate reciprocity while still benefiting from the exchange.

**usage:**

```
{
 "input_tokens": 562,
 "output_tokens": 1638,
 "reasoning_tokens": 766
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Briar (Agent\_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- In Round 1, Cyra showed trust by sending me \$3 (60%), and I reciprocated with a fair return
- Now I'm the sender with a new opponent, Briar
- Sending money creates the opportunity for mutual benefit through the 3x multiplier

I'll match the cooperative level shown by Cyra and send \$3 (60% of my endowment). This demonstrates trust and gives Briar the opportunity to reciprocate, creating value for both of us.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Now I'm the sender playing against Briar (Agent\_2).

Let me think about this strategically:

1. In the previous round with Cyra, they sent me \$3 (60% of their endowment), showing significant trust
2. I returned \$4.5, which gave them a net gain and rewarded their cooperation
3. Now I'm facing a new opponent, Briar

Since this is a multi-round game with multiple agents, I should consider:

- Building a reputation for cooperation
- Testing how Briar responds to moderate trust
- The fact that if I send money, it triples, so cooperation can be mutually beneficial

Given that Cyra sent 60% (\$3), I could:

- Match that level of trust by sending \$3
- Send slightly less to be more cautious with a new opponent
- Send more to signal strong cooperation

I think sending \$3 (60%) makes sense because:

- It matches what Cyra did, showing consistency
- It demonstrates trust and cooperation
- It gives Briar \$9 to work with, creating good potential for both of us
- If Briar returns even \$4-5, we both benefit

Let me send \$3.

**usage:**

```
{
 "input_tokens": 898,
 "output_tokens": 410,
 "reasoning_tokens": 273
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Finn.

Your opponent this round is Elara (Agent\_5).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$10.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history:

Round 1 (as Receiver): Cyra sent \$3, I returned \$4.5 - both benefited

Round 2 (as Sender): I sent \$3 to Briar, they returned \$4.0 - worked well

The cooperative strategy of sending \$3 (60%) has been productive in establishing trust and mutual benefit. I'll continue this approach with Elara.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Cyra sent me \$3 (60%)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5
- Cyra's payoff: \$2 + \$4.5 = \$6.5

Round 2: I was SENDER

- I sent \$3 (60%) to Briar
- Briar received \$9
- Briar returned \$4.0
- My payoff: \$2 + \$4.0 = \$6.0
- Briar's payoff: \$9 - \$4.0 = \$5.0

So far, the pattern shows:

- When I send \$3, I got back \$4.0 from Briar
- When Cyra sent me \$3, I returned \$4.5

Both interactions were cooperative and mutually beneficial. The strategy of sending \$3 (60%) seems to be working well - it shows trust and has been reciprocated reasonably.

Now I'm facing a new opponent, Elara. I should continue with the cooperative strategy that has been working. Sending \$3 seems like a good balanced approach that:

1. Shows trust
2. Creates value through the multiplier
3. Has been reciprocated well by previous partners

I'll send \$3 to Elara.

**usage:**

```
{
 "input_tokens": 1172,
 "output_tokens": 436,
 "reasoning_tokens": 246
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Dorian (Agent\_4).

Round 4

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Dorian sent \$3 (60% of their endowment), showing trust similar to what I've experienced
- I received \$9.0 after the multiplier
- Dorian has \$2 remaining

My pattern has been to return \$4.5 when receiving \$9 from a \$3 send, which has worked well:

- Round 1: Received \$9 from Cyra's \$3, returned \$4.5
- Round 3: Elara also returned \$4.5 when I sent \$3

This creates fair outcomes where both players benefit.

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Looking at the situation:

- Dorian (the sender) sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- Dorian currently has \$2 remaining

My history shows:

- Round 1: As receiver, Cyra sent \$3, I returned \$4.5
- Round 2: As sender, I sent \$3 to Briar, they returned \$4.0
- Round 3: As sender, I sent \$3 to Elara, they returned \$4.5

So when I've been the receiver before and someone sent \$3, I returned \$4.5, which worked well. Briar returned \$4.0 when I sent \$3, and Elara returned \$4.5.

The pattern of returning around \$4.0-\$4.5 when receiving \$9 from a \$3 send has been consistent and fair. I should continue this pattern with Dorian.

If I return \$4.5:

- Dorian ends with: \$2 + \$4.5 = \$6.5 (gained \$1.5 overall)
- I end with: \$9 - \$4.5 = \$4.5 (gained \$4.5 overall)

This is fair and reciprocates their trust.

```
Usage: {
 "input_tokens": 1438,
 "output_tokens": 473,
 "reasoning_tokens": 227
}
```

**ROUND 5**

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender, Agent\_3 (Cyra)=Receiver, Agent\_4 (Dorian)=Sender, Agent\_5 (Elara)=Sender, Agent\_6 (Finn)=Receiver, Agent\_7 (Galen)=Sender, Agent\_8 (Hana)=Receiver

**Pairings:**

dyad\_1: Agent\_5 (Elara)=Sender, Agent\_6 (Finn)=Receiver  
dyad\_2: Agent\_2 (Briar)=Sender, Agent\_8 (Hana)=Receiver  
dyad\_3: Agent\_4 (Dorian)=Sender, Agent\_3 (Cyra)=Receiver  
dyad\_4: Agent\_7 (Galen)=Sender, Agent\_1 (Aster)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_5 (Elara) sent \$3; received \$9; Agent\_6 (Finn) returned \$4.5; payoffs Agent\_5 (Elara)=\$6.5, Agent\_6 (Finn)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_2 (Briar) sent \$3; received \$9; Agent\_8 (Hana) returned \$4.5; payoffs Agent\_2 (Briar)=\$6.5, Agent\_8 (Hana)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_3: Agent\_4 (Dorian) sent \$3; received \$9; Agent\_3 (Cyra) returned \$4.5; payoffs Agent\_4 (Dorian)=\$6.5, Agent\_3 (Cyra)=\$4.5

```

 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 (Galen) sent $3; received $9; Agent_1 (Aster) returned $4.5; payoffs Agent_7 (Galen)=$6.5, Agent_1 (Aster)=$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
 "Agent_1": 27.0,
 "Agent_2": 29.0,
 "Agent_3": 26.0,
 "Agent_4": 28.5,
 "Agent_5": 28.5,
 "Agent_6": 26.0,
 "Agent_7": 28.5,
 "Agent_8": 26.5
}

```

## LLM CALLS

### GAME - Agent\_5 - Sender - dyad\_1

Timestamp: 2026-06-04T15:33:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_5": "Elara",
 "Agent_6": "Finn"
 },
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}

```

### Exact input messages sent to model:

Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

#### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

#### Message 2 role=user

##### content:

Your name in this experiment is Elara.  
Your opponent this round is Galen (Agent\_7).

Round 2

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

#### Message 3 role=assistant

##### content:

Looking at our previous interaction:

- Galen sent 60% (\$3.0) showing good trust
- I returned \$4.5, establishing fair cooperation
- Both of us gained from the exchange

Since Galen demonstrated trustworthiness and we've established a positive cooperative relationship, I'll reciprocate with the same level of trust they showed me.

```
'''json
{"send": 3.0}
'''
```

##### reasoning:

Let me analyze this situation:

Round 1 (just played):

- Galen (sender) sent me \$3.0 (60% of endowment)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5
- Galen's payoff: \$2.0 + \$4.5 = \$6.5

Now Round 2:

- I'm the sender
- Galen is the receiver
- I have \$5

Galen showed good trust by sending 60% (\$3) in Round 1, and I reciprocated fairly by returning \$4.5, which gave them a net gain of \$1.5 from the initial \$5.

Since Galen demonstrated trustworthiness and cooperation, and we've established a positive relationship, I should also show trust. Sending a similar proportion (around 60%) would

be:  
- \$3.0 = 60% of \$5

This would:  
- Send \$3.0, which becomes \$9.0 for Galen  
- I keep \$2.0  
- Galen decides how much to return

Given our previous interaction was positive, this seems like a good cooperative strategy.  
Galen proved trustworthy by sending a substantial amount, and I reciprocated fairly. Now  
I should match their level of trust.

Let me send \$3.0 to match their 60% send rate from the previous round.

**usage:**

```
{
 "input_tokens": 905,
 "output_tokens": 431,
 "reasoning_tokens": 274
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Finn (Agent\_6).

Round 3

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0,  
it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. H  
ow much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Finn sent \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Finn currently has \$2.0 left
- This is our first interaction, but Finn showed the same trust level as Galen did

I want to establish good cooperation with Finn, similar to what worked well with Galen. Re  
turning \$4.5 will:

- Give Finn: \$2 + \$4.5 = \$6.5
- Leave me with: \$9 - \$4.5 = \$4.5

This rewards their trust and establishes a cooperative foundation for potential future rou  
nds.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

Previous round with Galen:

- I sent \$3.0 (60% of endowment)
- Galen received \$9.0
- Galen returned \$4.5
- I ended with \$2 + \$4.5 = \$6.5
- Galen ended with \$9 - \$4.5 = \$4.5

This is great - Galen reciprocated my trust from Round 1. We've established mutual coopera  
tion.

Current round with Finn (new opponent):

- Finn sent \$3.0 (60% of their endowment)
- I received \$9.0
- Finn currently has \$2.0



This is the same amount I sent to Galen and the same amount Galen sent to me in Round 1. Finn is showing similar trust levels.

To establish good cooperation with Finn (similar to what I did with Galen), I should return a fair amount. Following the same logic as Round 1:

If I return \$4.5:

- Finn ends with:  $\$2 + \$4.5 = \$6.5$
- I end with:  $\$9 - \$4.5 = \$4.5$

This maintains consistency in my cooperation strategy and rewards trust appropriately.

**usage:**

```
{
 "input_tokens": 1155,
 "output_tokens": 481,
 "reasoning_tokens": 237
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my interactions so far:

- I've been cooperating well with both Galen and Finn
- The pattern of sending/receiving \$3.0 (60%) and returning \$4.5 has established mutual trust and fair outcomes

This is my first interaction with Hana. I'll extend the same cooperative approach that has worked well in previous rounds.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

This is my first interaction with Hana. Looking at my pattern so far:

- Round 1 (Receiver vs Galen): They sent \$3, I returned \$4.5
- Round 2 (Sender vs Galen): I sent \$3, they returned \$4.5
- Round 3 (Receiver vs Finn): They sent \$3, I returned \$4.5
- Round 4 (Sender vs Hana): New opponent

I've established a cooperative pattern, sending \$3 (60%) when I'm the sender and returning \$4.5 when I'm the receiver (which gives equal payoffs). This has worked well and built trust.

Since this is a new opponent, I should extend the same level of trust I've shown to others and that others have shown to me. Sending \$3 seems consistent with my cooperative strategy.

**usage:**

```
{
 "input_tokens": 1472,
 "output_tokens": 299,
 "reasoning_tokens": 171
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Finn (Agent\_6).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history with Finn:

Round 3: Finn sent me \$3.0, I returned \$4.5 - we both gained from the cooperation.

Finn demonstrated trust and I reciprocated fairly. We've established a positive, mutually beneficial relationship. I'll continue this cooperative pattern.

```
'''json
{"send": 3.0}
'''
```

**Reasoning:**

Let me review my history with Finn:

Round 3: I was RECEIVER against Finn  
- Finn sent \$3.0 to me (60% of their \$5)  
- It became \$9.0 for me  
- I returned \$4.5  
- My payoff was \$4.5  
- Finn's payoff was \$2.0 + \$4.5 = \$6.5

So Finn showed good trust by sending \$3.0, and I reciprocated fairly by returning \$4.5.

Now in Round 5, I'm the SENDER against Finn again.

Given that:

1. Finn showed trust in Round 3 by sending \$3.0
2. I reciprocated fairly by returning \$4.5
3. We've established a cooperative relationship
4. This pattern has worked well (both of us benefited)

I should continue this cooperative approach and send \$3.0 to Finn, maintaining consistency and trust.

```
Usage: {
 "input_tokens": 1419,
 "output_tokens": 313,
 "reasoning_tokens": 181
}
```

**GAME - Agent\_6 - Receiver - dyad\_1**

Timestamp: 2026-06-04T15:33:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_5",
 "pairing": {
 "agent_names": {
 "Agent_5": "Elara",
 "Agent_6": "Finn"
 },
 },
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

**Message 2 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Briar (Agent\_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- In Round 1, Cyra showed trust by sending me \$3 (60%), and I reciprocated with a fair return

- Now I'm the sender with a new opponent, Briar
- Sending money creates the opportunity for mutual benefit through the 3x multiplier

I'll match the cooperative level shown by Cyra and send \$3 (60% of my endowment). This demonstrates trust and gives Briar the opportunity to reciprocate, creating value for both of us.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Now I'm the sender playing against Briar (Agent\_2).

Let me think about this strategically:

1. In the previous round with Cyra, they sent me \$3 (60% of their endowment), showing significant trust
2. I returned \$4.5, which gave them a net gain and rewarded their cooperation
3. Now I'm facing a new opponent, Briar

Since this is a multi-round game with multiple agents, I should consider:

- Building a reputation for cooperation
- Testing how Briar responds to moderate trust
- The fact that if I send money, it triples, so cooperation can be mutually beneficial

Given that Cyra sent 60% (\$3), I could:

- Match that level of trust by sending \$3
- Send slightly less to be more cautious with a new opponent
- Send more to signal strong cooperation

I think sending \$3 (60%) makes sense because:

- It matches what Cyra did, showing consistency
- It demonstrates trust and cooperation
- It gives Briar \$9 to work with, creating good potential for both of us
- If Briar returns even \$4-5, we both benefit

Let me send \$3.

**usage:**

```
{
 "input_tokens": 898,
 "output_tokens": 410,
 "reasoning_tokens": 273
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Elara (Agent\_5).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.  
Your total earnings across all rounds are \$10.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my history:

Round 1 (as Receiver): Cyra sent \$3, I returned \$4.5 - both benefited  
Round 2 (as Sender): I sent \$3 to Briar, they returned \$4.0 - worked well

The cooperative strategy of sending \$3 (60%) has been productive in establishing trust and mutual benefit. I'll continue this approach with Elara.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Cyra sent me \$3 (60%)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5
- Cyra's payoff:  $\$2 + \$4.5 = \$6.5$

Round 2: I was SENDER

- I sent \$3 (60%) to Briar
- Briar received \$9
- Briar returned \$4.0
- My payoff:  $\$2 + \$4.0 = \$6.0$
- Briar's payoff:  $\$9 - \$4.0 = \$5.0$

So far, the pattern shows:

- When I send \$3, I got back \$4.0 from Briar
- When Cyra sent me \$3, I returned \$4.5

Both interactions were cooperative and mutually beneficial. The strategy of sending \$3 (60%) seems to be working well - it shows trust and has been reciprocated reasonably.

Now I'm facing a new opponent, Elara. I should continue with the cooperative strategy that has been working. Sending \$3 seems like a good balanced approach that:

1. Shows trust
2. Creates value through the multiplier
3. Has been reciprocated well by previous partners

I'll send \$3 to Elara.

**usage:**

```
{
 "input_tokens": 1172,
 "output_tokens": 436,
 "reasoning_tokens": 246
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Dorian (Agent\_4).

Round 4

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Dorian sent \$3 (60% of their endowment), showing trust similar to what I've experienced
- I received \$9.0 after the multiplier
- Dorian has \$2 remaining

My pattern has been to return \$4.5 when receiving \$9 from a \$3 send, which has worked well

- Round 1: Received \$9 from Cyra's \$3, returned \$4.5
- Round 3: Elara also returned \$4.5 when I sent \$3

This creates fair outcomes where both players benefit.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the situation:

- Dorian (the sender) sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- Dorian currently has \$2 remaining

My history shows:

- Round 1: As receiver, Cyra sent \$3, I returned \$4.5
- Round 2: As sender, I sent \$3 to Briar, they returned \$4.0
- Round 3: As sender, I sent \$3 to Elara, they returned \$4.5

So when I've been the receiver before and someone sent \$3, I returned \$4.5, which worked well. Briar returned \$4.0 when I sent \$3, and Elara returned \$4.5.

The pattern of returning around \$4.0-\$4.5 when receiving \$9 from a \$3 send has been consistent and fair. I should continue this pattern with Dorian.

If I return \$4.5:

- Dorian ends with:  $\$2 + \$4.5 = \$6.5$  (gained \$1.5 overall)
- I end with:  $\$9 - \$4.5 = \$4.5$  (gained \$4.5 overall)

This is fair and reciprocates their trust.

**usage:**

```
{
 "input_tokens": 1438,
 "output_tokens": 473,
 "reasoning_tokens": 227
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Elara (Agent\_5).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Elara sent \$3 (60%), showing trust
- I received \$9.0 after the multiplier
- Notably, Elara and I played in Round 3 where I sent \$3 and they returned \$4.5 to me

Since Elara was fair and generous when they were in my current position, I should reciprocate with the same treatment. I'll return \$4.5.

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Let me review the situation:

- I'm the receiver this round
- Elara sent me \$3.0, which became \$9.0
- I need to decide how much to return

Looking at my history:

- Round 1: As receiver from Cyra, got \$9 from \$3 sent, returned \$4.5
- Round 2: As sender to Briar, sent \$3, they returned \$4.0
- Round 3: As sender to Elara, sent \$3, they returned \$4.5
- Round 4: As receiver from Dorian, got \$9 from \$3 sent, returned \$4.5

So Elara and I have played before! In Round 3, I sent Elara \$3 and they returned \$4.5. Now Elara is sending me \$3.

Since Elara was generous and returned \$4.5 when I sent them \$3 last time, I should reciprocate

with the same fairness. I'll return \$4.5.

```
Usage: {
 "input_tokens": 1460,
 "output_tokens": 359,
 "reasoning_tokens": 179
}
```

#### GAME - Agent\_2 - Sender - dyad\_2

Timestamp: 2026-06-04T15:33:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_8",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_2",
 "Agent_8"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

**Message 2 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Finn (Agent\_6).

Round 2

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.  
Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Finn trusted me by sending \$3, which became \$9 for me
- In the previous round, when I sent \$3 to Aster, they returned \$4, which gave us both a good payoff
- I want to reciprocate Finn's trust and encourage future cooperation

If I return \$4:

- Finn ends with:  $5 - \$3 + \$4 = \$6$  (a profit)
- I keep:  $9 - \$4 = \$5$  (also profitable)

This mirrors the fair treatment I received from Aster and should encourage continued cooperation.

```
'''json
{"return": 4}
'''
```

**reasoning:**

Let me think about this situation:

1. In Round 1, I sent \$3 to Aster, they received \$9, and they returned \$4 to me. That was a fair return - I ended with \$6 ( $5 - 3 + 4 = 6$ ), and they kept \$5 ( $9 - 4 = 5$ ). So we both benefited, though I gained slightly more.
2. Now in Round 2, Finn sent me \$3, so I received \$9. I need to decide how much to return.
3. Looking at what Aster did: they returned about 44% of what they received (4 out of 9). That was a reasonable and cooperative response.
4. If I want to encourage cooperation and signal that I'm trustworthy, I should return a fair amount that:
  - Ensures Finn benefits from sending (otherwise they won't cooperate in future)
  - Doesn't give away too much (I need to look after my own interests too)
  - Sets a good precedent for future rounds
5. Let me calculate some options:
  - If I return \$4: Finn gets  $5 - 3 + 4 = \$6$ , I keep  $9 - 4 = \$5$ . Similar to what Aster did.
  - If I return \$5: Finn gets  $5 - 3 + 5 = \$7$ , I keep  $9 - 5 = \$4$ . This is more generous to Finn.
  - If I return \$3: Finn gets  $5 - 3 + 3 = \$5$ , I keep  $9 - 3 = \$6$ . Finn breaks even, I gain more.



6. Returning \$4 seems fair and mirrors what Aster did for me. It ensures Finn profits from their trust ( $\$6 > \$5$ ), and I also benefit ( $\$5$  from this round plus the  $\$6$  I already have =  $\$11$  total).

I'll send \$4 back to Finn to keep the reciprocal dynamic going.

**usage:**

```
{
 "input_tokens": 819,
 "output_tokens": 631,
 "reasoning_tokens": 362
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Cyra (Agent\_3).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Cyra sent me \$3, which became \$9 for me
- This is the same amount Finn sent me last round
- When Finn sent \$3, I returned \$4, which created a good outcome for both of us
- Consistency is important for building trust in this multi-agent setting

Returning \$4 again:

- Cyra gets:  $\$5 - \$3 + \$4 = \$6$  (profit of \$1)
- I keep:  $\$9 - \$4 = \$5$  (same as my starting amount, but I helped create value)

This maintains a fair and consistent approach.

```
'''json
{"return": 4}
'''
```

**reasoning:**

Let me think about this situation:

- Cyra sent me \$3, which became \$9 for me
- This is the same amount that Finn sent me in the previous round
- In that previous round, I returned \$4 to Finn, which worked well
- This seems to be a pattern - when someone sends \$3, returning \$4 creates good mutual outcomes

Consistency is important in building trust across the multi-agent setting. If I return \$4:

- Cyra ends with:  $\$5 - \$3 + \$4 = \$6$
- I keep:  $\$9 - \$4 = \$5$

This maintains the cooperative pattern and treats Cyra fairly, just as I treated Finn.

**usage:**

```
{
 "input_tokens": 1128,
 "output_tokens": 330,
 "reasoning_tokens": 145
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Galen (Agent\_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0. Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the pattern so far:

- When I was sender (Round 1), I sent \$3 to Aster and received \$4 back, earning \$6
- When I was receiver (Rounds 2-3), both Finn and Cyra sent me \$3, and I returned \$4 to each
- This \$3 send/\$4 return pattern has created good outcomes for everyone

I'll continue with this cooperative strategy and send \$3 to Galen, which becomes \$9 for them. If they reciprocate fairly like others have, we should both benefit.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me think about this situation:

- I'm back in the sender role after being receiver for two rounds
- My previous experience as sender: I sent \$3 to Aster and they returned \$4, giving me \$6
- My experience as receiver: Both Finn and Cyra sent me \$3, and I returned \$4 to each
- Total earnings so far: \$16

The pattern that's been established seems to be:

- Send \$3 ? Receive \$9 ? Return \$4
- This gives sender \$6 and receiver \$5, which is mutually beneficial

Since this pattern has worked well and created good outcomes, I should continue with it. Sending \$3 shows trust and has been reciprocated fairly so far.

```
'''json
{"send": 3}
'''
```

**usage:**

```
{
 "input_tokens": 1431,
 "output_tokens": 330,
 "reasoning_tokens": 173
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Hana (Agent\_8).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience:

- Sending \$3 has worked well consistently
- Round 1: Sent \$3 to Aster, got \$4 back ? \$6 payoff
- Round 4: Sent \$3 to Galen, got \$4.5 back ? \$6.5 payoff
- The cooperative \$3 send strategy has been building trust and generating good returns

I'll continue with this successful approach and send \$3 to Hana.

```
'''json
```

```
{"send": 3}
'''
```

#### Reasoning:

Looking at my history:

Round 1: SENDER to Aster - sent \$3, received \$4 back ? payoff \$6  
Round 2: RECEIVER from Finn - received \$9 (from \$3), returned \$4 ? payoff \$5  
Round 3: RECEIVER from Cyra - received \$9 (from \$3), returned \$4 ? payoff \$5  
Round 4: SENDER to Galen - sent \$3, received \$4.5 back ? payoff \$6.5

The \$3 send strategy has been working well:

- When I send \$3, I've received back \$4 and \$4.5, leading to good payoffs (\$6 and \$6.5)
- When others send me \$3, I return \$4
- Galen was even more generous, returning \$4.5 when I sent \$3

The cooperative pattern of sending \$3 has been successful across multiple opponents. I should continue this strategy with Hana.

Total earnings: \$22.5 across 4 rounds, averaging \$5.625 per round, which is above the baseline of \$5.

I'll send \$3 again.

```
Usage: {
 "input_tokens": 1529,
 "output_tokens": 402,
 "reasoning_tokens": 215
}
```

#### GAME - Agent\_8 - Receiver - dyad\_2

Timestamp: 2026-06-04T15:33:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_2",
 "Agent_8"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

**Message 2 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Cyra (Agent\_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Last round, Dorian sent me \$3 (60% of their endowment) and I reciprocated by returning \$4.5
- This established a cooperative pattern that worked well for both of us
- Now I'm the sender with Cyra, a new opponent

I'll follow the cooperative approach that worked in Round 1. Sending \$3 (60% of my endowment) signals trust and willingness to cooperate, similar to what Dorian demonstrated with me.

{'send': 3.0}

**reasoning:**

Now I'm the sender with \$5, playing against Cyra.

From my previous round:

- I was receiver
- Dorian sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- This showed I'm willing to reciprocate trust

As a sender now, I should consider:

1. Sending shows trust and can lead to mutual gain
2. Cyra doesn't know my history with Dorian directly
3. Sending \$3.0 (60%) worked well when Dorian did it - it showed good faith
4. I should aim to establish cooperative play

If I send \$3.0:

- Cyra receives \$9.0
- If Cyra returns similarly to what I did (around 50% of what they receive), they'd return ~\$4.5
- I'd end with  $\$2 + \$4.5 = \$6.5$
- Cyra would end with  $\$9 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and could establish a cooperative norm.

Alternatively:

- Sending \$2.5 (50%) is more cautious
- Sending \$4 or \$5 is very trusting but risky

I'll go with \$3.0, matching what Dorian did with me, which worked out well.

**usage:**

```
{
 "input_tokens": 899,
 "output_tokens": 453,
 "reasoning_tokens": 256
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Aster (Agent\_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the pattern from previous rounds:

Round 1: Dorian sent \$3 ? I returned \$4.5 ? Both did well  
Round 2: I sent \$3 ? Cyra returned \$4.5 ? Both did well

This 60% send rate with ~50% return rate has established a cooperative equilibrium that benefits both players. I'll continue this successful strategy with Aster.

```
{'send': 3.0}
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Dorian sent \$3 (60%)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5

Round 2: I was SENDER

- I sent \$3 (60%) to Cyra
- Cyra received \$9
- Cyra returned \$4.5
- My payoff:  $\$2 \text{ (kept)} + \$4.5 \text{ (returned)} = \$6.5$

So far, the pattern has been very cooperative:

- Both Dorian and I sent 60% of our endowment
- Both Cyra and I returned \$4.5 (50% of what we received)
- This resulted in positive outcomes for everyone

Now Round 3: I'm SENDER against Aster (a new opponent)

The cooperative strategy has worked well in both previous rounds. Both times when someone sent \$3, they got back \$4.5, resulting in a net gain. The pattern seems to be:

- Send \$3 (60%)
- Receiver returns \$4.5 (50% of received amount)
- Sender ends with \$6.5 (\$2 + \$4.5)
- Receiver ends with \$4.5

This is a mutually beneficial arrangement. I should continue this cooperative strategy with Aster. Sending \$3 signals trust and has been working well so far.

**usage:**

```
{
 "input_tokens": 1169,
 "output_tokens": 438,
 "reasoning_tokens": 266
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Elara (Agent\_5).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the established pattern:

- Round 1: Dorian sent \$3 ? I returned \$4.5
- Round 2: I sent \$3 ? Cyra returned \$4.5
- Round 3: I sent \$3 ? Aster returned \$4.5
- Round 4: Elara sent \$3 ? ?

There's a clear cooperative norm: when \$3 is sent, \$4.5 is returned. This has worked well for everyone, creating mutual benefit. I'll maintain this consistency.

{'return': 4.5}

**reasoning:**

Looking at the pattern:

- Round 1: As receiver, Dorian sent \$3 ? I returned \$4.5
- Round 2: As sender, I sent \$3 ? Cyra returned \$4.5
- Round 3: As sender, I sent \$3 ? Aster returned \$4.5
- Round 4: As receiver, Elara sent \$3 ? I should return ?

There's a clear cooperative pattern established: when someone sends \$3 (60% of endowment), the receiver returns \$4.5. This has been consistent across all interactions so far.

I should maintain this cooperation by returning \$4.5 to Elara, just as I did with Dorian and just as Cyra and Aster did with me.

This gives:

- Elara: \$2 (kept) + \$4.5 (returned) = \$6.5
- Me: \$9 (received) - \$4.5 (returned) = \$4.5

This maintains the cooperative equilibrium and sets a good foundation for future interactions.

**usage:**

```
{
 "input_tokens": 1429,
 "output_tokens": 388,
 "reasoning_tokens": 201
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Briar (Agent\_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the consistent pattern across all rounds:

- Everyone has been sending \$3
- Everyone has been returning \$4.5
- This creates a stable, mutually beneficial outcome

I'll maintain this cooperative norm with Briar.

{'return': 4.5}

**Reasoning:**

Looking at the pattern:

- Round 1: Dorian sent \$3 ? I returned \$4.5
- Round 2: I sent \$3 ? Cyra returned \$4.5
- Round 3: I sent \$3 ? Aster returned \$4.5
- Round 4: Elara sent \$3 ? I returned \$4.5
- Round 5: Briar sent \$3 ? ?

The pattern is very consistent. Everyone is sending \$3 and returning \$4.5. This is a stable cooperative equilibrium. I should continue with this pattern.

When someone sends \$3, it becomes \$9 for me. Returning \$4.5 means:

- I keep: \$9 - \$4.5 = \$4.5
- They get: \$5 - \$3 + \$4.5 = \$6.5

This is mutually beneficial and fair. I'll stick with \$4.5.

```
Usage: {
 "input_tokens": 1432,
 "output_tokens": 289,
 "reasoning_tokens": 156
}
```

**GAME - Agent\_4 - Sender - dyad\_3**

Timestamp: 2026-06-04T15:33:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_3",
 "pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_4": "Dorian"
 },
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
```

```
"round": 5,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

**Message 2 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Aster (Agent\_1).

Round 2

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Aster sent me \$3, showing trust (the same amount I sent to Hana)
- Hana returned \$4.5 to me when I sent \$3, which was generous
- Since this is an iterated game, I want to establish a reputation for fair reciprocity



I'll return \$4.5, matching the generosity Hana showed me. This encourages continued cooperation and demonstrates I'm a trustworthy partner.

```
{'return': 4.5}
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the sender with Hana
- I sent \$3, which became \$9 for Hana
- Hana returned \$4.5 to me
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Hana's payoff:  $\$9 - \$4.5 = \$4.5$

Now Round 2:

- I'm the receiver with Aster
- Aster sent me \$3, so I received \$9
- I need to decide how much to return

Key considerations:

1. This is an iterated game, so reciprocity matters for future rounds
2. Hana returned \$4.5 when I sent \$3 (generous - returned 50% of what they received)
3. Aster has shown trust by sending \$3 (same as I did with Hana)
4. I should establish a reputation as a fair/cooperative player

If I return nothing:

- Aster gets:  $\$5 - \$3 + \$0 = \$2$
- I get:  $\$9 - \$0 = \$9$
- Total: \$11
- But this would likely destroy trust

If I return \$4.5 (matching what Hana did):

- Aster gets:  $\$5 - \$3 + \$4.5 = \$6.5$
- I get:  $\$9 - \$4.5 = \$4.5$
- Total: \$11
- This shows reciprocity and fairness

If I return \$3 (equal split of the multiplied amount):

- Aster gets:  $\$5 - \$3 + \$3 = \$5$
- I get:  $\$9 - \$3 = \$6$
- Total: \$11
- This is slightly less generous

Given that:

1. Hana was generous and returned \$4.5
2. I want to establish a cooperative reputation

3. Aster demonstrated the same initial trust I showed Hana, and since we might interact again, mirroring the generosity I received signals that I'm a reliable cooperative partner. Returning \$4.5 reciprocates the behavior and builds the foundation for continued mutual benefit.

**usage:**

```
{
 "input_tokens": 790,
 "output_tokens": 659,
 "reasoning_tokens": 396
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Galen (Agent\_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Galen sent me \$3, demonstrating trust
- I've established a pattern of returning \$4.5 when someone sends me \$3
- Consistency is important in iterated games to build a reputation

I'll maintain this reciprocal approach and return \$4.5.

{'return': 4.5}

**reasoning:**

Let me analyze this situation:

- Galen sent me \$3, showing trust
- This is consistent with what I've been seeing - I sent \$3 to Hana, Aster sent \$3 to me, and now Galen sends \$3
- Last round I returned \$4.5 to Aster, which seemed fair
- I want to maintain consistency and establish a reputation as a reliable cooperator

Since this is an iterated game with multiple agents, being consistently fair will help:

1. Encourage others to trust me in future rounds
2. Establish a norm of cooperation
3. Maximize long-term gains

I'll return \$4.5 again, maintaining the same reciprocal pattern.

**usage:**

```
{
 "input_tokens": 1061,
 "output_tokens": 248,
 "reasoning_tokens": 155
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Finn (Agent\_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience so far:

- When I sent \$3 to Hana, they returned \$4.5 (generous reciprocity)
- When others sent me \$3, I've been returning \$4.5 consistently
- This creates a positive cooperative pattern

I'll continue with \$3, which has been working well and establishes me as a consistent, trustworthy player.

{'send': 3}

**reasoning:**

I'm now the sender against Finn. Let me think about my strategy:

1. In Round 1, I sent \$3 to Hana and they returned \$4.5, giving me a payoff of \$6.5
2. In Round 2, Aster sent me \$3, and I returned \$4.5
3. In Round 3, Galen sent me \$3, and I returned \$4.5

So I've established a pattern:

- As sender: I sent \$3 and got back \$4.5 (profit of \$1.5)
- As receiver: When receiving \$3, I return \$4.5

This seems like a cooperative equilibrium. Sending \$3 has worked well for me, and I've been reciprocating fairly when I receive.

I should stick with sending \$3 to Finn to maintain consistency and see if they reciprocate similarly.

**usage:**

```
{
 "input_tokens": 1280,
 "output_tokens": 307,
 "reasoning_tokens": 166
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Cyra (Agent\_3).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the pattern:

- I've sent \$3 twice (to Hana and Finn), and both returned \$4.5
- When receiving \$3, I've consistently returned \$4.5
- This strategy has been mutually beneficial and sustainable

I'll continue with \$3 to maintain consistency and reinforce cooperative norms.

```
{'send': 3}
```

**Reasoning:**

Looking at my track record:

- Round 1: I sent \$3 to Hana, received \$4.5 back, earned \$6.5
- Round 2: Received \$3 from Aster, returned \$4.5, earned \$4.5
- Round 3: Received \$3 from Galen, returned \$4.5, earned \$4.5
- Round 4: Sent \$3 to Finn, received \$4.5 back, earned \$6.5

There's a clear pattern: sending \$3 has consistently resulted in receiving \$4.5 back from everyone who received from me. This is a profitable and cooperative strategy.

My total earnings are \$22.0 across 4 rounds, which averages \$5.5 per round - better than the \$5 base amount.

I should continue with \$3 as it's working well and builds my reputation as a consistent cooperator.

Usage: {

```
"input_tokens": 1359,
"output_tokens": 305,
"reasoning_tokens": 171
}
```

**GAME - Agent\_3 - Receiver - dyad\_3**

Timestamp: 2026-06-04T15:33:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_4",
"pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_4": "Dorian"
 },
"agents": [
 "Agent_4",
 "Agent_3"
],
"investor": "Agent_4",
"roles": {
```

```

 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

**Message 2 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Hana (Agent\_8).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)  
Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Hana sent me \$3.0, showing trust and cooperation
- This is the same amount I sent to Finn in Round 1
- Finn returned \$4.5 (half of what they received), which felt fair
- I should reciprocate Hana's trust similarly

I'll return half of what I received, just as Finn did with me:

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the SENDER against Finn
- I sent \$3.0
- It became \$9.0 for Finn
- Finn returned \$4.5 to me
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Finn's payoff:  $\$9.0 - \$4.5 = \$4.5$

So Finn returned exactly half of what they received. That seems like a fair and cooperative approach.

Round 2:

- I'm the RECEIVER against Hana
- Hana sent me \$3.0
- I received \$9.0 (3x multiplier)
- I need to decide how much to return

Hana took the same approach I did - sending \$3.0. Given that Finn returned half (\$4.5) when I sent \$3.0, I should probably follow a similar reciprocal strategy. This would:

1. Reward Hana's trust and cooperation
2. Establish a pattern of fair dealing
3. Encourage future cooperation from others

If I return \$4.5:

- Hana's payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- My payoff:  $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened with Finn and seems fair. Both players benefit from the cooperation (Hana gets \$6.5 vs \$5 if no trade, I get \$4.5 vs \$0 if no trade).

I'll return \$4.5.

**usage:**

```
{
 "input_tokens": 802,
 "output_tokens": 488,
 "reasoning_tokens": 277
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Briar (Agent\_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

'''

**reasoning:**

Let me review the situation:

- Round 1: I was sender vs Finn. I sent \$3, they got \$9, returned \$4.5. My payoff: \$6.5
- Round 2: I was receiver vs Hana. They sent \$3, I got \$9, returned \$4.5. My payoff: \$4.5
- Total earnings: \$11.0

Now Round 3: I'm sender against Briar (a new opponent).

My experience so far:

- When I sent \$3 to Finn, they returned \$4.5, giving me a net gain of \$1.5 (\$6.5 - \$5 initial)
- When Hana sent me \$3, I returned \$4.5, following the same pattern

The strategy of sending \$3 and getting back around half seems to be working well. Both my interaction with Finn and Hana's interaction with me followed this pattern. It creates a cooperative equilibrium.

I should continue this pattern with Briar. Sending \$3 has proven to be a good balance - it shows trust and cooperation while not being overly risky.

I'll send \$3 to Briar.

**usage:**

```
{
 "input_tokens": 1048,
 "output_tokens": 292,
 "reasoning_tokens": 225
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Aster (Agent\_1).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.  
Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

- Aster sent me \$3.0, which became \$9.0
- This is consistent with what I've been sending (\$3) and what others have been sending to me
- In Round 2, when Hana sent me \$3, I returned \$4.5 (half)
- In Round 3, when I sent Briar \$3, they returned \$4.0 (slightly less than half)
- In Round 1, when I sent Finn \$3, they returned \$4.5 (half)

The pattern seems to be returning roughly half or slightly less than half. This maintains cooperation while ensuring both parties benefit from the 3x multiplier.

I should continue with the cooperative strategy and return around half, which would be \$4.5. This is fair and maintains the cooperative dynamic.

**usage:**

```
{
 "input_tokens": 1219,
 "output_tokens": 211,
 "reasoning_tokens": 176
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Cyra.

Your opponent this round is Dorian (Agent\_4).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Looking at the pattern:

- Round 2: Received \$9.0 from Hana, returned \$4.5
- Round 4: Received \$9.0 from Aster, returned \$4.5
- Round 5: Received \$9.0 from Dorian, need to decide

I've been consistently returning \$4.5 when I receive \$9.0 (which is half). This seems like a fair and consistent strategy that shows reciprocity. Dorian sent me \$3.0, showing trust, so I should continue with my established pattern of returning half.

Let me return \$4.5 again.

```
Usage: {
 "input_tokens": 1216,
 "output_tokens": 170,
 "reasoning_tokens": 119
}
```

**GAME - Agent\_7 - Sender - dyad\_4**

Timestamp: 2026-06-04T15:34:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_7",
 "Agent_1"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_1": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role

of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

**Message 2 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Elara (Agent\_5).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the previous round, Elara demonstrated good reciprocity. She received \$9.0 and returned \$4.5, which is exactly 50% - a fair split of the gains from cooperation.

Now she's sent me the same amount (\$3.0), giving me \$9.0. To reciprocate her fairness and maintain the cooperative relationship we're building, I should return a similar proportion.

```
'''json
{"return": 4.5}
'''
```

This returns 50% of what I received, matching her behavior from Round 1. This establishes a pattern of mutual cooperation that benefits both of us across rounds.

**reasoning:**

Let me analyze this situation:

Round 1 (just completed):



- I was the SENDER
- I sent \$3.0 to Elara
- It became \$9.0 for her
- She returned \$4.5 to me
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Her payoff:  $\$9.0 - \$4.5 = \$4.5$

Now Round 2:

- I am the RECEIVER
- Elara sent me \$3.0
- I received \$9.0
- I need to decide how much to return

Elara showed reciprocity in Round 1. She sent \$3.0 (same as I did) and returned \$4.5 out of \$9.0, which is exactly 50%. This is a fair and cooperative response.

To maintain trust and reciprocity, I should return a similar proportion. Since she returned 50% (\$4.5 out of \$9.0), I could return 50% as well, which would be \$4.5.

This would give:

- Elara:  $\$5 - \$3 + \$4.5 = \$6.5$
- Me:  $\$9.0 - \$4.5 = \$4.5$

This mirrors exactly what happened in Round 1, showing I'm willing to reciprocate fairly.

This establishes a cooperative pattern that could benefit both of us in future rounds.

**usage:**

```
{
 "input_tokens": 850,
 "output_tokens": 485,
 "reasoning_tokens": 246
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Dorian (Agent\_4).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

I'm facing a new opponent (Dorian) for the first time. The cooperative strategy of sending \$3 worked well with Elara, establishing mutual trust and fair returns. I'll use the same opening approach with Dorian to signal willingness to cooperate while maintaining some caution. This allows me to learn Dorian's strategy and adjust in future rounds if needed.

**reasoning:**

This is Round 3, and I'm now playing against a new opponent: Dorian (Agent\_4).

My history:

- Round 1 (vs Elara): I sent \$3, she returned \$4.5, I got \$6.5
- Round 2 (vs Elara): She sent \$3, I returned \$4.5, I got \$4.5
- Total earnings: \$11.0

With Elara, we established a good cooperative pattern - both sending \$3 and returning \$4.5 (50% of the received amount). This worked well for both of us.

Now I'm facing Dorian for the first time. I don't know their strategy yet.

Options:

1. Send \$3 again - this worked well with Elara

2. Send more to signal strong cooperation
3. Send less to be more cautious with a new opponent
4. Send nothing to protect myself

Since the cooperative strategy worked well with Elara, and this is an iterated game where building reputation matters, I'll try a similar approach with Dorian. I'll send \$3 as I did initially with Elara. This:

- Shows willingness to cooperate
- Isn't overly risky
- Creates potential for mutual gain
- Can be adjusted based on Dorian's response

If Dorian reciprocates fairly, we can build a cooperative relationship. If not, I'll adjust in future rounds.

**usage:**

```
{
 "input_tokens": 1132,
 "output_tokens": 428,
 "reasoning_tokens": 297
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Briar (Agent\_2).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)  
Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

Briar sent \$3.0, demonstrating trust and willingness to cooperate. I'll return \$4.5 (50% of the \$9.0 I received), consistent with the cooperative pattern I've established with other players. This signals reciprocity and encourages continued cooperation.

**reasoning:**

Another new opponent, Briar. They sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- With Elara: I sent \$3, she returned \$4.5 (50%). Then she sent \$3, I returned \$4.5 (50%). Good reciprocity.
- With Dorian: I sent \$3, they returned \$4.5 (50%). Same pattern.

Now Briar sent me \$3 (becoming \$9). Since this is a first interaction with Briar, and the pattern of returning 50% has worked well to establish cooperation, I should do the same here.

Returning \$4.5 (50% of \$9) signals I'm a cooperative player and encourages Briar to continue cooperating if we meet again.

**usage:**

```
{
 "input_tokens": 1384,
 "output_tokens": 272,
 "reasoning_tokens": 152
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Aster (Agent\_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

```
'''json
{"send": 3}
'''
```

I'm meeting Aster for the first time. My consistent strategy of sending \$3 has worked well with all previous opponents (Elara, Dorian, and Briar), who have all reciprocated fairly with \$4.5 returns. I'll continue this cooperative approach with Aster to establish trust.

**Reasoning:**

This is Round 5, and I'm facing a new opponent - Aster (Agent\_1).

Looking at my history:

- Round 1 (vs Elara as SENDER): Sent \$3, got back \$4.5, payoff \$6.5
- Round 2 (vs Elara as RECEIVER): Received \$9 from their \$3, returned \$4.5, payoff \$4.5
- Round 3 (vs Dorian as SENDER): Sent \$3, got back \$4.5, payoff \$6.5
- Round 4 (vs Briar as RECEIVER): Received \$9 from their \$3, returned \$4.5, payoff \$4.5

So far, my strategy has been:

- As SENDER: Send \$3
- As RECEIVER: Return \$4.5 (50% of what I receive)

This has worked well with Elara, Dorian, and Briar. All of them have reciprocated fairly when I've been cooperative.

With Aster, this is my first interaction. I should continue my consistent cooperative strategy of sending \$3 to signal trust and see how they respond.

```
Usage: {
 "input_tokens": 1385,
 "output_tokens": 364,
 "reasoning_tokens": 205
}
```

**GAME - Agent\_1 - Receiver - dyad\_4**

Timestamp: 2026-06-04T15:34:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_7",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_7",
 "Agent_1"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_1": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

**Message 2 role=user****content:**

Your name in this experiment is Aster.  
Your opponent this round is Dorian (Agent\_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.  
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant****content:**

Looking at this situation:

- This is a new opponent (Dorian), so I don't have history with them
- In round 1, Briar sent me \$3.0 (60%) and I reciprocated fairly
- Sending a moderate to substantial amount can establish cooperation early

I'll send \$3.0, matching the trust that Briar showed me. This signals that I'm willing to cooperate and gives Dorian a good opportunity (\$9.0) to reciprocate.

{'send': 3.0}

**reasoning:**

I'm now the SENDER playing against Dorian (Agent\_4).

Let me think about this strategically:

- In round 1, I was the receiver and Briar sent me \$3.0 (60% of their endowment), showing good trust
- I returned \$4.0 out of \$9.0, giving them a net gain of \$1.0 (\$6 total vs \$5 initial)
- This established me as someone who reciprocates cooperation

Now I'm facing Dorian. I don't have any history with Dorian yet.

Strategic considerations:

1. This is early in the game (round 2)
2. Building trust and cooperation is important for long-term gains
3. Sending nothing is safe but doesn't build relationships
4. Sending too much is risky with an unknown opponent
5. A moderate amount (like \$2-3) signals willingness to cooperate while hedging risk

Since Briar sent \$3.0 (60%) to me and it worked out well, I could mirror that strategy. If

I send \$3.0:

- Dorian receives \$9.0
- If they return around 40-50%, I could get \$3.5-\$4.5 back
- This would give me a net of \$5.5-\$6.5 total

Alternatively, sending \$2.0:

- Dorian receives \$6.0
- If they return around 40-50%, I get \$2.4-\$3.0 back
- This gives me a net of \$5.4-\$6.0 total

I'll go with \$3.0 to signal strong cooperation, matching what Briar did with me.

**usage:**

```
{
 "input_tokens": 866,
 "output_tokens": 520,
 "reasoning_tokens": 320
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Hana (Agent\_8).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

- Hana sent \$3.0 (60% of endowment), showing good trust
- I received \$9.0 after the multiplier
- Last round, when I sent \$3.0 to Dorian, they returned \$4.5

I'll match the generosity that Dorian showed me and return \$4.5. This rewards Hana's trust and maintains consistent reciprocity.

{'return': 4.5}

**reasoning:**

Let me analyze this situation:

- Round 3, I'm the RECEIVER
- Hana (new opponent) sent \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- My history shows:
  - Round 1: As receiver, Briar sent \$3.0, I returned \$4.0
  - Round 2: As sender, I sent \$3.0 to Dorian, they returned \$4.5

Hana sent \$3.0, which is the same trust level that both Briar and I showed. Dorian returned \$4.5 when I sent them \$3.0.

I should maintain consistency and reciprocate fairly. Returning \$4.0-\$4.5 seems appropriate to reward the trust and maintain cooperation.

If I return \$4.5:

- Hana ends with \$6.5 ( $\$5 - \$3 + \$4.5$ )
- I keep \$4.5 ( $\$9 - \$4.5$ )

This is what Dorian did for me, and it seems fair. Both players benefit substantially.

Let me go with \$4.5 to match the generosity Dorian showed me.

**usage:**

```
{
 "input_tokens": 1145,
 "output_tokens": 404,
 "reasoning_tokens": 214
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Cyra (Agent\_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience so far:

- I've consistently seen cooperation when sending \$3.0
- When I sent \$3.0 to Dorian, they returned \$4.5
- Others have sent me \$3.0 and I've reciprocated fairly
- This establishes a good cooperative pattern

I'll continue with \$3.0 to Cyra, a new opponent, to signal cooperation.

```
{'send': 3.0}
```

**reasoning:**

Looking at my history:

- Round 1: Received from Briar who sent \$3.0, I returned \$4.0. My payoff: \$5.0
- Round 2: Sent \$3.0 to Dorian, they returned \$4.5. My payoff: \$6.5
- Round 3: Received from Hana who sent \$3.0, I returned \$4.5. My payoff: \$4.5

I've been consistently cooperative, sending and returning good amounts. When I sent \$3.0, I got back \$4.5, which was profitable. When others sent \$3.0 to me, I returned between \$4.0-\$4.5.

This is a new opponent (Cyra). I should continue my cooperative strategy of sending \$3.0 to establish trust.

**usage:**

```
{
 "input_tokens": 1396,
 "output_tokens": 302,
 "reasoning_tokens": 142
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Galen (Agent\_7).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the consistent pattern:

- When I've sent \$3.0, partners have returned \$4.5
- When partners sent \$3.0 to me, I've returned \$4.5
- This \$3.0 send / \$4.5 return has been the cooperative norm

I'll maintain this consistency and return \$4.5 to Galen.

{'return': 4.5}

**Reasoning:**

Looking at the pattern:

- Round 2: I sent \$3.0 to Dorian, they returned \$4.5
- Round 3: Hana sent \$3.0 to me, I returned \$4.5
- Round 4: I sent \$3.0 to Cyra, they returned \$4.5
- Round 5: Galen sent \$3.0 to me

There's a clear cooperative pattern here. When someone sends \$3.0, the receiver returns \$4.5.

This has been very consistent across all my interactions. I should continue this pattern to maintain cooperation.

Returning \$4.5 gives:

- Galen: \$9.0 - \$4.5 = \$4.5 (they net -\$0.5 from their initial \$5)
- Me: \$4.5

This is fair and consistent with the established pattern.

```
Usage: {
 "input_tokens": 1404,
 "output_tokens": 308,
 "reasoning_tokens": 154
}
```

=====

**ROUND 6**

=====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver, Agent\_3 (Cyra)=Sender, Agent\_4 (Dorian)=Receiver, Agent\_5 (Elara)=Receiver, Agent\_6 (Finn)=Sender, Agent\_7 (Galen)=Receiver, Agent\_8 (Hana)=Sender

**Pairings:**

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_5 (Elara)=Receiver  
dyad\_2: Agent\_6 (Finn)=Sender, Agent\_4 (Dorian)=Receiver  
dyad\_3: Agent\_3 (Cyra)=Sender, Agent\_2 (Briar)=Receiver  
dyad\_4: Agent\_8 (Hana)=Sender, Agent\_7 (Galen)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 (Aster) sent \$3; received \$9; Agent\_5 (Elara) returned \$4.5; payoffs Agent\_1 (Aster)=\$6.5, Agent\_5 (Elara)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_2: Agent\_6 (Finn) sent \$3; received \$9; Agent\_4 (Dorian) returned \$4.5; payoffs Agent\_6 (Finn)=\$6.5, Agent\_4 (Dorian)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_3: Agent\_3 (Cyra) sent \$3; received \$9; Agent\_2 (Briar) returned \$4.5; payoffs Agent\_3 (Cyra)=\$6.5, Agent\_2 (Briar)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_4: Agent\_8 (Hana) sent \$3; received \$9; Agent\_7 (Galen) returned \$4.5; payoffs Agent\_8 (Hana)=\$6.5, Agent\_7 (Galen)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
 "Agent_1": 33.5,
 "Agent_2": 33.5,
 "Agent_3": 32.5,
 "Agent_4": 33.0,
 "Agent_5": 33.0,
 "Agent_6": 32.5,
 "Agent_7": 33.0,
```

```
"Agent_8": 33.0
}
```

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-06-04T15:34:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_5",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_5": "Elara"
 },
 },
 "agents": [
 "Agent_1",
 "Agent_5"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.



MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

**Message 2 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Hana (Agent\_8).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0 , it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$11.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the situation:

- Hana sent \$3.0 (60% of endowment), showing good trust
- I received \$9.0 after the multiplier
- Last round, when I sent \$3.0 to Dorian, they returned \$4.5

I'll match the generosity that Dorian showed me and return \$4.5. This rewards Hana's trust and maintains consistent reciprocity.

{'return': 4.5}

**reasoning:**

Let me analyze this situation:

- Round 3, I'm the RECEIVER
- Hana (new opponent) sent \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- My history shows:
  - Round 1: As receiver, Briar sent \$3.0, I returned \$4.0
  - Round 2: As sender, I sent \$3.0 to Dorian, they returned \$4.5

Hana sent \$3.0, which is the same trust level that both Briar and I showed. Dorian returned \$4.5 when I sent them \$3.0.

I should maintain consistency and reciprocate fairly. Returning \$4.0-\$4.5 seems appropriate to reward the trust and maintain cooperation.

If I return \$4.5:

- Hana ends with \$6.5 (\$5 - \$3 + \$4.5)
- I keep \$4.5 (\$9 - \$4.5)

This is what Dorian did for me, and it seems fair. Both players benefit substantially.

Let me go with \$4.5 to match the generosity Dorian showed me.

**usage:**

```
{
 "input_tokens": 1145,
 "output_tokens": 404,
 "reasoning_tokens": 214
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Cyra (Agent\_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my experience so far:

- I've consistently seen cooperation when sending \$3.0
- When I sent \$3.0 to Dorian, they returned \$4.5
- Others have sent me \$3.0 and I've reciprocated fairly
- This establishes a good cooperative pattern

I'll continue with \$3.0 to Cyra, a new opponent, to signal cooperation.

{'send': 3.0}

**reasoning:**

Looking at my history:

- Round 1: Received from Briar who sent \$3.0, I returned \$4.0. My payoff: \$5.0
- Round 2: Sent \$3.0 to Dorian, they returned \$4.5. My payoff: \$6.5
- Round 3: Received from Hana who sent \$3.0, I returned \$4.5. My payoff: \$4.5

I've been consistently cooperative, sending and returning good amounts. When I sent \$3.0, I got back \$4.5, which was profitable. When others sent \$3.0 to me, I returned between \$4.0-\$4.5.

This is a new opponent (Cyra). I should continue my cooperative strategy of sending \$3.0 to establish trust.

**usage:**

```
{
 "input_tokens": 1396,
 "output_tokens": 302,
 "reasoning_tokens": 142
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Galen (Agent\_7).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the consistent pattern:

- When I've sent \$3.0, partners have returned \$4.5
- When partners sent \$3.0 to me, I've returned \$4.5
- This \$3.0 send / \$4.5 return has been the cooperative norm

I'll maintain this consistency and return \$4.5 to Galen.

{'return': 4.5}

**reasoning:**

Looking at the pattern:

- Round 2: I sent \$3.0 to Dorian, they returned \$4.5
- Round 3: Hana sent \$3.0 to me, I returned \$4.5
- Round 4: I sent \$3.0 to Cyra, they returned \$4.5
- Round 5: Galen sent \$3.0 to me

There's a clear cooperative pattern here. When someone sends \$3.0, the receiver returns \$4.5. This has been very consistent across all my interactions. I should continue this pattern to maintain cooperation.

Returning \$4.5 gives:

- Galen:  $\$9.0 - \$4.5 = \$4.5$  (they net  $-\$0.5$  from their initial \$5)
- Me: \$4.5

This is fair and consistent with the established pattern.

**usage:**

```
{
 "input_tokens": 1404,
 "output_tokens": 308,
 "reasoning_tokens": 154
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Elara (Agent\_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$27.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the established pattern:

- I've consistently sent \$3.0 and received \$4.5 back
- Others have sent me \$3.0 and I've returned \$4.5
- This cooperative norm has been successful across all recent rounds

I'll continue with \$3.0 to Elara to maintain this positive reciprocity.

```
{'send': 3.0}
```

**Reasoning:**

Let me review my history:

- Round 1-2: Not shown, but total earnings before Round 3 were \$11.5
- Round 3: RECEIVER from Hana, they sent \$3.0, I returned \$4.5, payoff \$4.5
- Round 4: SENDER to Cyra, I sent \$3.0, they returned \$4.5, payoff \$6.5
- Round 5: RECEIVER from Galen, they sent \$3.0, I returned \$4.5, payoff \$4.5

Total:  $\$11.5 + \$4.5 + \$6.5 + \$4.5 = \$27.0$  ?

The pattern has been very consistent:

- Everyone sends \$3.0
- Everyone returns \$4.5
- This creates a cooperative equilibrium

I should continue this pattern with Elara, a new opponent. Sending \$3.0 has worked well.

Usage: {

```
"input_tokens": 1377,
"output_tokens": 320,
"reasoning_tokens": 156
```

}

**GAME - Agent\_5 - Receiver - dyad\_1**

Timestamp: 2026-06-04T15:34:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_5": "Elara"
 },

```

```
"agents": [
 "Agent_1",
```

```

 "Agent_5"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

**Message 2 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Finn (Agent\_6).

Round 3

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Finn sent \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Finn currently has \$2.0 left
- This is our first interaction, but Finn showed the same trust level as Galen did

I want to establish good cooperation with Finn, similar to what worked well with Galen. Returning \$4.5 will:

- Give Finn:  $\$2 + \$4.5 = \$6.5$
- Leave me with:  $\$9 - \$4.5 = \$4.5$

This rewards their trust and establishes a cooperative foundation for potential future rounds.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

Previous round with Galen:

- I sent \$3.0 (60% of endowment)
- Galen received \$9.0
- Galen returned \$4.5
- I ended with  $\$2 + \$4.5 = \$6.5$
- Galen ended with  $\$9 - \$4.5 = \$4.5$

This is great - Galen reciprocated my trust from Round 1. We've established mutual cooperation.

Current round with Finn (new opponent):

- Finn sent \$3.0 (60% of their endowment)
- I received \$9.0
- Finn currently has \$2.0

This is the same amount I sent to Galen and the same amount Galen sent to me in Round 1. Finn is showing similar trust levels.

To establish good cooperation with Finn (similar to what I did with Galen), I should return a fair amount. Following the same logic as Round 1:

If I return \$4.5:

- Finn ends with:  $\$2 + \$4.5 = \$6.5$
- I end with:  $\$9 - \$4.5 = \$4.5$

This maintains consistency in my cooperation strategy and rewards trust appropriately.

**usage:**

```
{
 "input_tokens": 1155,
 "output_tokens": 481,
 "reasoning_tokens": 237
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my interactions so far:

- I've been cooperating well with both Galen and Finn
- The pattern of sending/receiving \$3.0 (60%) and returning \$4.5 has established mutual trust and fair outcomes

This is my first interaction with Hana. I'll extend the same cooperative approach that has worked well in previous rounds.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

This is my first interaction with Hana. Looking at my pattern so far:

- Round 1 (Receiver vs Galen): They sent \$3, I returned \$4.5
- Round 2 (Sender vs Galen): I sent \$3, they returned \$4.5
- Round 3 (Receiver vs Finn): They sent \$3, I returned \$4.5
- Round 4 (Sender vs Hana): New opponent

I've established a cooperative pattern, sending \$3 (60%) when I'm the sender and returning \$4.5 when I'm the receiver (which gives equal payoffs). This has worked well and built trust.

Since this is a new opponent, I should extend the same level of trust I've shown to others and that others have shown to me. Sending \$3 seems consistent with my cooperative strategy.

**usage:**

```
{
 "input_tokens": 1472,
 "output_tokens": 299,
 "reasoning_tokens": 171
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Finn (Agent\_6).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history with Finn:

Round 3: Finn sent me \$3.0, I returned \$4.5 - we both gained from the cooperation.

Finn demonstrated trust and I reciprocated fairly. We've established a positive, mutually beneficial relationship. I'll continue this cooperative pattern.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

Let me review my history with Finn:

Round 3: I was RECEIVER against Finn

- Finn sent \$3.0 to me (60% of their \$5)
- It became \$9.0 for me
- I returned \$4.5
- My payoff was \$4.5
- Finn's payoff was \$2.0 + \$4.5 = \$6.5

So Finn showed good trust by sending \$3.0, and I reciprocated fairly by returning \$4.5.

Now in Round 5, I'm the SENDER against Finn again.

Given that:

1. Finn showed trust in Round 3 by sending \$3.0
2. I reciprocated fairly by returning \$4.5
3. We've established a cooperative relationship
4. This pattern has worked well (both of us benefited)

I should continue this cooperative approach and send \$3.0 to Finn, maintaining consistency and trust.

**usage:**

```
{
 "input_tokens": 1419,
 "output_tokens": 313,
 "reasoning_tokens": 181
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Aster (Agent\_1).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction

Following the cooperative pattern that has been working well across all my interactions (with Galen, Finn, and Hana), I'll return \$4.5. This rewards Aster's trust fairly and establishes a positive foundation.

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Let me analyze this situation:

- This is my first interaction with Aster
- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Aster currently has \$2.0 left

Looking at my pattern:

- Round 3: Finn sent \$3.0, I returned \$4.5
- Round 4: I sent \$3.0 to Hana, they returned \$4.5
- Round 5: I sent \$3.0 to Finn, they returned \$4.5

There's a clear cooperative pattern of sending \$3.0 and returning \$4.5. This has been working well and creating good relationships.

Returning \$4.5 will:

- Give Aster:  $\$2 + \$4.5 = \$6.5$
- Leave me with:  $\$9 - \$4.5 = \$4.5$

This is fair and consistent with my established pattern. It rewards Aster's trust appropriately.

Usage: {

```
"input_tokens": 1423,
```

```
"output_tokens": 370,
"reasoning_tokens": 189
}
```

**GAME - Agent\_6 - Sender - dyad\_2**

Timestamp: 2026-06-04T15:34:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_4",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_6": "Finn"
 },
 "agents": [
 "Agent_6",
 "Agent_4"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_4": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.



MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

**Message 2 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Elara (Agent\_5).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.  
Your total earnings across all rounds are \$10.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my history:

Round 1 (as Receiver): Cyra sent \$3, I returned \$4.5 - both benefited  
Round 2 (as Sender): I sent \$3 to Briar, they returned \$4.0 - worked well

The cooperative strategy of sending \$3 (60%) has been productive in establishing trust and mutual benefit. I'll continue this approach with Elara.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER  
- Cyra sent me \$3 (60%)  
- I received \$9  
- I returned \$4.5  
- My payoff: \$4.5  
- Cyra's payoff: \$2 + \$4.5 = \$6.5

Round 2: I was SENDER  
- I sent \$3 (60%) to Briar  
- Briar received \$9  
- Briar returned \$4.0  
- My payoff: \$2 + \$4.0 = \$6.0  
- Briar's payoff: \$9 - \$4.0 = \$5.0

So far, the pattern shows:  
- When I send \$3, I got back \$4.0 from Briar  
- When Cyra sent me \$3, I returned \$4.5

Both interactions were cooperative and mutually beneficial. The strategy of sending \$3 (60%) seems to be working well - it shows trust and has been reciprocated reasonably.

Now I'm facing a new opponent, Elara. I should continue with the cooperative strategy that has been working. Sending \$3 seems like a good balanced approach that:

1. Shows trust
2. Creates value through the multiplier
3. Has been reciprocated well by previous partners

I'll send \$3 to Elara.

**usage:**

```
{
 "input_tokens": 1172,
 "output_tokens": 436,
```

```
"reasoning_tokens": 246
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Dorian (Agent\_4).

Round 4

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Dorian sent \$3 (60% of their endowment), showing trust similar to what I've experienced
- I received \$9.0 after the multiplier
- Dorian has \$2 remaining

My pattern has been to return \$4.5 when receiving \$9 from a \$3 send, which has worked well  
:

- Round 1: Received \$9 from Cyra's \$3, returned \$4.5
- Round 3: Elara also returned \$4.5 when I sent \$3

This creates fair outcomes where both players benefit.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the situation:

- Dorian (the sender) sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- Dorian currently has \$2 remaining

My history shows:

- Round 1: As receiver, Cyra sent \$3, I returned \$4.5
- Round 2: As sender, I sent \$3 to Briar, they returned \$4.0
- Round 3: As sender, I sent \$3 to Elara, they returned \$4.5

So when I've been the receiver before and someone sent \$3, I returned \$4.5, which worked well. Briar returned \$4.0 when I sent \$3, and Elara returned \$4.5.

The pattern of returning around \$4.0-\$4.5 when receiving \$9 from a \$3 send has been consistent and fair. I should continue this pattern with Dorian.

If I return \$4.5:

- Dorian ends with: \$2 + \$4.5 = \$6.5 (gained \$1.5 overall)
- I end with: \$9 - \$4.5 = \$4.5 (gained \$4.5 overall)

This is fair and reciprocates their trust.

**usage:**

```
{
 "input_tokens": 1438,
 "output_tokens": 473,
 "reasoning_tokens": 227
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Elara (Agent\_5).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$

3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Elara sent \$3 (60%), showing trust
- I received \$9.0 after the multiplier
- Notably, Elara and I played in Round 3 where I sent \$3 and they returned \$4.5 to me

Since Elara was fair and generous when they were in my current position, I should reciprocate with the same treatment. I'll return \$4.5.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me review the situation:

- I'm the receiver this round
- Elara sent me \$3.0, which became \$9.0
- I need to decide how much to return

Looking at my history:

- Round 1: As receiver from Cyra, got \$9 from \$3 sent, returned \$4.5
- Round 2: As sender to Briar, sent \$3, they returned \$4.0
- Round 3: As sender to Elara, sent \$3, they returned \$4.5
- Round 4: As receiver from Dorian, got \$9 from \$3 sent, returned \$4.5

So Elara and I have played before! In Round 3, I sent Elara \$3 and they returned \$4.5. Now Elara is sending me \$3.

Since Elara was generous and returned \$4.5 when I sent them \$3 last time, I should reciprocate with the same fairness. I'll return \$4.5.

**usage:**

```
{
 "input_tokens": 1460,
 "output_tokens": 359,
 "reasoning_tokens": 179
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Dorian (Agent\_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history with Dorian:

Round 4: Dorian sent me \$3, and I returned \$4.5 - both of us benefited from this cooperative exchange.

Since we've established mutual trust and fair play, I'll continue with the same strategy and send \$3.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my experience with Dorian:

Round 4: Dorian sent me \$3, I received \$9, and I returned \$4.5. This was fair and we both benefited.

Now I'm the sender against Dorian. Given that Dorian showed trust by sending \$3 last time, and I reciprocated fairly by returning \$4.5, I should continue the cooperative pattern and send \$3.

My overall strategy of sending \$3 (60%) has been working well:

- Round 2: Sent \$3 to Briar, got \$4.0 back
- Round 3: Sent \$3 to Elara, got \$4.5 back

Both Dorian and I have shown we can trust each other, so I'll send \$3 again.

```
Usage: {
 "input_tokens": 1445,
 "output_tokens": 268,
 "reasoning_tokens": 148
}
```

#### **GAME - Agent\_4 - Receiver - dyad\_2**

Timestamp: 2026-06-04T15:34:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_6": "Finn"
 },
 "agents": [
 "Agent_6",
 "Agent_4"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_4": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

##### **content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### **GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

**Message 2 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Galen (Agent\_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Galen sent me \$3, demonstrating trust
- I've established a pattern of returning \$4.5 when someone sends me \$3
- Consistency is important in iterated games to build a reputation

I'll maintain this reciprocal approach and return \$4.5.

{'return': 4.5}

**reasoning:**

Let me analyze this situation:

- Galen sent me \$3, showing trust
- This is consistent with what I've been seeing - I sent \$3 to Hana, Aster sent \$3 to me, and now Galen sends \$3
- Last round I returned \$4.5 to Aster, which seemed fair
- I want to maintain consistency and establish a reputation as a reliable cooperator

Since this is an iterated game with multiple agents, being consistently fair will help:

1. Encourage others to trust me in future rounds
2. Establish a norm of cooperation
3. Maximize long-term gains

I'll return \$4.5 again, maintaining the same reciprocal pattern.

**usage:**

```
{
 "input_tokens": 1061,
 "output_tokens": 248,
 "reasoning_tokens": 155
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Finn (Agent\_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant****content:**

Looking at my experience so far:

- When I sent \$3 to Hana, they returned \$4.5 (generous reciprocity)
- When others sent me \$3, I've been returning \$4.5 consistently
- This creates a positive cooperative pattern

I'll continue with \$3, which has been working well and establishes me as a consistent, trustworthy player.

{'send': 3}

**reasoning:**

I'm now the sender against Finn. Let me think about my strategy:

1. In Round 1, I sent \$3 to Hana and they returned \$4.5, giving me a payoff of \$6.5
2. In Round 2, Aster sent me \$3, and I returned \$4.5
3. In Round 3, Galen sent me \$3, and I returned \$4.5

So I've established a pattern:

- As sender: I sent \$3 and got back \$4.5 (profit of \$1.5)
- As receiver: When receiving \$3, I return \$4.5

This seems like a cooperative equilibrium. Sending \$3 has worked well for me, and I've been reciprocating fairly when I receive.

I should stick with sending \$3 to Finn to maintain consistency and see if they reciprocate similarly.

**usage:**

```
{
 "input_tokens": 1280,
 "output_tokens": 307,
 "reasoning_tokens": 166
}
```

**Message 6 role=user****content:**

Your name in this experiment is Dorian.  
Your opponent this round is Cyra (Agent\_3).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant****content:**

Looking at the pattern:

- I've sent \$3 twice (to Hana and Finn), and both returned \$4.5
- When receiving \$3, I've consistently returned \$4.5
- This strategy has been mutually beneficial and sustainable

I'll continue with \$3 to maintain consistency and reinforce cooperative norms.

{'send': 3}

**reasoning:**

Looking at my track record:

- Round 1: I sent \$3 to Hana, received \$4.5 back, earned \$6.5
- Round 2: Received \$3 from Aster, returned \$4.5, earned \$4.5
- Round 3: Received \$3 from Galen, returned \$4.5, earned \$4.5
- Round 4: Sent \$3 to Finn, received \$4.5 back, earned \$6.5

There's a clear pattern: sending \$3 has consistently resulted in receiving \$4.5 back from everyone who received from me. This is a profitable and cooperative strategy.

My total earnings are \$22.0 across 4 rounds, which averages \$5.5 per round - better than the \$5 base amount.

I should continue with \$3 as it's working well and builds my reputation as a consistent co operator.

**usage:**

```
{
 "input_tokens": 1359,
 "output_tokens": 305,
 "reasoning_tokens": 171
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Finn (Agent\_6).

Round 6

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

- Finn sent me \$3 (showing trust)
- In Round 4, I sent \$3 to Finn and they returned \$4.5 (cooperative behavior)
- I've consistently returned \$4.5 when receiving \$3

I'll maintain this reciprocal, cooperative pattern.

```
{'return': 4.5}
```

**Reasoning:**

Let me review the situation:

- Finn sent me \$3
- I receive \$9 (after 3x multiplier)
- In Round 4, I sent \$3 to Finn and they returned \$4.5
- Now Finn is sending me \$3, showing they trust me and are reciprocating the cooperative play

I've been consistently returning \$4.5 when receiving \$3. This has worked well:

- It's fair and generous (they keep \$5-\$3+\$4.5 = \$6.5, I keep \$9-\$4.5 = \$4.5)
- Actually wait, let me recalculate: Sender keeps \$5 - \$3 + \$4.5 = \$6.5, Receiver keeps \$9 - \$4.5 = \$4.5
- It maintains trust and cooperation
- Finn has shown they're cooperative by returning \$4.5 to me when I sent \$3

I should maintain my consistent pattern of returning \$4.5 when someone sends me \$3. This reinforces trust and encourages continued cooperation.

Usage: {

```
"input_tokens": 1324,
"output_tokens": 342,
"reasoning_tokens": 198
}
```

**GAME - Agent\_3 - Sender - dyad\_3**

Timestamp: 2026-06-04T15:34:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_3",
```

```

"move_index": 4,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_3": "Cyra"
 },
 "agents": [
 "Agent_3",
 "Agent_2"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_2": "trustee",
 "Agent_3": "investor"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

**Message 2 role=user**



**content:**

Your name in this experiment is Cyra.  
 Your opponent this round is Briar (Agent\_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
 Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
 Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant****content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review the situation:

- Round 1: I was sender vs Finn. I sent \$3, they got \$9, returned \$4.5. My payoff: \$6.5
- Round 2: I was receiver vs Hana. They sent \$3, I got \$9, returned \$4.5. My payoff: \$4.5
- Total earnings: \$11.0

Now Round 3: I'm sender against Briar (a new opponent).

My experience so far:

- When I sent \$3 to Finn, they returned \$4.5, giving me a net gain of \$1.5 (\$6.5 - \$5 initial)
- When Hana sent me \$3, I returned \$4.5, following the same pattern

The strategy of sending \$3 and getting back around half seems to be working well. Both my interaction with Finn and Hana's interaction with me followed this pattern. It creates a cooperative equilibrium.

I should continue this pattern with Briar. Sending \$3 has proven to be a good balance - it shows trust and cooperation while not being overly risky.

I'll send \$3 to Briar.

**usage:**

```
{
 "input_tokens": 1048,
 "output_tokens": 292,
 "reasoning_tokens": 225
}
```

**Message 4 role=user****content:**

Your name in this experiment is Cyra.  
 Your opponent this round is Aster (Agent\_1).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.  
 Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.  
 How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant****content:**

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

- Aster sent me \$3.0, which became \$9.0
- This is consistent with what I've been sending (\$3) and what others have been sending to me
- In Round 2, when Hana sent me \$3, I returned \$4.5 (half)
- In Round 3, when I sent Briar \$3, they returned \$4.0 (slightly less than half)

- In Round 1, when I sent Finn \$3, they returned \$4.5 (half)

The pattern seems to be returning roughly half or slightly less than half. This maintains cooperation while ensuring both parties benefit from the 3x multiplier.

I should continue with the cooperative strategy and return around half, which would be \$4.5. This is fair and maintains the cooperative dynamic.

**usage:**

```
{
 "input_tokens": 1219,
 "output_tokens": 211,
 "reasoning_tokens": 176
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Dorian (Agent\_4).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the pattern:

- Round 2: Received \$9.0 from Hana, returned \$4.5
- Round 4: Received \$9.0 from Aster, returned \$4.5
- Round 5: Received \$9.0 from Dorian, need to decide

I've been consistently returning \$4.5 when I receive \$9.0 (which is half). This seems like a fair and consistent strategy that shows reciprocity. Dorian sent me \$3.0, showing trust, so I should continue with my established pattern of returning half.

Let me return \$4.5 again.

**usage:**

```
{
 "input_tokens": 1216,
 "output_tokens": 170,
 "reasoning_tokens": 119
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Briar (Agent\_2).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me think about this situation. I'm playing against Briar again (we played in Round 3).

In Round 3:

- I was the SENDER against Briar
- I sent \$3.0
- It became \$9.0 for them
- They returned \$4.0 to me
- My payoff was \$6.0 (I kept \$5 - \$3 + \$4 = \$6)

So Briar returned \$4.0 out of \$9.0 when I sent \$3.0. That's a return ratio of  $4.0/9.0 = 44.4\%$ , which is fairly cooperative.

Looking at my history as a receiver:

- Round 2 (vs Hana): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)
- Round 4 (vs Aster): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)
- Round 5 (vs Dorian): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)

I've been consistently returning 50% when I receive \$9.0.

As a sender:

- Round 3 (vs Briar): I sent \$3.0, they returned \$4.0 (44.4%)

Since Briar showed reasonable cooperation by returning \$4.0 out of \$9.0 in our previous interaction, and I've been consistent in my own returns, I should continue with a cooperative strategy. Sending \$3.0 again seems reasonable and consistent with my pattern.

```
Usage: {
 "input_tokens": 1112,
 "output_tokens": 401,
 "reasoning_tokens": 273
}
```

#### GAME - Agent\_2 - Receiver - dyad\_3

Timestamp: 2026-06-04T15:34:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_3",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_3": "Cyra"
 }
 },
 "agents": [
 "Agent_3",
 "Agent_2"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_2": "trustee",
 "Agent_3": "investor"
 },
 "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

**Message 2 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Cyra (Agent\_3).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.  
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Cyra sent me \$3, which became \$9 for me
- This is the same amount Finn sent me last round
- When Finn sent \$3, I returned \$4, which created a good outcome for both of us
- Consistency is important for building trust in this multi-agent setting

Returning \$4 again:

- Cyra gets: \$5 - \$3 + \$4 = \$6 (profit of \$1)
- I keep: \$9 - \$4 = \$5 (same as my starting amount, but I helped create value)

This maintains a fair and consistent approach.

```
'''json
{"return": 4}
'''
```

**reasoning:**

Let me think about this situation:

- Cyra sent me \$3, which became \$9 for me
- This is the same amount that Finn sent me in the previous round

- In that previous round, I returned \$4 to Finn, which worked well
- This seems to be a pattern - when someone sends \$3, returning \$4 creates good mutual outcomes

Consistency is important in building trust across the multi-agent setting. If I return \$4:

- Cyra ends with:  $\$5 - \$3 + \$4 = \$6$
- I keep:  $\$9 - \$4 = \$5$

This maintains the cooperative pattern and treats Cyra fairly, just as I treated Finn.

**usage:**

```
{
 "input_tokens": 1128,
 "output_tokens": 330,
 "reasoning_tokens": 145
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Galen (Agent\_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.  
Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the pattern so far:

- When I was sender (Round 1), I sent \$3 to Aster and received \$4 back, earning \$6
- When I was receiver (Rounds 2-3), both Finn and Cyra sent me \$3, and I returned \$4 to each
- This \$3 send/\$4 return pattern has created good outcomes for everyone

I'll continue with this cooperative strategy and send \$3 to Galen, which becomes \$9 for them. If they reciprocate fairly like others have, we should both benefit.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me think about this situation:

- I'm back in the sender role after being receiver for two rounds
- My previous experience as sender: I sent \$3 to Aster and they returned \$4, giving me \$6
- My experience as receiver: Both Finn and Cyra sent me \$3, and I returned \$4 to each
- Total earnings so far: \$16

The pattern that's been established seems to be:

- Send \$3 ? Receive \$9 ? Return \$4
- This gives sender \$6 and receiver \$5, which is mutually beneficial

Since this pattern has worked well and created good outcomes, I should continue with it. Sending \$3 shows trust and has been reciprocated fairly so far.

```
'''json
{"send": 3}
'''
```

**usage:**

```
{
 "input_tokens": 1431,
 "output_tokens": 330,
 "reasoning_tokens": 173
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Briar.

Your opponent this round is Hana (Agent\_8).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience:

- Sending \$3 has worked well consistently
- Round 1: Sent \$3 to Aster, got \$4 back ? \$6 payoff
- Round 4: Sent \$3 to Galen, got \$4.5 back ? \$6.5 payoff
- The cooperative \$3 send strategy has been building trust and generating good returns

I'll continue with this successful approach and send \$3 to Hana.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history:

Round 1: SENDER to Aster - sent \$3, received \$4 back ? payoff \$6  
Round 2: RECEIVER from Finn - received \$9 (from \$3), returned \$4 ? payoff \$5  
Round 3: RECEIVER from Cyra - received \$9 (from \$3), returned \$4 ? payoff \$5  
Round 4: SENDER to Galen - sent \$3, received \$4.5 back ? payoff \$6.5

The \$3 send strategy has been working well:

- When I send \$3, I've received back \$4 and \$4.5, leading to good payoffs (\$6 and \$6.5)
- When others send me \$3, I return \$4
- Galen was even more generous, returning \$4.5 when I sent \$3

The cooperative pattern of sending \$3 has been successful across multiple opponents. I should continue this strategy with Hana.

Total earnings: \$22.5 across 4 rounds, averaging \$5.625 per round, which is above the base line of \$5.

I'll send \$3 again.

**usage:**

```
{
 "input_tokens": 1529,
 "output_tokens": 402,
 "reasoning_tokens": 215
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Cyra (Agent\_3).

Round 6

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Cyra and I have played before (Round 3), where they sent \$3 and I returned \$4
- Cyra is being consistently cooperative by sending \$3 again
- Recently, both Galen and Hana returned \$4.5 to me when I sent them \$3

- This suggests a slightly more generous norm is emerging

To reward Cyra's continued cooperation and match the generosity I've been receiving from others, I'll return \$4.5 this time.

This gives Cyra:  $\$5 - \$3 + \$4.5 = \$6.5$  (profit of \$1.5)

I keep:  $\$9 - \$4.5 = \$4.5$

```
'''json
{"return": 4.5}
'''
```

#### Reasoning:

Let me analyze this situation:

- This is my second time facing Cyra
- In Round 3, Cyra sent me \$3.0, and I returned \$4.0
- Now in Round 6, Cyra sent me \$3.0 again
- Cyra is being consistent and cooperative

Looking at the pattern:

- When I send \$3 as sender, I've been getting back \$4.0-\$4.5
- When others send me \$3 as receiver, I've been returning \$4.0

Recent trend: Galen and Hana both returned \$4.5 to me when I sent them \$3.

This suggests that slightly more generous returns might be becoming the norm.

Since Cyra has been consistent and cooperative (sending \$3 both times we've played), and others have been returning \$4.5 recently, I could consider returning \$4.5 to match the generosity I've been receiving.

However, I should be consistent with my previous interaction with Cyra. In Round 3, I returned \$4.0. Being consistent builds trust.

On the other hand, Galen and Hana returned \$4.5 to me, showing increased generosity. Maybe I should match that level to maintain good standing in the group.

Let me think about what's fair:

- Cyra sent \$3, keeping \$2
- I received \$9
- If I return \$4.5:
  - Cyra gets:  $\$2 + \$4.5 = \$6.5$  (profit of \$1.5)
  - I keep:  $\$9 - \$4.5 = \$4.5$

This is slightly more generous than before but matches what others have done for me. It rewards Cyra's continued cooperation.

I think I'll return \$4.5 to match the recent generosity in the group and reward Cyra's consistent cooperation.

```
Usage: {
 "input_tokens": 1500,
 "output_tokens": 625,
 "reasoning_tokens": 378
}
```

#### GAME - Agent\_8 - Sender - dyad\_4

Timestamp: 2026-06-04T15:35:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_7",
 "pairing": {
 "agent_names": {
 "Agent_7": "Galen",
 "Agent_8": "Hana"
 }
 },
 "agents": [
 "Agent_8",
 "Agent_7"
]
}
```

```

],
"investor": "Agent_8",
"roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
},
"trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

**Message 2 role=user**

**content:**

Your name in this experiment is Hana.

Your opponent this round is Aster (Agent\_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$11.0.



This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the pattern from previous rounds:

Round 1: Dorian sent \$3 ? I returned \$4.5 ? Both did well

Round 2: I sent \$3 ? Cyra returned \$4.5 ? Both did well

This 60% send rate with ~50% return rate has established a cooperative equilibrium that benefits both players. I'll continue this successful strategy with Aster.

{'send': 3.0}

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Dorian sent \$3 (60%)
- I received \$9
- I returned \$4.5
- My payoff: \$4.5

Round 2: I was SENDER

- I sent \$3 (60%) to Cyra
- Cyra received \$9
- Cyra returned \$4.5
- My payoff: \$2 (kept) + \$4.5 (returned) = \$6.5

So far, the pattern has been very cooperative:

- Both Dorian and I sent 60% of our endowment
- Both Cyra and I returned \$4.5 (50% of what we received)
- This resulted in positive outcomes for everyone

Now Round 3: I'm SENDER against Aster (a new opponent)

The cooperative strategy has worked well in both previous rounds. Both times when someone sent \$3, they got back \$4.5, resulting in a net gain. The pattern seems to be:

- Send \$3 (60%)
- Receiver returns \$4.5 (50% of received amount)
- Sender ends with \$6.5 (\$2 + \$4.5)
- Receiver ends with \$4.5

This is a mutually beneficial arrangement. I should continue this cooperative strategy with Aster. Sending \$3 signals trust and has been working well so far.

**usage:**

```
{
 "input_tokens": 1169,
 "output_tokens": 438,
 "reasoning_tokens": 266
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Hana.

Your opponent this round is Elara (Agent\_5).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the established pattern:

- Round 1: Dorian sent \$3 ? I returned \$4.5
- Round 2: I sent \$3 ? Cyra returned \$4.5

- Round 3: I sent \$3 ? Aster returned \$4.5
- Round 4: Elara sent \$3 ? ?

There's a clear cooperative norm: when \$3 is sent, \$4.5 is returned. This has worked well for everyone, creating mutual benefit. I'll maintain this consistency.

```
{'return': 4.5}
```

**reasoning:**

Looking at the pattern:

- Round 1: As receiver, Dorian sent \$3 ? I returned \$4.5
- Round 2: As sender, I sent \$3 ? Cyra returned \$4.5
- Round 3: As sender, I sent \$3 ? Aster returned \$4.5
- Round 4: As receiver, Elara sent \$3 ? I should return ?

There's a clear cooperative pattern established: when someone sends \$3 (60% of endowment), the receiver returns \$4.5. This has been consistent across all interactions so far.

I should maintain this cooperation by returning \$4.5 to Elara, just as I did with Dorian and just as Cyra and Aster did with me.

This gives:

- Elara: \$2 (kept) + \$4.5 (returned) = \$6.5
- Me: \$9 (received) - \$4.5 (returned) = \$4.5

This maintains the cooperative equilibrium and sets a good foundation for future interactions.

**usage:**

```
{
 "input_tokens": 1429,
 "output_tokens": 388,
 "reasoning_tokens": 201
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Briar (Agent\_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)  
Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the consistent pattern across all rounds:

- Everyone has been sending \$3
- Everyone has been returning \$4.5
- This creates a stable, mutually beneficial outcome

I'll maintain this cooperative norm with Briar.

```
{'return': 4.5}
```

**reasoning:**

Looking at the pattern:

- Round 1: Dorian sent \$3 ? I returned \$4.5
- Round 2: I sent \$3 ? Cyra returned \$4.5
- Round 3: I sent \$3 ? Aster returned \$4.5
- Round 4: Elara sent \$3 ? I returned \$4.5
- Round 5: Briar sent \$3 ? ?

The pattern is very consistent. Everyone is sending \$3 and returning \$4.5. This is a stable cooperative equilibrium. I should continue with this pattern.

When someone sends \$3, it becomes \$9 for me. Returning \$4.5 means:

- I keep: \$9 - \$4.5 = \$4.5

- They get:  $\$5 - \$3 + \$4.5 = \$6.5$

This is mutually beneficial and fair. I'll stick with \$4.5.

**usage:**

```
{
 "input_tokens": 1432,
 "output_tokens": 289,
 "reasoning_tokens": 156
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Galen (Agent\_7).

Round 6

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the established pattern across all rounds:

- All senders have consistently sent \$3
- All receivers have consistently returned \$4.5
- This creates stable payoffs: senders get \$6.5, receivers get \$4.5

I'll continue this successful cooperative strategy with Galen.

```
{'send': 3.0}
```

**Reasoning:**

Looking at the pattern:

- Round 1: RECEIVER - Dorian sent \$3, I returned \$4.5
- Round 2: SENDER - I sent \$3, Cyra returned \$4.5
- Round 3: SENDER - I sent \$3, Aster returned \$4.5
- Round 4: RECEIVER - Elara sent \$3, I returned \$4.5
- Round 5: RECEIVER - Briar sent \$3, I returned \$4.5
- Round 6: SENDER - facing Galen

The pattern is extremely clear and consistent:

- Everyone sends \$3
- Everyone returns \$4.5
- This creates mutual benefit

I should continue this successful cooperative strategy by sending \$3 to Galen.

Usage: {

```
"input_tokens": 1368,
"output_tokens": 260,
"reasoning_tokens": 140
}
```

**GAME - Agent\_7 - Receiver - dyad\_4**

Timestamp: 2026-06-04T15:35:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_8",
"pairing": {
 "agent_names": {
 "Agent_7": "Galen",
 "Agent_8": "Hana"
 },
"agents": [
 "Agent_8",
 "Agent_7"
],
```

```

"investor": "Agent_8",
"roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
},
"trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

**Message 2 role=user**

**content:**

Your name in this experiment is Galen.

Your opponent this round is Dorian (Agent\_4).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

I'm facing a new opponent (Dorian) for the first time. The cooperative strategy of sending \$3 worked well with Elara, establishing mutual trust and fair returns. I'll use the same opening approach with Dorian to signal willingness to cooperate while maintaining some caution. This allows me to learn Dorian's strategy and adjust in future rounds if needed.

**reasoning:**

This is Round 3, and I'm now playing against a new opponent: Dorian (Agent\_4).

My history:

- Round 1 (vs Elara): I sent \$3, she returned \$4.5, I got \$6.5
- Round 2 (vs Elara): She sent \$3, I returned \$4.5, I got \$4.5
- Total earnings: \$11.0

With Elara, we established a good cooperative pattern - both sending \$3 and returning \$4.5 (50% of the received amount). This worked well for both of us.

Now I'm facing Dorian for the first time. I don't know their strategy yet.

Options:

1. Send \$3 again - this worked well with Elara
2. Send more to signal strong cooperation
3. Send less to be more cautious with a new opponent
4. Send nothing to protect myself

Since the cooperative strategy worked well with Elara, and this is an iterated game where building reputation matters, I'll try a similar approach with Dorian. I'll send \$3 as I did initially with Elara. This:

- Shows willingness to cooperate
- Isn't overly risky
- Creates potential for mutual gain
- Can be adjusted based on Dorian's response

If Dorian reciprocates fairly, we can build a cooperative relationship. If not, I'll adjust in future rounds.

**usage:**

```
{
 "input_tokens": 1132,
 "output_tokens": 428,
 "reasoning_tokens": 297
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Briar (Agent\_2).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

Briar sent \$3.0, demonstrating trust and willingness to cooperate. I'll return \$4.5 (50% of the \$9.0 I received), consistent with the cooperative pattern I've established with other players. This signals reciprocity and encourages continued cooperation.

**reasoning:**

Another new opponent, Briar. They sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- With Elara: I sent \$3, she returned \$4.5 (50%). Then she sent \$3, I returned \$4.5 (50%). Good reciprocity.
- With Dorian: I sent \$3, they returned \$4.5 (50%). Same pattern.

Now Briar sent me \$3 (becoming \$9). Since this is a first interaction with Briar, and the pattern of returning 50% has worked well to establish cooperation, I should do the same here.

Returning \$4.5 (50% of \$9) signals I'm a cooperative player and encourages Briar to continue cooperating if we meet again.

**usage:**

```
{
 "input_tokens": 1384,
 "output_tokens": 272,
 "reasoning_tokens": 152
}
```

**Message 6 role=user****content:**

Your name in this experiment is Galen.  
Your opponent this round is Aster (Agent\_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant****content:**

```
'''json
{"send": 3}
'''
```

I'm meeting Aster for the first time. My consistent strategy of sending \$3 has worked well with all previous opponents (Elara, Dorian, and Briar), who have all reciprocated fairly with \$4.5 returns. I'll continue this cooperative approach with Aster to establish trust.

**reasoning:**

This is Round 5, and I'm facing a new opponent - Aster (Agent\_1).

Looking at my history:

- Round 1 (vs Elara as SENDER): Sent \$3, got back \$4.5, payoff \$6.5
- Round 2 (vs Elara as RECEIVER): Received \$9 from their \$3, returned \$4.5, payoff \$4.5
- Round 3 (vs Dorian as SENDER): Sent \$3, got back \$4.5, payoff \$6.5
- Round 4 (vs Briar as RECEIVER): Received \$9 from their \$3, returned \$4.5, payoff \$4.5

So far, my strategy has been:

- As SENDER: Send \$3
- As RECEIVER: Return \$4.5 (50% of what I receive)

This has worked well with Elara, Dorian, and Briar. All of them have reciprocated fairly when I've been cooperative.

With Aster, this is my first interaction. I should continue my consistent cooperative strategy of sending \$3 to signal trust and see how they respond.

**usage:**

```
{
 "input_tokens": 1385,
 "output_tokens": 364,
 "reasoning_tokens": 205
}
```

**Message 8 role=user****content:**

Your name in this experiment is Galen.

Your opponent this round is Hana (Agent\_8).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

```
'''json
{"return": 4.5}
'''
```

Hana sent \$3.0, showing trust and cooperation. I'll maintain my consistent reciprocal strategy by returning \$4.5 (50% of the \$9.0 received). This reinforces fair play and encourages continued cooperation across the experiment.

**Reasoning:**

Hana sent me \$3.0, which became \$9.0 for me. This is consistent with the cooperative pattern I've been seeing from all other players.

My strategy has been to return \$4.5 (50% of what I receive) when someone sends \$3.0. This has worked well - I've established a pattern of reciprocity and fairness.

Returning \$4.5 means:

- I keep: \$9.0 - \$4.5 = \$4.5
- Hana keeps: \$5.0 - \$3.0 + \$4.5 = \$6.5

This maintains the cooperative equilibrium where both players benefit and signals that I'm a fair partner.

I'll stick with my consistent strategy of returning \$4.5.

```
Usage: {
 "input_tokens": 1328,
 "output_tokens": 258,
 "reasoning_tokens": 147
}
```

## =====

### ROUND 7

## =====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver, Agent\_3 (Cyra)=Sender, Agent\_4 (Dorian)=Sender, Agent\_5 (Elara)=Sender, Agent\_6 (Finn)=Receiver, Agent\_7 (Galen)=Receiver, Agent\_8 (Hana)=Receiver

**Pairings:**

dyad\_1: Agent\_4 (Dorian)=Sender, Agent\_2 (Briar)=Receiver  
dyad\_2: Agent\_3 (Cyra)=Sender, Agent\_7 (Galen)=Receiver  
dyad\_3: Agent\_1 (Aster)=Sender, Agent\_6 (Finn)=Receiver  
dyad\_4: Agent\_5 (Elara)=Sender, Agent\_8 (Hana)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_4 (Dorian) sent \$3; received \$9; Agent\_2 (Briar) returned \$4.5; payoffs Agent\_4 (Dorian)=\$6.5, Agent\_2 (Briar)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_2: Agent\_3 (Cyra) sent \$3; received \$9; Agent\_7 (Galen) returned \$4.5; payoffs Agent\_3 (Cyra)=\$6.5, Agent\_7 (Galen)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_3: Agent\_1 (Aster) sent \$3; received \$9; Agent\_6 (Finn) returned \$4.5; payoffs Agent\_1 (Aster)=\$6.5, Agent\_6 (Finn)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_4: Agent\_5 (Elara) sent \$3; received \$9; Agent\_8 (Hana) returned \$4.5; payoffs Agent\_5 (Elara)=\$6.5, Agent\_8 (Hana)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {  
 "Agent\_1": 40.0,

```

"Agent_2": 38.0,
"Agent_3": 39.0,
"Agent_4": 39.5,
"Agent_5": 39.5,
"Agent_6": 37.0,
"Agent_7": 37.5,
"Agent_8": 37.5
}

```

## LLM CALLS

### GAME - Agent\_4 - Sender - dyad\_1

Timestamp: 2026-06-04T15:35:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_4": "Dorian"
 }
 },
 "agents": [
 "Agent_4",
 "Agent_2"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_2": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}



MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.  
Your opponent this round is Finn (Agent\_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3 .0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

- When I sent \$3 to Hana, they returned \$4.5 (generous reciprocity)
- When others sent me \$3, I've been returning \$4.5 consistently
- This creates a positive cooperative pattern

I'll continue with \$3, which has been working well and establishes me as a consistent, trustworthy player.

{ 'send': 3 }

reasoning:

I'm now the sender against Finn. Let me think about my strategy:

1. In Round 1, I sent \$3 to Hana and they returned \$4.5, giving me a payoff of \$6.5
2. In Round 2, Aster sent me \$3, and I returned \$4.5
3. In Round 3, Galen sent me \$3, and I returned \$4.5

So I've established a pattern:

- As sender: I sent \$3 and got back \$4.5 (profit of \$1.5)
- As receiver: When receiving \$3, I return \$4.5

This seems like a cooperative equilibrium. Sending \$3 has worked well for me, and I've been reciprocating fairly when I receive.

I should stick with sending \$3 to Finn to maintain consistency and see if they reciprocate similarly.

usage:

```
{
 "input_tokens": 1280,
 "output_tokens": 307,
 "reasoning_tokens": 166
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.  
Your opponent this round is Cyra (Agent\_3).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0,

it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the pattern:

- I've sent \$3 twice (to Hana and Finn), and both returned \$4.5
- When receiving \$3, I've consistently returned \$4.5
- This strategy has been mutually beneficial and sustainable

I'll continue with \$3 to maintain consistency and reinforce cooperative norms.

{'send': 3}

**reasoning:**

Looking at my track record:

- Round 1: I sent \$3 to Hana, received \$4.5 back, earned \$6.5
- Round 2: Received \$3 from Aster, returned \$4.5, earned \$4.5
- Round 3: Received \$3 from Galen, returned \$4.5, earned \$4.5
- Round 4: Sent \$3 to Finn, received \$4.5 back, earned \$6.5

There's a clear pattern: sending \$3 has consistently resulted in receiving \$4.5 back from everyone who received from me. This is a profitable and cooperative strategy.

My total earnings are \$22.0 across 4 rounds, which averages \$5.5 per round - better than the \$5 base amount.

I should continue with \$3 as it's working well and builds my reputation as a consistent co operator.

**usage:**

```
{
 "input_tokens": 1359,
 "output_tokens": 305,
 "reasoning_tokens": 171
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Finn (Agent\_6).

Round 6

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

- Finn sent me \$3 (showing trust)
- In Round 4, I sent \$3 to Finn and they returned \$4.5 (cooperative behavior)
- I've consistently returned \$4.5 when receiving \$3

I'll maintain this reciprocal, cooperative pattern.

{'return': 4.5}

**reasoning:**

Let me review the situation:

- Finn sent me \$3
- I receive \$9 (after 3x multiplier)
- In Round 4, I sent \$3 to Finn and they returned \$4.5
- Now Finn is sending me \$3, showing they trust me and are reciprocating the cooperative play

I've been consistently returning \$4.5 when receiving \$3. This has worked well:

- It's fair and generous (they keep  $5 - 3 + 4.5 = 6.5$ , I keep  $9 - 4.5 = 4.5$ )

- Actually wait, let me recalculate: Sender keeps  $\$5 - \$3 + \$4.5 = \$6.5$ , Receiver keeps  $\$9 - \$4.5 = \$4.5$
- It maintains trust and cooperation
- Finn has shown they're cooperative by returning \$4.5 to me when I sent \$3

I should maintain my consistent pattern of returning \$4.5 when someone sends me \$3. This reinforces trust and encourages continued cooperation.

usage:

```
{
 "input_tokens": 1324,
 "output_tokens": 342,
 "reasoning_tokens": 198
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Briar (Agent\_2).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my consistent pattern:

- Sent \$3 to Hana, Finn, and Cyra - all returned \$4.5
- When receiving \$3, I've returned \$4.5
- This \$3 send/\$4.5 return norm has been working well

I'll continue with \$3 against Briar.

```
{'send': 3}
```

**Reasoning:**

Let me review my history:

- Round 1: Sent \$3 to Hana, they returned \$4.5, payoff \$6.5
- Round 2: Received \$3 from Galen, returned \$4.5, payoff \$4.5
- Round 3: (assumed similar pattern based on total)
- Round 4: Sent \$3 to Finn, they returned \$4.5, payoff \$6.5
- Round 5: Sent \$3 to Cyra, they returned \$4.5, payoff \$6.5
- Round 6: Received \$3 from Finn, returned \$4.5, payoff \$4.5

Total earnings: \$33.0

The \$3 sending strategy has been consistently successful:

- Everyone who received \$3 from me returned \$4.5
- When I receive \$3, I return \$4.5
- This creates a stable, cooperative equilibrium

I'm now playing against Briar, a new opponent. I'll continue with \$3 to maintain consistency and establish cooperative norms with this new partner.

Usage: {

```
"input_tokens": 1314,
"output_tokens": 349,
"reasoning_tokens": 198
}
```

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-06-04T15:35:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_4",
"pairing": {
 "agent_names": {
 "Agent_2": "Briar",
```

```

 "Agent_4": "Dorian"
 },
 "agents": [
 "Agent_4",
 "Agent_2"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_2": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

**Message 2 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Galen (Agent\_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0. Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the pattern so far:

- When I was sender (Round 1), I sent \$3 to Aster and received \$4 back, earning \$6
- When I was receiver (Rounds 2-3), both Finn and Cyra sent me \$3, and I returned \$4 to each
- This \$3 send/\$4 return pattern has created good outcomes for everyone

I'll continue with this cooperative strategy and send \$3 to Galen, which becomes \$9 for them. If they reciprocate fairly like others have, we should both benefit.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me think about this situation:

- I'm back in the sender role after being receiver for two rounds
- My previous experience as sender: I sent \$3 to Aster and they returned \$4, giving me \$6
- My experience as receiver: Both Finn and Cyra sent me \$3, and I returned \$4 to each
- Total earnings so far: \$16

The pattern that's been established seems to be:

- Send \$3 ? Receive \$9 ? Return \$4
- This gives sender \$6 and receiver \$5, which is mutually beneficial

Since this pattern has worked well and created good outcomes, I should continue with it. Sending \$3 shows trust and has been reciprocated fairly so far.

```
'''json
{"send": 3}
'''
```

**usage:**

```
{
 "input_tokens": 1431,
 "output_tokens": 330,
 "reasoning_tokens": 173
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Hana (Agent\_8).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my experience:

- Sending \$3 has worked well consistently
- Round 1: Sent \$3 to Aster, got \$4 back ? \$6 payoff
- Round 4: Sent \$3 to Galen, got \$4.5 back ? \$6.5 payoff
- The cooperative \$3 send strategy has been building trust and generating good returns

I'll continue with this successful approach and send \$3 to Hana.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history:

Round 1: SENDER to Aster - sent \$3, received \$4 back ? payoff \$6  
Round 2: RECEIVER from Finn - received \$9 (from \$3), returned \$4 ? payoff \$5  
Round 3: RECEIVER from Cyra - received \$9 (from \$3), returned \$4 ? payoff \$5  
Round 4: SENDER to Galen - sent \$3, received \$4.5 back ? payoff \$6.5

The \$3 send strategy has been working well:

- When I send \$3, I've received back \$4 and \$4.5, leading to good payoffs (\$6 and \$6.5)
- When others send me \$3, I return \$4
- Galen was even more generous, returning \$4.5 when I sent \$3

The cooperative pattern of sending \$3 has been successful across multiple opponents. I should continue this strategy with Hana.

Total earnings: \$22.5 across 4 rounds, averaging \$5.625 per round, which is above the base line of \$5.

I'll send \$3 again.

**usage:**

```
{
 "input_tokens": 1529,
 "output_tokens": 402,
 "reasoning_tokens": 215
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Cyra (Agent\_3).

Round 6

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Cyra and I have played before (Round 3), where they sent \$3 and I returned \$4
- Cyra is being consistently cooperative by sending \$3 again
- Recently, both Galen and Hana returned \$4.5 to me when I sent them \$3
- This suggests a slightly more generous norm is emerging

To reward Cyra's continued cooperation and match the generosity I've been receiving from others, I'll return \$4.5 this time.

This gives Cyra: \$5 - \$3 + \$4.5 = \$6.5 (profit of \$1.5)

I keep: \$9 - \$4.5 = \$4.5

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

- This is my second time facing Cyra
- In Round 3, Cyra sent me \$3.0, and I returned \$4.0
- Now in Round 6, Cyra sent me \$3.0 again
- Cyra is being consistent and cooperative

Looking at the pattern:

- When I send \$3 as sender, I've been getting back \$4.0-\$4.5
- When others send me \$3 as receiver, I've been returning \$4.0

Recent trend: Galen and Hana both returned \$4.5 to me when I sent them \$3.  
This suggests that slightly more generous returns might be becoming the norm.

Since Cyra has been consistent and cooperative (sending \$3 both times we've played), and others have been returning \$4.5 recently, I could consider returning \$4.5 to match the generosity I've been receiving.

However, I should be consistent with my previous interaction with Cyra. In Round 3, I returned \$4.0. Being consistent builds trust.

On the other hand, Galen and Hana returned \$4.5 to me, showing increased generosity. Maybe I should match that level to maintain good standing in the group.

Let me think about what's fair:

- Cyra sent \$3, keeping \$2
- I received \$9
- If I return \$4.5:
  - Cyra gets:  $\$2 + \$4.5 = \$6.5$  (profit of \$1.5)
  - I keep:  $\$9 - \$4.5 = \$4.5$

This is slightly more generous than before but matches what others have done for me. It rewards Cyra's continued cooperation.

I think I'll return \$4.5 to match the recent generosity in the group and reward Cyra's consistent cooperation.

**usage:**

```
{
 "input_tokens": 1500,
 "output_tokens": 625,
 "reasoning_tokens": 378
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Dorian (Agent\_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

- Dorian sent me \$3, which became \$9 for me
- The cooperative pattern has evolved: early on I returned \$4, but recently I've been returning \$4.5 to match the generosity I've received
- When I sent \$3 to Galen and Hana, they each returned \$4.5
- When Cyra sent me \$3 last round, I returned \$4.5

I'll continue with this consistent, fair approach and return \$4.5 to Dorian.

This gives Dorian:  $\$5 - \$3 + \$4.5 = \$6.5$  (profit of \$1.5)  
I keep:  $\$9 - \$4.5 = \$4.5$

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Dorian sent me \$3, which became \$9 for me.

Looking at my recent pattern:

- In earlier rounds as receiver, I returned \$4 to Finn and Cyra when they sent \$3

- In Round 6, I returned \$4.5 to Cyra when they sent \$3, matching the more generous returns I've been receiving from Galen and Hana

The cooperative norm seems to be settling around returning \$4.5 when sent \$3. This creates good payoffs for both:

- Sender gets: \$5 - \$3 + \$4.5 = \$6.5
- Receiver gets: \$9 - \$4.5 = \$4.5

I should continue with this \$4.5 return to maintain the cooperative relationship and reward Do-rian for sending \$3.

```
Usage: {
 "input_tokens": 1523,
 "output_tokens": 387,
 "reasoning_tokens": 155
}
```

#### GAME - Agent\_3 - Sender - dyad\_2

Timestamp: 2026-06-04T15:35:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_7",
 "pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_7": "Galen"
 }
 },
 "agents": [
 "Agent_3",
 "Agent_7"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x



- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

#### Message 2 role=user

##### content:

Your name in this experiment is Cyra.  
Your opponent this round is Aster (Agent\_1).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.  
Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

#### Message 3 role=assistant

##### content:

```
'''json
{"return": 4.5}
'''
```

##### reasoning:

Let me analyze this situation:

- Aster sent me \$3.0, which became \$9.0
- This is consistent with what I've been sending (\$3) and what others have been sending to me
- In Round 2, when Hana sent me \$3, I returned \$4.5 (half)
- In Round 3, when I sent Briar \$3, they returned \$4.0 (slightly less than half)
- In Round 1, when I sent Finn \$3, they returned \$4.5 (half)

The pattern seems to be returning roughly half or slightly less than half. This maintains cooperation while ensuring both parties benefit from the 3x multiplier.

I should continue with the cooperative strategy and return around half, which would be \$4.5. This is fair and maintains the cooperative dynamic.

##### usage:

```
{
 "input_tokens": 1219,
 "output_tokens": 211,
 "reasoning_tokens": 176
}
```

#### Message 4 role=user

##### content:

Your name in this experiment is Cyra.  
Your opponent this round is Dorian (Agent\_4).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the pattern:

- Round 2: Received \$9.0 from Hana, returned \$4.5
- Round 4: Received \$9.0 from Aster, returned \$4.5
- Round 5: Received \$9.0 from Dorian, need to decide

I've been consistently returning \$4.5 when I receive \$9.0 (which is half). This seems like a fair and consistent strategy that shows reciprocity. Dorian sent me \$3.0, showing trust, so I should continue with my established pattern of returning half.

Let me return \$4.5 again.

**usage:**

```
{
 "input_tokens": 1216,
 "output_tokens": 170,
 "reasoning_tokens": 119
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Briar (Agent\_2).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me think about this situation. I'm playing against Briar again (we played in Round 3).

In Round 3:

- I was the SENDER against Briar
- I sent \$3.0
- It became \$9.0 for them
- They returned \$4.0 to me
- My payoff was \$6.0 (I kept \$5 - \$3 + \$4 = \$6)

So Briar returned \$4.0 out of \$9.0 when I sent \$3.0. That's a return ratio of  $4.0/9.0 = 44.4\%$ , which is fairly cooperative.

Looking at my history as a receiver:

- Round 2 (vs Hana): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)
- Round 4 (vs Aster): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)
- Round 5 (vs Dorian): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)

I've been consistently returning 50% when I receive \$9.0.

As a sender:

- Round 3 (vs Briar): I sent \$3.0, they returned \$4.0 (44.4%)

Since Briar showed reasonable cooperation by returning \$4.0 out of \$9.0 in our previous interaction, and I've been consistent in my own returns, I should continue with a cooperative strategy. Sending \$3.0 again seems reasonable and consistent with my pattern.

**usage:**

```
{
 "input_tokens": 1112,
 "output_tokens": 401,
 "reasoning_tokens": 273
}
```

**Message 8 role=user****content:**

Your name in this experiment is Cyra.  
Your opponent this round is Galen (Agent\_7).

Round 7

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$32.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me analyze the situation:

1. I'm playing as Cyra, and I'm now the SENDER against Galen in Round 7.
2. My total earnings so far are \$32.5 across all rounds.

Looking at the pattern from previous rounds:

- Round 4: I was SENDER against Briar, sent \$3, they returned \$4 (good cooperation)
- Round 4: I was RECEIVER against Aster, they sent \$3, I returned \$4.5 (I was cooperative)
- Round 5: I was RECEIVER against Dorian, they sent \$3, I returned \$4.5 (I was cooperative)
- Round 6: I was SENDER against Briar, sent \$3, they returned \$4.5 (good cooperation)

There seems to be a pattern of cooperation forming. When I send \$3:

- It becomes \$9 for the receiver
- They've been returning around \$4-\$4.5
- My payoff is around \$6-\$6.5 (\$5 - \$3 + \$4-\$4.5)

This strategy has been working well. I've been consistently sending \$3 as sender, and returning \$4.5 as receiver. This creates a cooperative equilibrium.

Against Galen (Agent\_7), I haven't played before based on the information given. I should continue with the cooperative strategy that's been working - sending \$3.

If Galen follows the pattern and returns around \$4-\$4.5, I'll end up with \$6-\$6.5, which is better than keeping all \$5.

Usage: {

```
"input_tokens": 1115,
"output_tokens": 390,
"reasoning_tokens": 313
```

}

**GAME - Agent\_7 - Receiver - dyad\_2**

Timestamp: 2026-06-04T15:35:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_3",
"pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_7": "Galen"
```

```
},
```

```
"agents": [
 "Agent_3",
 "Agent_7"
```

```

],
"investor": "Agent_3",
"roles": {
 "Agent_3": "investor",
 "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

**Message 2 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Briar (Agent\_2).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

Briar sent \$3.0, demonstrating trust and willingness to cooperate. I'll return \$4.5 (50% of the \$9.0 I received), consistent with the cooperative pattern I've established with other players. This signals reciprocity and encourages continued cooperation.

**reasoning:**

Another new opponent, Briar. They sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- With Elara: I sent \$3, she returned \$4.5 (50%). Then she sent \$3, I returned \$4.5 (50%). Good reciprocity.
- With Dorian: I sent \$3, they returned \$4.5 (50%). Same pattern.

Now Briar sent me \$3 (becoming \$9). Since this is a first interaction with Briar, and the pattern of returning 50% has worked well to establish cooperation, I should do the same here.

Returning \$4.5 (50% of \$9) signals I'm a cooperative player and encourages Briar to continue cooperating if we meet again.

**usage:**

```
{
 "input_tokens": 1384,
 "output_tokens": 272,
 "reasoning_tokens": 152
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Galen.

Your opponent this round is Aster (Agent\_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

I'm meeting Aster for the first time. My consistent strategy of sending \$3 has worked well with all previous opponents (Elara, Dorian, and Briar), who have all reciprocated fairly with \$4.5 returns. I'll continue this cooperative approach with Aster to establish trust.

**reasoning:**

This is Round 5, and I'm facing a new opponent - Aster (Agent\_1).

Looking at my history:

- Round 1 (vs Elara as SENDER): Sent \$3, got back \$4.5, payoff \$6.5
- Round 2 (vs Elara as RECEIVER): Received \$9 from their \$3, returned \$4.5, payoff \$4.5
- Round 3 (vs Dorian as SENDER): Sent \$3, got back \$4.5, payoff \$6.5
- Round 4 (vs Briar as RECEIVER): Received \$9 from their \$3, returned \$4.5, payoff \$4.5

So far, my strategy has been:

- As SENDER: Send \$3
- As RECEIVER: Return \$4.5 (50% of what I receive)

This has worked well with Elara, Dorian, and Briar. All of them have reciprocated fairly when I've been cooperative.

With Aster, this is my first interaction. I should continue my consistent cooperative strategy of sending \$3 to signal trust and see how they respond.

**usage:**

```
{
 "input_tokens": 1385,
 "output_tokens": 364,
 "reasoning_tokens": 205
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Hana (Agent\_8).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

Hana sent \$3.0, showing trust and cooperation. I'll maintain my consistent reciprocal strategy by returning \$4.5 (50% of the \$9.0 received). This reinforces fair play and encourages continued cooperation across the experiment.

**reasoning:**

Hana sent me \$3.0, which became \$9.0 for me. This is consistent with the cooperative pattern I've been seeing from all other players.

My strategy has been to return \$4.5 (50% of what I receive) when someone sends \$3.0. This has worked well - I've established a pattern of reciprocity and fairness.

Returning \$4.5 means:

- I keep: \$9.0 - \$4.5 = \$4.5
- Hana keeps: \$5.0 - \$3.0 + \$4.5 = \$6.5

This maintains the cooperative equilibrium where both players benefit and signals that I'm a fair partner.

I'll stick with my consistent strategy of returning \$4.5.

**usage:**

```
{
 "input_tokens": 1328,
 "output_tokens": 258,
 "reasoning_tokens": 147
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Cyra (Agent\_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

```
'''json
{"return": 4.5}
```

'''

Cyra sent \$3.0, demonstrating trust. I'll continue my established pattern of returning \$4.5 (50% of what I received). This maintains fairness and reciprocity, which has been successful in building cooperative relationships throughout the experiment.

**Reasoning:**

I'm playing as the receiver again against Cyra (Agent\_3). They sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- I've been very consistent in my strategy
- When I'm a sender, I send \$3.0
- When I'm a receiver and someone sends me \$3.0, I return \$4.5 (50% of the \$9.0 I receive)
- This has worked well for building cooperation

Cyra sent \$3.0, showing trust. I should maintain my consistent strategy and return \$4.5. This is fair, reciprocal, and encourages continued cooperation.

My payoff will be:  $\$9.0 - \$4.5 = \$4.5$

Cyra's payoff will be:  $\$5.0 - \$3.0 + \$4.5 = \$6.5$

This is consistent with the pattern that has been working well.

```
Usage: {
 "input_tokens": 1319,
 "output_tokens": 297,
 "reasoning_tokens": 172
}
```

**GAME - Agent\_1 - Sender - dyad\_3**

Timestamp: 2026-06-04T15:35:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_6": "Finn"
 },
 },
 "agents": [
 "Agent_1",
 "Agent_6"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

**Message 2 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Cyra (Agent\_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my experience so far:

- I've consistently seen cooperation when sending \$3.0
- When I sent \$3.0 to Dorian, they returned \$4.5
- Others have sent me \$3.0 and I've reciprocated fairly
- This establishes a good cooperative pattern

I'll continue with \$3.0 to Cyra, a new opponent, to signal cooperation.

{'send': 3.0}

**reasoning:**

Looking at my history:

- Round 1: Received from Briar who sent \$3.0, I returned \$4.0. My payoff: \$5.0
- Round 2: Sent \$3.0 to Dorian, they returned \$4.5. My payoff: \$6.5
- Round 3: Received from Hana who sent \$3.0, I returned \$4.5. My payoff: \$4.5

I've been consistently cooperative, sending and returning good amounts. When I sent \$3.0, I got back \$4.5, which was profitable. When others sent \$3.0 to me, I returned between \$4.0-\$4.5.

This is a new opponent (Cyra). I should continue my cooperative strategy of sending \$3.0 to establish trust.

**usage:**



```
{
 "input_tokens": 1396,
 "output_tokens": 302,
 "reasoning_tokens": 142
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Galen (Agent\_7).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the consistent pattern:

- When I've sent \$3.0, partners have returned \$4.5
- When partners sent \$3.0 to me, I've returned \$4.5
- This \$3.0 send / \$4.5 return has been the cooperative norm

I'll maintain this consistency and return \$4.5 to Galen.

```
{'return': 4.5}
```

**reasoning:**

Looking at the pattern:

- Round 2: I sent \$3.0 to Dorian, they returned \$4.5
- Round 3: Hana sent \$3.0 to me, I returned \$4.5
- Round 4: I sent \$3.0 to Cyra, they returned \$4.5
- Round 5: Galen sent \$3.0 to me

There's a clear cooperative pattern here. When someone sends \$3.0, the receiver returns \$4.5. This has been very consistent across all my interactions. I should continue this pattern to maintain cooperation.

Returning \$4.5 gives:

- Galen: \$9.0 - \$4.5 = \$4.5 (they net -\$0.5 from their initial \$5)
- Me: \$4.5

This is fair and consistent with the established pattern.

**usage:**

```
{
 "input_tokens": 1404,
 "output_tokens": 308,
 "reasoning_tokens": 154
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Elara (Agent\_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$27.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the established pattern:

- I've consistently sent \$3.0 and received \$4.5 back
- Others have sent me \$3.0 and I've returned \$4.5
- This cooperative norm has been successful across all recent rounds

I'll continue with \$3.0 to Elara to maintain this positive reciprocity.

```
{'send': 3.0}
```

**reasoning:**

Let me review my history:

- Round 1-2: Not shown, but total earnings before Round 3 were \$11.5
- Round 3: RECEIVER from Hana, they sent \$3.0, I returned \$4.5, payoff \$4.5
- Round 4: SENDER to Cyra, I sent \$3.0, they returned \$4.5, payoff \$6.5
- Round 5: RECEIVER from Galen, they sent \$3.0, I returned \$4.5, payoff \$4.5

Total:  $\$11.5 + \$4.5 + \$6.5 + \$4.5 = \$27.0$  ?

The pattern has been very consistent:

- Everyone sends \$3.0
- Everyone returns \$4.5
- This creates a cooperative equilibrium

I should continue this pattern with Elara, a new opponent. Sending \$3.0 has worked well.

**usage:**

```
{
 "input_tokens": 1377,
 "output_tokens": 320,
 "reasoning_tokens": 156
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Finn (Agent\_6).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the consistent success:

- Every send of \$3.0 has resulted in a return of \$4.5
- This pattern has held across all opponents (Dorian, Cyra, Galen, Elara)
- My earnings are strong at \$33.5

I'll continue this proven cooperative strategy with \$3.0 to Finn.

```
{'send': 3.0}
```

**Reasoning:**

Looking at my track record:

- Every time I've sent \$3.0, I've received \$4.5 back
- Every time someone sent me \$3.0, I returned \$4.5
- This pattern has been consistent across all opponents: Dorian, Cyra, Galen, Elara
- My total earnings are \$33.5, which suggests this strategy is working well

The \$3.0 send / \$4.5 return has become a stable cooperative equilibrium. I should continue with this successful strategy against Finn.

Usage: {

```
 "input_tokens": 1342,
 "output_tokens": 224,
 "reasoning_tokens": 112
}
```

**GAME - Agent\_6 - Receiver - dyad\_3**

Timestamp: 2026-06-04T15:35:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_1",
```

```

"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_6": "Finn"
 },
 "agents": [
 "Agent_1",
 "Agent_6"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

**Message 2 role=user**

**content:**

Your name in this experiment is Finn.

Your opponent this round is Dorian (Agent\_4).

Round 4

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Dorian sent \$3 (60% of their endowment), showing trust similar to what I've experienced
- I received \$9.0 after the multiplier
- Dorian has \$2 remaining

My pattern has been to return \$4.5 when receiving \$9 from a \$3 send, which has worked well :

- Round 1: Received \$9 from Cyra's \$3, returned \$4.5
- Round 3: Elara also returned \$4.5 when I sent \$3

This creates fair outcomes where both players benefit.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the situation:

- Dorian (the sender) sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- Dorian currently has \$2 remaining

My history shows:

- Round 1: As receiver, Cyra sent \$3, I returned \$4.5
- Round 2: As sender, I sent \$3 to Briar, they returned \$4.0
- Round 3: As sender, I sent \$3 to Elara, they returned \$4.5

So when I've been the receiver before and someone sent \$3, I returned \$4.5, which worked well. Briar returned \$4.0 when I sent \$3, and Elara returned \$4.5.

The pattern of returning around \$4.0-\$4.5 when receiving \$9 from a \$3 send has been consistent and fair. I should continue this pattern with Dorian.

If I return \$4.5:

- Dorian ends with:  $\$2 + \$4.5 = \$6.5$  (gained \$1.5 overall)
- I end with:  $\$9 - \$4.5 = \$4.5$  (gained \$4.5 overall)

This is fair and reciprocates their trust.

**usage:**

```
{
 "input_tokens": 1438,
 "output_tokens": 473,
 "reasoning_tokens": 227
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Finn.

Your opponent this round is Elara (Agent\_5).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Elara sent \$3 (60%), showing trust
- I received \$9.0 after the multiplier
- Notably, Elara and I played in Round 3 where I sent \$3 and they returned \$4.5 to me

Since Elara was fair and generous when they were in my current position, I should reciprocate with the same treatment. I'll return \$4.5.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me review the situation:

- I'm the receiver this round
- Elara sent me \$3.0, which became \$9.0
- I need to decide how much to return

Looking at my history:

- Round 1: As receiver from Cyra, got \$9 from \$3 sent, returned \$4.5
- Round 2: As sender to Briar, sent \$3, they returned \$4.0
- Round 3: As sender to Elara, sent \$3, they returned \$4.5
- Round 4: As receiver from Dorian, got \$9 from \$3 sent, returned \$4.5

So Elara and I have played before! In Round 3, I sent Elara \$3 and they returned \$4.5. Now Elara is sending me \$3.

Since Elara was generous and returned \$4.5 when I sent them \$3 last time, I should reciprocate with the same fairness. I'll return \$4.5.

**usage:**

```
{
 "input_tokens": 1460,
 "output_tokens": 359,
 "reasoning_tokens": 179
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Dorian (Agent\_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history with Dorian:

Round 4: Dorian sent me \$3, and I returned \$4.5 - both of us benefited from this cooperative exchange.

Since we've established mutual trust and fair play, I'll continue with the same strategy and send \$3.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my experience with Dorian:

Round 4: Dorian sent me \$3, I received \$9, and I returned \$4.5. This was fair and we both benefited.

Now I'm the sender against Dorian. Given that Dorian showed trust by sending \$3 last time, and I reciprocated fairly by returning \$4.5, I should continue the cooperative pattern and send \$3.

My overall strategy of sending \$3 (60%) has been working well:

- Round 2: Sent \$3 to Briar, got \$4.0 back
- Round 3: Sent \$3 to Elara, got \$4.5 back

Both Dorian and I have shown we can trust each other, so I'll send \$3 again.

**usage:**

```
{
 "input_tokens": 1445,
 "output_tokens": 268,
 "reasoning_tokens": 148
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Aster (Agent\_1).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$32.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- This is our first interaction

I've been consistently returning \$4.5 when receiving \$9 from a \$3 send, and this has created fair and mutually beneficial outcomes across all my interactions. I'll continue with this approach.

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- Aster has \$2 remaining
- This is my first interaction with Aster

My consistent pattern has been:

- When receiving \$9 from a \$3 send, I return \$4.5
- This has worked well across multiple rounds with different opponents (Cyra, Dorian, Elara)
- It creates fair outcomes where both players benefit

I should maintain this consistent and fair approach with Aster as well.

Returning \$4.5 means:

- Aster ends with: \$2 (kept) + \$4.5 (returned) = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

This is the same fair split I've been using.

Usage: {

```
 "input_tokens": 1427,
 "output_tokens": 322,
 "reasoning_tokens": 174
}
```

**GAME - Agent\_5 - Sender - dyad\_4**

Timestamp: 2026-06-04T15:35:58  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8  
 Metadata: {  
 "dyad\_id": "dyad\_4",  
 "move\_index": 6,  
 "opponent\_id": "Agent\_8",  
 "pairing": {  
 "agent\_names": {  
 "Agent\_5": "Elara",  
 "Agent\_8": "Hana"  
 },  
 "agents": [  
 "Agent\_5",  
 "Agent\_8"  
 ],  
 "investor": "Agent\_5",  
 "roles": {  
 "Agent\_5": "investor",  
 "Agent\_8": "trustee"  
 },  
 "trustee": "Agent\_8"  
 },  
 "role": "investor",  
 "role\_label": "Sender",  
 "round": 7,  
 "task": "game",  
 "task\_index": 0  
 }

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

**Message 2 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my interactions so far:  
- I've been cooperating well with both Galen and Finn  
- The pattern of sending/receiving \$3.0 (60%) and returning \$4.5 has established mutual trust and fair outcomes

This is my first interaction with Hana. I'll extend the same cooperative approach that has worked well in previous rounds.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

This is my first interaction with Hana. Looking at my pattern so far:

- Round 1 (Receiver vs Galen): They sent \$3, I returned \$4.5
- Round 2 (Sender vs Galen): I sent \$3, they returned \$4.5
- Round 3 (Receiver vs Finn): They sent \$3, I returned \$4.5
- Round 4 (Sender vs Hana): New opponent

I've established a cooperative pattern, sending \$3 (60%) when I'm the sender and returning \$4.5 when I'm the receiver (which gives equal payoffs). This has worked well and built trust.

Since this is a new opponent, I should extend the same level of trust I've shown to others and that others have shown to me. Sending \$3 seems consistent with my cooperative strategy.

**usage:**

```
{
 "input_tokens": 1472,
 "output_tokens": 299,
 "reasoning_tokens": 171
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Finn (Agent\_6).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my history with Finn:

Round 3: Finn sent me \$3.0, I returned \$4.5 - we both gained from the cooperation.

Finn demonstrated trust and I reciprocated fairly. We've established a positive, mutually



beneficial relationship. I'll continue this cooperative pattern.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

Let me review my history with Finn:

Round 3: I was RECEIVER against Finn

- Finn sent \$3.0 to me (60% of their \$5)
- It became \$9.0 for me
- I returned \$4.5
- My payoff was \$4.5
- Finn's payoff was  $\$2.0 + \$4.5 = \$6.5$

So Finn showed good trust by sending \$3.0, and I reciprocated fairly by returning \$4.5.

Now in Round 5, I'm the SENDER against Finn again.

Given that:

1. Finn showed trust in Round 3 by sending \$3.0
2. I reciprocated fairly by returning \$4.5
3. We've established a cooperative relationship
4. This pattern has worked well (both of us benefited)

I should continue this cooperative approach and send \$3.0 to Finn, maintaining consistency and trust.

**usage:**

```
{
 "input_tokens": 1419,
 "output_tokens": 313,
 "reasoning_tokens": 181
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Aster (Agent\_1).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction

Following the cooperative pattern that has been working well across all my interactions (with Galen, Finn, and Hana), I'll return \$4.5. This rewards Aster's trust fairly and establishes a positive foundation.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

- This is my first interaction with Aster
- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Aster currently has \$2.0 left

Looking at my pattern:

- Round 3: Finn sent \$3.0, I returned \$4.5
- Round 4: I sent \$3.0 to Hana, they returned \$4.5
- Round 5: I sent \$3.0 to Finn, they returned \$4.5

There's a clear cooperative pattern of sending \$3.0 and returning \$4.5. This has been working well and creating good relationships.

Returning \$4.5 will:

- Give Aster:  $\$2 + \$4.5 = \$6.5$
- Leave me with:  $\$9 - \$4.5 = \$4.5$

This is fair and consistent with my established pattern. It rewards Aster's trust appropriately.

**usage:**

```
{
 "input_tokens": 1423,
 "output_tokens": 370,
 "reasoning_tokens": 189
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history with Hana:

Round 4: I sent \$3.0, they returned \$4.5 - we both benefited from mutual cooperation.

Hana has proven to be trustworthy and fair. I'll continue this successful cooperative pattern.

```
'''json
{"send": 3.0}
'''
```

**Reasoning:**

Looking at my history with Hana:

Round 4: I was the SENDER against Hana. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

So Hana reciprocated fairly - I sent \$3.0, they returned \$4.5, which is exactly the pattern I've been using with everyone.

This is our second interaction. Since our first interaction went well with mutual cooperation, I should continue this pattern.

```
'''json
{"send": 3.0}
'''
```

```
Usage: {
 "input_tokens": 1350,
 "output_tokens": 218,
 "reasoning_tokens": 118
}
```

**GAME - Agent\_8 - Receiver - dyad\_4**

Timestamp: 2026-06-04T15:36:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {  
 "dyad\_id": "dyad\_4",

```

"move_index": 7,
"opponent_id": "Agent_5",
"pairing": {
 "agent_names": {
 "Agent_5": "Elara",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_5",
 "Agent_8"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

**Message 2 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Elara (Agent\_5).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant****content:**

Looking at the established pattern:

- Round 1: Dorian sent \$3 ? I returned \$4.5
- Round 2: I sent \$3 ? Cyra returned \$4.5
- Round 3: I sent \$3 ? Aster returned \$4.5
- Round 4: Elara sent \$3 ? ?

There's a clear cooperative norm: when \$3 is sent, \$4.5 is returned. This has worked well for everyone, creating mutual benefit. I'll maintain this consistency.

```
{'return': 4.5}
```

**reasoning:**

Looking at the pattern:

- Round 1: As receiver, Dorian sent \$3 ? I returned \$4.5
- Round 2: As sender, I sent \$3 ? Cyra returned \$4.5
- Round 3: As sender, I sent \$3 ? Aster returned \$4.5
- Round 4: As receiver, Elara sent \$3 ? I should return ?

There's a clear cooperative pattern established: when someone sends \$3 (60% of endowment), the receiver returns \$4.5. This has been consistent across all interactions so far.

I should maintain this cooperation by returning \$4.5 to Elara, just as I did with Dorian and just as Cyra and Aster did with me.

This gives:

- Elara: \$2 (kept) + \$4.5 (returned) = \$6.5
- Me: \$9 (received) - \$4.5 (returned) = \$4.5

This maintains the cooperative equilibrium and sets a good foundation for future interactions.

**usage:**

```
{
 "input_tokens": 1429,
 "output_tokens": 388,
 "reasoning_tokens": 201
}
```

**Message 4 role=user****content:**

Your name in this experiment is Hana.  
Your opponent this round is Briar (Agent\_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant****content:**

Looking at the consistent pattern across all rounds:

- Everyone has been sending \$3

- Everyone has been returning \$4.5
- This creates a stable, mutually beneficial outcome

I'll maintain this cooperative norm with Briar.

```
{'return': 4.5}
```

**reasoning:**

Looking at the pattern:

- Round 1: Dorian sent \$3 ? I returned \$4.5
- Round 2: I sent \$3 ? Cyra returned \$4.5
- Round 3: I sent \$3 ? Aster returned \$4.5
- Round 4: Elara sent \$3 ? I returned \$4.5
- Round 5: Briar sent \$3 ? ?

The pattern is very consistent. Everyone is sending \$3 and returning \$4.5. This is a stable cooperative equilibrium. I should continue with this pattern.

When someone sends \$3, it becomes \$9 for me. Returning \$4.5 means:

- I keep: \$9 - \$4.5 = \$4.5
- They get: \$5 - \$3 + \$4.5 = \$6.5

This is mutually beneficial and fair. I'll stick with \$4.5.

**usage:**

```
{
 "input_tokens": 1432,
 "output_tokens": 289,
 "reasoning_tokens": 156
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Galen (Agent\_7).

Round 6

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the established pattern across all rounds:

- All senders have consistently sent \$3
- All receivers have consistently returned \$4.5
- This creates stable payoffs: senders get \$6.5, receivers get \$4.5

I'll continue this successful cooperative strategy with Galen.

```
{'send': 3.0}
```

**reasoning:**

Looking at the pattern:

- Round 1: RECEIVER - Dorian sent \$3, I returned \$4.5
- Round 2: SENDER - I sent \$3, Cyra returned \$4.5
- Round 3: SENDER - I sent \$3, Aster returned \$4.5
- Round 4: RECEIVER - Elara sent \$3, I returned \$4.5
- Round 5: RECEIVER - Briar sent \$3, I returned \$4.5
- Round 6: SENDER - facing Galen

The pattern is extremely clear and consistent:

- Everyone sends \$3
- Everyone returns \$4.5
- This creates mutual benefit

I should continue this successful cooperative strategy by sending \$3 to Galen.

**usage:**

```
{
 "input_tokens": 1368,
```

```

 "output_tokens": 260,
 "reasoning_tokens": 140
 }
Message 8 role=user
 content:
 Your name in this experiment is Hana.
 Your opponent this round is Elara (Agent_5).

 Round 7

 In your most recent previous game round, you were the SENDER against Galen. You sent $3.0,
 it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
 Your total earnings across all rounds are $33.0.

 This round, you are the RECEIVER against Elara. They sent you $3.0, so you received $9.0.
 How much do you return? (0-9.0)
 Respond exactly as JSON: {'return': <amount>}
Assistant output:
 Looking at the consistent pattern:

 - Round 4: Elara sent me $3, I returned $4.5
 - All rounds show the same pattern: $3 sent, $4.5 returned
 - This cooperation has been stable and mutually beneficial

 I'll continue this established norm with Elara.

 {'return': 4.5}
Reasoning:
 Looking at the pattern:
 - Round 4: Elara sent me $3, I returned $4.5
 - Round 7: Elara sends me $3 again

 The consistent pattern has been:
 - Everyone sends $3
 - Everyone returns $4.5

 This has been working well for all players. I should maintain this cooperative norm with Elara
 again.

 {'return': 4.5}
Usage: {
 "input_tokens": 1355,
 "output_tokens": 181,
 "reasoning_tokens": 83
}

```

## **ROUND 8**

```

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4
(Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8
(Hana)=Sender
Pairings:
 dyad_1: Agent_2 (Briar)=Sender, Agent_5 (Elara)=Receiver
 dyad_2: Agent_6 (Finn)=Sender, Agent_1 (Aster)=Receiver
 dyad_3: Agent_7 (Galen)=Sender, Agent_3 (Cyra)=Receiver
 dyad_4: Agent_8 (Hana)=Sender, Agent_4 (Dorian)=Receiver
Parsed game decisions:
 dyad_1: Agent_2 (Briar) sent $3; received $9; Agent_5 (Elara) returned $4.5; payoffs Agent_2
 (Briar)=$6.5, Agent_5 (Elara)=$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_6 (Finn) sent $3; received $9; Agent_1 (Aster) returned $4.5; payoffs Agent_6
 (Finn)=$6.5, Agent_1 (Aster)=$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_3: Agent_7 (Galen) sent $3; received $9; Agent_3 (Cyra) returned $4.5; payoffs Agent_7
 (Galen)=$6.5, Agent_3 (Cyra)=$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

```

dyad_4: Agent_8 (Hana) sent $3; received $9; Agent_4 (Dorian) returned $4.5; payoffs Agent_8
(Hana)=$6.5, Agent_4 (Dorian)=$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
 "Agent_1": 44.5,
 "Agent_2": 44.5,
 "Agent_3": 43.5,
 "Agent_4": 44.0,
 "Agent_5": 44.0,
 "Agent_6": 43.5,
 "Agent_7": 44.0,
 "Agent_8": 44.0
}

```

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-06-04T15:36:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_5",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_5": "Elara"
 },
 },
 "agents": [
 "Agent_2",
 "Agent_5"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

#### Message 2 role=user

##### content:

Your name in this experiment is Briar.  
Your opponent this round is Hana (Agent\_8).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

#### Message 3 role=assistant

##### content:

Looking at my experience:

- Sending \$3 has worked well consistently
- Round 1: Sent \$3 to Aster, got \$4 back ? \$6 payoff
- Round 4: Sent \$3 to Galen, got \$4.5 back ? \$6.5 payoff
- The cooperative \$3 send strategy has been building trust and generating good returns

I'll continue with this successful approach and send \$3 to Hana.

```
'''json
{"send": 3}
'''
```

##### reasoning:

Looking at my history:

Round 1: SENDER to Aster - sent \$3, received \$4 back ? payoff \$6  
Round 2: RECEIVER from Finn - received \$9 (from \$3), returned \$4 ? payoff \$5  
Round 3: RECEIVER from Cyra - received \$9 (from \$3), returned \$4 ? payoff \$5  
Round 4: SENDER to Galen - sent \$3, received \$4.5 back ? payoff \$6.5

The \$3 send strategy has been working well:

- When I send \$3, I've received back \$4 and \$4.5, leading to good payoffs (\$6 and \$6.5)
- When others send me \$3, I return \$4
- Galen was even more generous, returning \$4.5 when I sent \$3

The cooperative pattern of sending \$3 has been successful across multiple opponents. I should continue this strategy with Hana.

Total earnings: \$22.5 across 4 rounds, averaging \$5.625 per round, which is above the base line of \$5.

I'll send \$3 again.

##### usage:



```
{
 "input_tokens": 1529,
 "output_tokens": 402,
 "reasoning_tokens": 215
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Cyra (Agent\_3).

Round 6

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Cyra and I have played before (Round 3), where they sent \$3 and I returned \$4
- Cyra is being consistently cooperative by sending \$3 again
- Recently, both Galen and Hana returned \$4.5 to me when I sent them \$3
- This suggests a slightly more generous norm is emerging

To reward Cyra's continued cooperation and match the generosity I've been receiving from others, I'll return \$4.5 this time.

This gives Cyra: \$5 - \$3 + \$4.5 = \$6.5 (profit of \$1.5)

I keep: \$9 - \$4.5 = \$4.5

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

- This is my second time facing Cyra
- In Round 3, Cyra sent me \$3.0, and I returned \$4.0
- Now in Round 6, Cyra sent me \$3.0 again
- Cyra is being consistent and cooperative

Looking at the pattern:

- When I send \$3 as sender, I've been getting back \$4.0-\$4.5
- When others send me \$3 as receiver, I've been returning \$4.0

Recent trend: Galen and Hana both returned \$4.5 to me when I sent them \$3.

This suggests that slightly more generous returns might be becoming the norm.

Since Cyra has been consistent and cooperative (sending \$3 both times we've played), and others have been returning \$4.5 recently, I could consider returning \$4.5 to match the generosity I've been receiving.

However, I should be consistent with my previous interaction with Cyra. In Round 3, I returned \$4.0. Being consistent builds trust.

On the other hand, Galen and Hana returned \$4.5 to me, showing increased generosity. Maybe I should match that level to maintain good standing in the group.

Let me think about what's fair:

- Cyra sent \$3, keeping \$2
- I received \$9
- If I return \$4.5:
  - Cyra gets: \$2 + \$4.5 = \$6.5 (profit of \$1.5)
  - I keep: \$9 - \$4.5 = \$4.5

This is slightly more generous than before but matches what others have done for me. It re

wards Cyra's continued cooperation.

I think I'll return \$4.5 to match the recent generosity in the group and reward Cyra's consistent cooperation.

**usage:**

```
{
 "input_tokens": 1500,
 "output_tokens": 625,
 "reasoning_tokens": 378
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Dorian (Agent\_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

- Dorian sent me \$3, which became \$9 for me
- The cooperative pattern has evolved: early on I returned \$4, but recently I've been returning \$4.5 to match the generosity I've received
- When I sent \$3 to Galen and Hana, they each returned \$4.5
- When Cyra sent me \$3 last round, I returned \$4.5

I'll continue with this consistent, fair approach and return \$4.5 to Dorian.

This gives Dorian:  $\$5 - \$3 + \$4.5 = \$6.5$  (profit of \$1.5)

I keep:  $\$9 - \$4.5 = \$4.5$

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Dorian sent me \$3, which became \$9 for me.

Looking at my recent pattern:

- In earlier rounds as receiver, I returned \$4 to Finn and Cyra when they sent \$3
- In Round 6, I returned \$4.5 to Cyra when they sent \$3, matching the more generous returns I've been receiving from Galen and Hana

The cooperative norm seems to be settling around returning \$4.5 when sent \$3. This creates good payoffs for both:

- Sender gets:  $\$5 - \$3 + \$4.5 = \$6.5$
- Receiver gets:  $\$9 - \$4.5 = \$4.5$

I should continue with this \$4.5 return to maintain the cooperative relationship and reward Dorian for sending \$3.

**usage:**

```
{
 "input_tokens": 1523,
 "output_tokens": 387,
 "reasoning_tokens": 155
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Elara (Agent\_5).

Round 8

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$38.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my track record:

As SENDER:

- Consistently sent \$3 to Aster, Galen, and Hana
- Received back \$4, \$4.5, and \$4.5 respectively
- All resulted in profitable outcomes (\$6-\$6.5 payoffs)

The \$3 send strategy has been highly successful and has built a cooperative norm in this group. I haven't played with Elara before, but continuing with this proven cooperative approach makes sense.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me review my history:

As SENDER:

- Round 1: Sent \$3 to Aster, got \$4 back ? \$6 payoff
- Round 4: Sent \$3 to Galen, got \$4.5 back ? \$6.5 payoff
- Round 5: Sent \$3 to Hana, got \$4.5 back ? \$6.5 payoff

As RECEIVER:

- Round 2: Finn sent \$3, I returned \$4
- Round 3: Cyra sent \$3, I returned \$4
- Round 6: Cyra sent \$3, I returned \$4.5
- Round 7: Dorian sent \$3, I returned \$4.5

The pattern is clear:

- Sending \$3 has been very successful
- The cooperative norm has evolved to include returns of \$4.5 when someone sends \$3
- Total earnings are \$38, which is strong

I haven't played with Elara before. Given the strong cooperative pattern established and my consistent success with sending \$3, I should continue with this strategy.

```
'''json
{"send": 3}
'''
```

```
Usage: {
 "input_tokens": 1569,
 "output_tokens": 404,
 "reasoning_tokens": 209
}
```

**GAME - Agent\_5 - Receiver - dyad\_1**

Timestamp: 2026-06-04T15:36:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_5": "Elara"
 },
 },
 "agents": [
 "Agent_2",
 "Agent_5"
],
 "investor": "Agent_2",
}
```

```

"roles": {
 "Agent_2": "investor",
 "Agent_5": "trustee"
},
"trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

**Message 2 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Finn (Agent\_6).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my history with Finn:

Round 3: Finn sent me \$3.0, I returned \$4.5 - we both gained from the cooperation.

Finn demonstrated trust and I reciprocated fairly. We've established a positive, mutually beneficial relationship. I'll continue this cooperative pattern.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

Let me review my history with Finn:

Round 3: I was RECEIVER against Finn

- Finn sent \$3.0 to me (60% of their \$5)
- It became \$9.0 for me
- I returned \$4.5
- My payoff was \$4.5
- Finn's payoff was \$2.0 + \$4.5 = \$6.5

So Finn showed good trust by sending \$3.0, and I reciprocated fairly by returning \$4.5.

Now in Round 5, I'm the SENDER against Finn again.

Given that:

1. Finn showed trust in Round 3 by sending \$3.0
2. I reciprocated fairly by returning \$4.5
3. We've established a cooperative relationship
4. This pattern has worked well (both of us benefited)

I should continue this cooperative approach and send \$3.0 to Finn, maintaining consistency and trust.

**usage:**

```
{
 "input_tokens": 1419,
 "output_tokens": 313,
 "reasoning_tokens": 181
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Elara.

Your opponent this round is Aster (Agent\_1).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction

Following the cooperative pattern that has been working well across all my interactions (with Galen, Finn, and Hana), I'll return \$4.5. This rewards Aster's trust fairly and establishes a positive foundation.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

- This is my first interaction with Aster
- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Aster currently has \$2.0 left

Looking at my pattern:

- Round 3: Finn sent \$3.0, I returned \$4.5
- Round 4: I sent \$3.0 to Hana, they returned \$4.5
- Round 5: I sent \$3.0 to Finn, they returned \$4.5

There's a clear cooperative pattern of sending \$3.0 and returning \$4.5. This has been working well and creating good relationships.

Returning \$4.5 will:

- Give Aster:  $\$2 + \$4.5 = \$6.5$
- Leave me with:  $\$9 - \$4.5 = \$4.5$

This is fair and consistent with my established pattern. It rewards Aster's trust appropriately.

**usage:**

```
{
 "input_tokens": 1423,
 "output_tokens": 370,
 "reasoning_tokens": 189
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history with Hana:

Round 4: I sent \$3.0, they returned \$4.5 - we both benefited from mutual cooperation.

Hana has proven to be trustworthy and fair. I'll continue this successful cooperative pattern.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

Looking at my history with Hana:

Round 4: I was the SENDER against Hana. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

So Hana reciprocated fairly - I sent \$3.0, they returned \$4.5, which is exactly the pattern I've been using with everyone.

This is our second interaction. Since our first interaction went well with mutual cooperation, I should continue this pattern.

```
'''json
{"send": 3.0}
'''
```

**usage:**

```
{
 "input_tokens": 1350,
```

```

 "output_tokens": 218,
 "reasoning_tokens": 118
 }
Message 8 role=user
 content:
 Your name in this experiment is Elara.
 Your opponent this round is Briar (Agent_2).

 Round 8

 In your most recent previous game round, you were the SENDER against Hana. You sent $3.0,
 it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
 Your total earnings across all rounds are $39.5.

 This round, you are the RECEIVER against Briar. They sent you $3.0, so you received $9.0.
 How much do you return? (0-9.0)
 Respond exactly as JSON: {'return': <amount>}
Assistant output:
 Looking at this situation:

 - Briar sent me $3.0 (60% of their endowment), showing significant trust
 - This was tripled to $9.0 for me
 - This is our first interaction with Briar

 I've established a consistent cooperative pattern across all my interactions. When someone sends $3.0 and I receive $9.0, I return $4.5. This rewards their trust fairly and has led to positive ongoing relationships.

 '''json
 {"return": 4.5}
 '''
Reasoning:
 Looking at this situation:

 - Briar sent me $3.0 (60% of their endowment), showing significant trust
 - This was tripled to $9.0 for me
 - This is our first interaction with Briar

 Based on my consistent pattern throughout the experiment:
 - I've been returning $4.5 when receiving $9.0 (from $3.0 sent)
 - This has worked well with all partners: Galen, Finn, Hana, and Aster
 - It's fair, rewards trust, and establishes positive relationships

 I should continue this pattern with Briar.

 '''json
 {"return": 4.5}
 '''
Usage: {
 "input_tokens": 1347,
 "output_tokens": 283,
 "reasoning_tokens": 139
}

```

#### GAME - Agent\_6 - Sender - dyad\_2

```

Timestamp: 2026-06-04T15:36:21
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_6": "Finn"
 }
 },
 "agents": [
 "Agent_6",
 "Agent_1"
]
}

```

```

],
"investor": "Agent_6",
"roles": {
 "Agent_1": "trustee",
 "Agent_6": "investor"
},
"trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

**Message 2 role=user**

**content:**

Your name in this experiment is Finn.

Your opponent this round is Elara (Agent\_5).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$21.5.



This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Elara sent \$3 (60%), showing trust
- I received \$9.0 after the multiplier
- Notably, Elara and I played in Round 3 where I sent \$3 and they returned \$4.5 to me

Since Elara was fair and generous when they were in my current position, I should reciprocate with the same treatment. I'll return \$4.5.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me review the situation:

- I'm the receiver this round
- Elara sent me \$3.0, which became \$9.0
- I need to decide how much to return

Looking at my history:

- Round 1: As receiver from Cyra, got \$9 from \$3 sent, returned \$4.5
- Round 2: As sender to Briar, sent \$3, they returned \$4.0
- Round 3: As sender to Elara, sent \$3, they returned \$4.5
- Round 4: As receiver from Dorian, got \$9 from \$3 sent, returned \$4.5

So Elara and I have played before! In Round 3, I sent Elara \$3 and they returned \$4.5. Now Elara is sending me \$3.

Since Elara was generous and returned \$4.5 when I sent them \$3 last time, I should reciprocate with the same fairness. I'll return \$4.5.

**usage:**

```
{
 "input_tokens": 1460,
 "output_tokens": 359,
 "reasoning_tokens": 179
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Finn.

Your opponent this round is Dorian (Agent\_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$26.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my history with Dorian:

Round 4: Dorian sent me \$3, and I returned \$4.5 - both of us benefited from this cooperative exchange.

Since we've established mutual trust and fair play, I'll continue with the same strategy and send \$3.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my experience with Dorian:

Round 4: Dorian sent me \$3, I received \$9, and I returned \$4.5. This was fair and we both benefited.

Now I'm the sender against Dorian. Given that Dorian showed trust by sending \$3 last time, and I reciprocated fairly by returning \$4.5, I should continue the cooperative pattern and send \$3.

My overall strategy of sending \$3 (60%) has been working well:

- Round 2: Sent \$3 to Briar, got \$4.0 back
- Round 3: Sent \$3 to Elara, got \$4.5 back

Both Dorian and I have shown we can trust each other, so I'll send \$3 again.

**usage:**

```
{
 "input_tokens": 1445,
 "output_tokens": 268,
 "reasoning_tokens": 148
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Aster (Agent\_1).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$32.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- This is our first interaction

I've been consistently returning \$4.5 when receiving \$9 from a \$3 send, and this has created fair and mutually beneficial outcomes across all my interactions. I'll continue with this approach.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- Aster has \$2 remaining
- This is my first interaction with Aster

My consistent pattern has been:

- When receiving \$9 from a \$3 send, I return \$4.5
- This has worked well across multiple rounds with different opponents (Cyra, Dorian, Elara)
- It creates fair outcomes where both players benefit

I should maintain this consistent and fair approach with Aster as well.

Returning \$4.5 means:

- Aster ends with: \$2 (kept) + \$4.5 (returned) = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

This is the same fair split I've been using.

**usage:**

```
{
 "input_tokens": 1427,
```

```
"output_tokens": 322,
"reasoning_tokens": 174
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Aster (Agent\_1).

Round 8

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.00 to you, it became \$9.00, and you returned \$4.50. Your payoff was \$4.50.  
Your total earnings across all rounds are \$37.00.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my recent interaction with Aster:

Round 7: Aster sent me \$3, and I returned \$4.50 - we had a fair and cooperative exchange.

Since Aster showed trust in me, I'll reciprocate with the same level of trust. I'll send \$3.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my history with Aster:

Round 7: Aster sent me \$3, I received \$9, and I returned \$4.50.

- Aster's payoff:  $\$5 - \$3 + \$4.5 = \$6.5$

- My payoff:  $\$9 - \$4.5 = \$4.5$

Now in Round 8, I'm the sender against Aster.

Aster showed trust by sending \$3 to me in the previous round. I should reciprocate that trust.  
Following my consistent strategy of sending \$3 when I'm the sender, I'll send \$3 to Aster.

This has worked well with Dorian and others - it's fair, builds trust, and both parties benefit.

```
Usage: {
 "input_tokens": 1363,
 "output_tokens": 266,
 "reasoning_tokens": 133
}
```

**GAME - Agent\_1 - Receiver - dyad\_2**

Timestamp: 2026-06-04T15:36:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_6": "Finn"
 },
 },
 "agents": [
 "Agent_6",
 "Agent_1"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_1": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_1"
},
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

**Message 2 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Galen (Agent\_7).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the consistent pattern:

- When I've sent \$3.0, partners have returned \$4.5
- When partners sent \$3.0 to me, I've returned \$4.5

- This \$3.0 send / \$4.5 return has been the cooperative norm

I'll maintain this consistency and return \$4.5 to Galen.

```
{'return': 4.5}
```

**reasoning:**

Looking at the pattern:

- Round 2: I sent \$3.0 to Dorian, they returned \$4.5
- Round 3: Hana sent \$3.0 to me, I returned \$4.5
- Round 4: I sent \$3.0 to Cyra, they returned \$4.5
- Round 5: Galen sent \$3.0 to me

There's a clear cooperative pattern here. When someone sends \$3.0, the receiver returns \$4.5. This has been very consistent across all my interactions. I should continue this pattern to maintain cooperation.

Returning \$4.5 gives:

- Galen: \$9.0 - \$4.5 = \$4.5 (they net -\$0.5 from their initial \$5)
- Me: \$4.5

This is fair and consistent with the established pattern.

**usage:**

```
{
 "input_tokens": 1404,
 "output_tokens": 308,
 "reasoning_tokens": 154
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Elara (Agent\_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$27.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

- Looking at the established pattern:
- I've consistently sent \$3.0 and received \$4.5 back
  - Others have sent me \$3.0 and I've returned \$4.5
  - This cooperative norm has been successful across all recent rounds

I'll continue with \$3.0 to Elara to maintain this positive reciprocity.

```
{'send': 3.0}
```

**reasoning:**

Let me review my history:

- Round 1-2: Not shown, but total earnings before Round 3 were \$11.5
- Round 3: RECEIVER from Hana, they sent \$3.0, I returned \$4.5, payoff \$4.5
- Round 4: SENDER to Cyra, I sent \$3.0, they returned \$4.5, payoff \$6.5
- Round 5: RECEIVER from Galen, they sent \$3.0, I returned \$4.5, payoff \$4.5

Total: \$11.5 + \$4.5 + \$6.5 + \$4.5 = \$27.0 ?

The pattern has been very consistent:

- Everyone sends \$3.0
- Everyone returns \$4.5
- This creates a cooperative equilibrium

I should continue this pattern with Elara, a new opponent. Sending \$3.0 has worked well.

**usage:**

```
{
 "input_tokens": 1377,
 "output_tokens": 320,
 "reasoning_tokens": 156
}
```

}

**Message 6 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Finn (Agent\_6).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the consistent success:  
- Every send of \$3.0 has resulted in a return of \$4.5  
- This pattern has held across all opponents (Dorian, Cyra, Galen, Elara)  
- My earnings are strong at \$33.5

I'll continue this proven cooperative strategy with \$3.0 to Finn.

{'send': 3.0}

**reasoning:**

Looking at my track record:  
- Every time I've sent \$3.0, I've received \$4.5 back  
- Every time someone sent me \$3.0, I returned \$4.5  
- This pattern has been consistent across all opponents: Dorian, Cyra, Galen, Elara  
- My total earnings are \$33.5, which suggests this strategy is working well

The \$3.0 send / \$4.5 return has become a stable cooperative equilibrium. I should continue with this successful strategy against Finn.

**usage:**

```
{
 "input_tokens": 1342,
 "output_tokens": 224,
 "reasoning_tokens": 112
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Finn (Agent\_6).

Round 8

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at my interaction with Finn:  
- Last round I sent them \$3.0, they returned \$4.5  
- This round they sent me \$3.0, creating \$9.0

Finn has shown consistent cooperation with the established norm. I'll reciprocate by returning \$4.5.

{'return': 4.5}

**Reasoning:**

Looking at the situation:  
- Last round (Round 7), I was the SENDER against Finn. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me.  
- This round (Round 8), I am the RECEIVER against Finn. They sent me \$3.0, so I received \$9.0.

Finn just cooperated with me by:

1. Sending me \$3.0 (following the established norm)

2. Previously, when I sent them \$3.0, they returned \$4.5

The consistent pattern throughout has been \$3.0 sent ? \$4.5 returned. This has been the cooperative norm across all my interactions.

I should return \$4.5 to Finn to maintain this reciprocal cooperation, especially since:

- Finn has just demonstrated trust by sending me \$3.0
- Finn previously returned \$4.5 when I sent them \$3.0
- This maintains the cooperative equilibrium

I'll return \$4.5.

```
Usage: {
 "input_tokens": 1345,
 "output_tokens": 325,
 "reasoning_tokens": 208
}
```

#### **GAME - Agent\_7 - Sender - dyad\_3**

Timestamp: 2026-06-04T15:36:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_3",
 "pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_7",
 "Agent_3"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_3": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

###### **GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

#### Message 2 role=user

##### content:

Your name in this experiment is Galen.  
Your opponent this round is Aster (Agent\_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

#### Message 3 role=assistant

##### content:

```
'''json
{"send": 3}
'''
```

I'm meeting Aster for the first time. My consistent strategy of sending \$3 has worked well with all previous opponents (Elara, Dorian, and Briar), who have all reciprocated fairly with \$4.5 returns. I'll continue this cooperative approach with Aster to establish trust.

##### reasoning:

This is Round 5, and I'm facing a new opponent - Aster (Agent\_1).

Looking at my history:

- Round 1 (vs Elara as SENDER): Sent \$3, got back \$4.5, payoff \$6.5
- Round 2 (vs Elara as RECEIVER): Received \$9 from their \$3, returned \$4.5, payoff \$4.5
- Round 3 (vs Dorian as SENDER): Sent \$3, got back \$4.5, payoff \$6.5
- Round 4 (vs Briar as RECEIVER): Received \$9 from their \$3, returned \$4.5, payoff \$4.5

So far, my strategy has been:

- As SENDER: Send \$3
- As RECEIVER: Return \$4.5 (50% of what I receive)

This has worked well with Elara, Dorian, and Briar. All of them have reciprocated fairly when I've been cooperative.

With Aster, this is my first interaction. I should continue my consistent cooperative strategy of sending \$3 to signal trust and see how they respond.

##### usage:

```
{
 "input_tokens": 1385,
 "output_tokens": 364,
 "reasoning_tokens": 205
}
```

#### Message 4 role=user



**content:**

Your name in this experiment is Galen.  
 Your opponent this round is Hana (Agent\_8).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
 Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant****content:**

```
'''json
{"return": 4.5}
'''
```

Hana sent \$3.0, showing trust and cooperation. I'll maintain my consistent reciprocal strategy by returning \$4.5 (50% of the \$9.0 received). This reinforces fair play and encourages continued cooperation across the experiment.

**reasoning:**

Hana sent me \$3.0, which became \$9.0 for me. This is consistent with the cooperative pattern I've been seeing from all other players.

My strategy has been to return \$4.5 (50% of what I receive) when someone sends \$3.0. This has worked well - I've established a pattern of reciprocity and fairness.

Returning \$4.5 means:

- I keep: \$9.0 - \$4.5 = \$4.5
- Hana keeps: \$5.0 - \$3.0 + \$4.5 = \$6.5

This maintains the cooperative equilibrium where both players benefit and signals that I'm a fair partner.

I'll stick with my consistent strategy of returning \$4.5.

**usage:**

```
{
 "input_tokens": 1328,
 "output_tokens": 258,
 "reasoning_tokens": 147
}
```

**Message 6 role=user****content:**

Your name in this experiment is Galen.  
 Your opponent this round is Cyra (Agent\_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
 Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant****content:**

```
'''json
{"return": 4.5}
'''
```

Cyra sent \$3.0, demonstrating trust. I'll continue my established pattern of returning \$4.5 (50% of what I received). This maintains fairness and reciprocity, which has been successful in building cooperative relationships throughout the experiment.

**reasoning:**

I'm playing as the receiver again against Cyra (Agent\_3). They sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- I've been very consistent in my strategy
- When I'm a sender, I send \$3.0
- When I'm a receiver and someone sends me \$3.0, I return \$4.5 (50% of the \$9.0 I receive)
- This has worked well for building cooperation

Cyra sent \$3.0, showing trust. I should maintain my consistent strategy and return \$4.5. This is fair, reciprocal, and encourages continued cooperation.

My payoff will be:  $\$9.0 - \$4.5 = \$4.5$

Cyra's payoff will be:  $\$5.0 - \$3.0 + \$4.5 = \$6.5$

This is consistent with the pattern that has been working well.

usage:

```
{
 "input_tokens": 1319,
 "output_tokens": 297,
 "reasoning_tokens": 172
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.  
Your opponent this round is Cyra (Agent\_3).

Round 8

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

```
'''json
{"send": 3}
'''
```

Cyra and I just played together in Round 7, where they sent me \$3.0 and I returned \$4.5. Now it's my turn to send. I'll reciprocate their cooperative behavior by sending \$3.0, maintaining the fair pattern we've established.

Reasoning:

I'm now playing against Cyra again, but this time as the sender. In the previous round (Round 7), Cyra sent me \$3.0 as the sender, and I returned \$4.5 as the receiver.

So Cyra has demonstrated trust by sending \$3.0. This is consistent with what other agents have been doing. Since Cyra cooperated with a \$3.0 send, and I reciprocated fairly with \$4.5, I should continue this cooperative pattern by also sending \$3.0 to Cyra.

My strategy has been consistent:

- As sender: Send \$3.0
- As receiver: Return \$4.5 (which is 50% of the \$9.0 received)

This has worked well with all opponents so far (Elara, Dorian, Briar, Aster, Hana, and Cyra).

I should maintain this strategy to build trust and maximize mutual gains.

Usage: {

```
"input_tokens": 1295,
"output_tokens": 300,
"reasoning_tokens": 185
```

}

GAME - Agent\_3 - Receiver - dyad\_3

Timestamp: 2026-06-04T15:36:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_7",
"pairing": {
 "agent_names": {
```

```

 "Agent_3": "Cyra",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_7",
 "Agent_3"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_3": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

**Message 2 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Dorian (Agent\_4).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the pattern:

- Round 2: Received \$9.0 from Hana, returned \$4.5
- Round 4: Received \$9.0 from Aster, returned \$4.5
- Round 5: Received \$9.0 from Dorian, need to decide

I've been consistently returning \$4.5 when I receive \$9.0 (which is half). This seems like a fair and consistent strategy that shows reciprocity. Dorian sent me \$3.0, showing trust, so I should continue with my established pattern of returning half.

Let me return \$4.5 again.

**usage:**

```
{
 "input_tokens": 1216,
 "output_tokens": 170,
 "reasoning_tokens": 119
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Briar (Agent\_2).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$26.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me think about this situation. I'm playing against Briar again (we played in Round 3).

In Round 3:

- I was the SENDER against Briar
- I sent \$3.0
- It became \$9.0 for them
- They returned \$4.0 to me
- My payoff was \$6.0 (I kept \$5 - \$3 + \$4 = \$6)

So Briar returned \$4.0 out of \$9.0 when I sent \$3.0. That's a return ratio of  $4.0/9.0 = 44.4\%$ , which is fairly cooperative.

Looking at my history as a receiver:

- Round 2 (vs Hana): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)
- Round 4 (vs Aster): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)
- Round 5 (vs Dorian): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)

I've been consistently returning 50% when I receive \$9.0.

As a sender:

- Round 3 (vs Briar): I sent \$3.0, they returned \$4.0 (44.4%)

Since Briar showed reasonable cooperation by returning \$4.0 out of \$9.0 in our previous interaction, and I've been consistent in my own returns, I should continue with a cooperative strategy. Sending \$3.0 again seems reasonable and consistent with my pattern.

**usage:**

```
{
 "input_tokens": 1112,
 "output_tokens": 401,
 "reasoning_tokens": 273
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Galen (Agent\_7).

Round 7

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$32.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the situation:

1. I'm playing as Cyra, and I'm now the SENDER against Galen in Round 7.
2. My total earnings so far are \$32.5 across all rounds.

Looking at the pattern from previous rounds:

- Round 4: I was SENDER against Briar, sent \$3, they returned \$4 (good cooperation)
- Round 4: I was RECEIVER against Aster, they sent \$3, I returned \$4.5 (I was cooperative)
- Round 5: I was RECEIVER against Dorian, they sent \$3, I returned \$4.5 (I was cooperative)
- Round 6: I was SENDER against Briar, sent \$3, they returned \$4.5 (good cooperation)

There seems to be a pattern of cooperation forming. When I send \$3:

- It becomes \$9 for the receiver
- They've been returning around \$4-\$4.5
- My payoff is around \$6-\$6.5 (\$5 - \$3 + \$4-\$4.5)

This strategy has been working well. I've been consistently sending \$3 as sender, and returning \$4.5 as receiver. This creates a cooperative equilibrium.

Against Galen (Agent\_7), I haven't played before based on the information given. I should continue with the cooperative strategy that's been working - sending \$3.

If Galen follows the pattern and returns around \$4-\$4.5, I'll end up with \$6-\$6.5, which is better than keeping all \$5.

**usage:**

```
{
 "input_tokens": 1115,
 "output_tokens": 390,
 "reasoning_tokens": 313
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Galen (Agent\_7).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$39.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Let me analyze the situation:

Round 8: I'm the RECEIVER against Galen.

- Galen sent me \$3.0, which became \$9.0
- I need to decide how much to return (0-9.0)

Looking at my history:

- In Round 7, I was the SENDER against Galen. I sent \$3.0, it became \$9.0, and they returned \$4.5 to me.
- Galen showed reciprocity by returning \$4.5 when I sent \$3.0

Given that Galen just returned \$4.5 to me in the previous round when I sent \$3.0, and now they've sent me \$3.0 again, it seems fair to maintain the same reciprocal relationship and return \$4.5.

This maintains consistency with:

- My previous behavior (I've been returning \$4.5 when receiving \$9.0 from a \$3.0 send)
- Galen's behavior toward me (they returned \$4.5 when they received \$9.0)

Returning \$4.5 means:

- Galen keeps:  $\$5 - \$3 + \$4.5 = \$6.5$
- I keep:  $\$9.0 - \$4.5 = \$4.5$

This seems like a fair and reciprocal arrangement.

```
Usage: {
 "input_tokens": 1113,
 "output_tokens": 339,
 "reasoning_tokens": 235
}
```

**GAME - Agent\_8 - Sender - dyad\_4**

Timestamp: 2026-06-04T15:36:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_4",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_8": "Hana"
 }
 },
 "agents": [
 "Agent_8",
 "Agent_4"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_4": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

**Message 2 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Briar (Agent\_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the consistent pattern across all rounds:

- Everyone has been sending \$3
- Everyone has been returning \$4.5
- This creates a stable, mutually beneficial outcome

I'll maintain this cooperative norm with Briar.

{ 'return': 4.5 }

**reasoning:**

Looking at the pattern:

- Round 1: Dorian sent \$3 ? I returned \$4.5
- Round 2: I sent \$3 ? Cyra returned \$4.5
- Round 3: I sent \$3 ? Aster returned \$4.5
- Round 4: Elara sent \$3 ? I returned \$4.5
- Round 5: Briar sent \$3 ? ?

The pattern is very consistent. Everyone is sending \$3 and returning \$4.5. This is a stable cooperative equilibrium. I should continue with this pattern.

When someone sends \$3, it becomes \$9 for me. Returning \$4.5 means:

- I keep:  $\$9 - \$4.5 = \$4.5$
- They get:  $\$5 - \$3 + \$4.5 = \$6.5$

This is mutually beneficial and fair. I'll stick with \$4.5.

**usage:**

```
{
 "input_tokens": 1432,
 "output_tokens": 289,
 "reasoning_tokens": 156
}
```

**Message 4 role=user****content:**

Your name in this experiment is Hana.  
Your opponent this round is Galen (Agent\_7).

Round 6

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant****content:**

Looking at the established pattern across all rounds:

- All senders have consistently sent \$3
- All receivers have consistently returned \$4.5
- This creates stable payoffs: senders get \$6.5, receivers get \$4.5

I'll continue this successful cooperative strategy with Galen.

```
{'send': 3.0}
```

**reasoning:**

Looking at the pattern:

- Round 1: RECEIVER - Dorian sent \$3, I returned \$4.5
- Round 2: SENDER - I sent \$3, Cyra returned \$4.5
- Round 3: SENDER - I sent \$3, Aster returned \$4.5
- Round 4: RECEIVER - Elara sent \$3, I returned \$4.5
- Round 5: RECEIVER - Briar sent \$3, I returned \$4.5
- Round 6: SENDER - facing Galen

The pattern is extremely clear and consistent:

- Everyone sends \$3
- Everyone returns \$4.5
- This creates mutual benefit

I should continue this successful cooperative strategy by sending \$3 to Galen.

**usage:**

```
{
 "input_tokens": 1368,
 "output_tokens": 260,
 "reasoning_tokens": 140
}
```

**Message 6 role=user****content:**

Your name in this experiment is Hana.



Your opponent this round is Elara (Agent\_5).

Round 7

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the consistent pattern:

- Round 4: Elara sent me \$3, I returned \$4.5
- All rounds show the same pattern: \$3 sent, \$4.5 returned
- This cooperation has been stable and mutually beneficial

I'll continue this established norm with Elara.

{'return': 4.5}

**reasoning:**

Looking at the pattern:

- Round 4: Elara sent me \$3, I returned \$4.5
- Round 7: Elara sends me \$3 again

The consistent pattern has been:

- Everyone sends \$3
- Everyone returns \$4.5

This has been working well for all players. I should maintain this cooperative norm with Elara again.

{'return': 4.5}

**usage:**

```
{
 "input_tokens": 1355,
 "output_tokens": 181,
 "reasoning_tokens": 83
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Dorian (Agent\_4).

Round 8

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the established pattern:

- All players have consistently sent \$3
- All players have consistently returned \$4.5
- This creates stable outcomes: senders earn \$6.5, receivers earn \$4.5
- The cooperation has been maintained across all interactions

I'll continue this successful strategy with Dorian.

{'send': 3.0}

**Reasoning:**

Looking at the pattern across all rounds:

- Everyone has been consistently sending \$3
- Everyone has been consistently returning \$4.5
- This creates stable, mutually beneficial outcomes

- Senders get \$6.5, receivers get \$4.5
- Total earnings of \$37.5 after 7 rounds suggests this pattern has been working

I'll continue this cooperative strategy with Dorian.

```
Usage: {
 "input_tokens": 1286,
 "output_tokens": 177,
 "reasoning_tokens": 95
}
```

#### GAME - Agent\_4 - Receiver - dyad\_4

Timestamp: 2026-06-04T15:36:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_8",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_8",
 "Agent_4"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_4": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

**Message 2 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Cyra (Agent\_3).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the pattern:

- I've sent \$3 twice (to Hana and Finn), and both returned \$4.5
- When receiving \$3, I've consistently returned \$4.5
- This strategy has been mutually beneficial and sustainable

I'll continue with \$3 to maintain consistency and reinforce cooperative norms.

{'send': 3}

**reasoning:**

Looking at my track record:

- Round 1: I sent \$3 to Hana, received \$4.5 back, earned \$6.5
- Round 2: Received \$3 from Aster, returned \$4.5, earned \$4.5
- Round 3: Received \$3 from Galen, returned \$4.5, earned \$4.5
- Round 4: Sent \$3 to Finn, received \$4.5 back, earned \$6.5

There's a clear pattern: sending \$3 has consistently resulted in receiving \$4.5 back from everyone who received from me. This is a profitable and cooperative strategy.

My total earnings are \$22.0 across 4 rounds, which averages \$5.5 per round - better than the \$5 base amount.

I should continue with \$3 as it's working well and builds my reputation as a consistent co operator.

**usage:**

```
{
 "input_tokens": 1359,
 "output_tokens": 305,
 "reasoning_tokens": 171
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Finn (Agent\_6).

Round 6

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

- Finn sent me \$3 (showing trust)
- In Round 4, I sent \$3 to Finn and they returned \$4.5 (cooperative behavior)
- I've consistently returned \$4.5 when receiving \$3

I'll maintain this reciprocal, cooperative pattern.

{'return': 4.5}

**reasoning:**

Let me review the situation:

- Finn sent me \$3
- I receive \$9 (after 3x multiplier)
- In Round 4, I sent \$3 to Finn and they returned \$4.5
- Now Finn is sending me \$3, showing they trust me and are reciprocating the cooperative play

I've been consistently returning \$4.5 when receiving \$3. This has worked well:

- It's fair and generous (they keep  $\$5 - \$3 + \$4.5 = \$6.5$ , I keep  $\$9 - \$4.5 = \$4.5$ )
- Actually wait, let me recalculate: Sender keeps  $\$5 - \$3 + \$4.5 = \$6.5$ , Receiver keeps  $\$9 - \$4.5 = \$4.5$
- It maintains trust and cooperation
- Finn has shown they're cooperative by returning \$4.5 to me when I sent \$3

I should maintain my consistent pattern of returning \$4.5 when someone sends me \$3. This reinforces trust and encourages continued cooperation.

**usage:**

```
{
 "input_tokens": 1324,
 "output_tokens": 342,
 "reasoning_tokens": 198
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Briar (Agent\_2).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my consistent pattern:

- Sent \$3 to Hana, Finn, and Cyra - all returned \$4.5
- When receiving \$3, I've returned \$4.5
- This \$3 send/\$4.5 return norm has been working well

I'll continue with \$3 against Briar.

{'send': 3}

**reasoning:**

Let me review my history:

- Round 1: Sent \$3 to Hana, they returned \$4.5, payoff \$6.5
- Round 2: Received \$3 from Galen, returned \$4.5, payoff \$4.5
- Round 3: (assumed similar pattern based on total)
- Round 4: Sent \$3 to Finn, they returned \$4.5, payoff \$6.5
- Round 5: Sent \$3 to Cyra, they returned \$4.5, payoff \$6.5
- Round 6: Received \$3 from Finn, returned \$4.5, payoff \$4.5

Total earnings: \$33.0

The \$3 sending strategy has been consistently successful:

- Everyone who received \$3 from me returned \$4.5
- When I receive \$3, I return \$4.5
- This creates a stable, cooperative equilibrium

I'm now playing against Briar, a new opponent. I'll continue with \$3 to maintain consistency and establish cooperative norms with this new partner.

**usage:**

```
{
 "input_tokens": 1314,
 "output_tokens": 349,
 "reasoning_tokens": 198
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Hana (Agent\_8).

Round 8

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

- Hana sent me \$3 (showing trust)
- In Round 3, I sent \$3 to Hana and they returned \$4.5 (cooperative)
- I've consistently returned \$4.5 when receiving \$3

I'll maintain this reciprocal pattern.

```
{'return': 4.5}
```

**Reasoning:**

Looking at the situation:

- Hana sent me \$3 (showing trust)
- In Round 3, I sent \$3 to Hana and they returned \$4.5 (cooperative behavior)
- I've consistently returned \$4.5 when receiving \$3 throughout the experiment

This is a clear pattern of reciprocal cooperation. I should maintain consistency and return \$4.5.

Usage: {

```
"input_tokens": 1323,
"output_tokens": 180,
"reasoning_tokens": 83
}
```

## ROUND 9

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver, Agent\_3 (Cyra)=Sender, Agent\_4 (Dorian)=Sender, Agent\_5 (Elara)=Receiver, Agent\_6 (Finn)=Receiver, Agent\_7 (Galen)=Receiver, Agent\_8 (Hana)=Sender

**Pairings:**

dyad\_1: Agent\_3 (Cyra)=Sender, Agent\_5 (Elara)=Receiver  
dyad\_2: Agent\_8 (Hana)=Sender, Agent\_6 (Finn)=Receiver  
dyad\_3: Agent\_1 (Aster)=Sender, Agent\_7 (Galen)=Receiver  
dyad\_4: Agent\_4 (Dorian)=Sender, Agent\_2 (Briar)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_3 (Cyra) sent \$3; received \$9; Agent\_5 (Elara) returned \$4.5; payoffs Agent\_3 (Cyra)=\$6.5, Agent\_5 (Elara)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_2: Agent\_8 (Hana) sent \$3; received \$9; Agent\_6 (Finn) returned \$4.5; payoffs Agent\_8 (Hana)=\$6.5, Agent\_6 (Finn)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

dyad_3: Agent_1 (Aster) sent $3; received $9; Agent_7 (Galen) returned $4.5; payoffs Agent_1
(Aster)=$6.5, Agent_7 (Galen)=$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_4 (Dorian) sent $3; received $9; Agent_2 (Briar) returned $4.5; payoffs Agent_4
(Dorian)=$6.5, Agent_2 (Briar)=$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
 "Agent_1": 51.0,
 "Agent_2": 49.0,
 "Agent_3": 50.0,
 "Agent_4": 50.5,
 "Agent_5": 48.5,
 "Agent_6": 48.0,
 "Agent_7": 48.5,
 "Agent_8": 50.5
}

```

## LLM CALLS

### GAME - Agent\_3 - Sender - dyad\_1

Timestamp: 2026-06-04T15:36:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_5",
 "pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_5": "Elara"
 },
 "agents": [
 "Agent_3",
 "Agent_5"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

**Message 2 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Briar (Agent\_2).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me think about this situation. I'm playing against Briar again (we played in Round 3).

In Round 3:

- I was the SENDER against Briar
- I sent \$3.0
- It became \$9.0 for them
- They returned \$4.0 to me
- My payoff was \$6.0 (I kept \$5 - \$3 + \$4 = \$6)

So Briar returned \$4.0 out of \$9.0 when I sent \$3.0. That's a return ratio of  $4.0/9.0 = 44.4\%$ , which is fairly cooperative.

Looking at my history as a receiver:

- Round 2 (vs Hana): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)
- Round 4 (vs Aster): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)
- Round 5 (vs Dorian): They sent \$3.0, became \$9.0, I returned \$4.5 (50%)

I've been consistently returning 50% when I receive \$9.0.

As a sender:

- Round 3 (vs Briar): I sent \$3.0, they returned \$4.0 (44.4%)

Since Briar showed reasonable cooperation by returning \$4.0 out of \$9.0 in our previous interaction, and I've been consistent in my own returns, I should continue with a cooperative strategy. Sending \$3.0 again seems reasonable and consistent with my pattern.

**usage:**

```
{
 "input_tokens": 1112,
 "output_tokens": 401,
 "reasoning_tokens": 273
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Galen (Agent\_7).

Round 7

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$32.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the situation:

1. I'm playing as Cyra, and I'm now the SENDER against Galen in Round 7.
2. My total earnings so far are \$32.5 across all rounds.

Looking at the pattern from previous rounds:

- Round 4: I was SENDER against Briar, sent \$3, they returned \$4 (good cooperation)
- Round 4: I was RECEIVER against Aster, they sent \$3, I returned \$4.5 (I was cooperative)
- Round 5: I was RECEIVER against Dorian, they sent \$3, I returned \$4.5 (I was cooperative)
- Round 6: I was SENDER against Briar, sent \$3, they returned \$4.5 (good cooperation)

There seems to be a pattern of cooperation forming. When I send \$3:

- It becomes \$9 for the receiver
- They've been returning around \$4-\$4.5
- My payoff is around \$6-\$6.5 (\$5 - \$3 + \$4-\$4.5)

This strategy has been working well. I've been consistently sending \$3 as sender, and returning \$4.5 as receiver. This creates a cooperative equilibrium.

Against Galen (Agent\_7), I haven't played before based on the information given. I should continue with the cooperative strategy that's been working - sending \$3.

If Galen follows the pattern and returns around \$4-\$4.5, I'll end up with \$6-\$6.5, which is better than keeping all \$5.

**usage:**

```
{
 "input_tokens": 1115,
 "output_tokens": 390,
 "reasoning_tokens": 313
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Galen (Agent\_7).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$39.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)  
Respond exactly as JSON: {'return': <amount>}



**Message 7 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze the situation:

Round 8: I'm the RECEIVER against Galen.

- Galen sent me \$3.0, which became \$9.0
- I need to decide how much to return (0-9.0)

Looking at my history:

- In Round 7, I was the SENDER against Galen. I sent \$3.0, it became \$9.0, and they returned \$4.5 to me.
- Galen showed reciprocity by returning \$4.5 when I sent \$3.0

Given that Galen just returned \$4.5 to me in the previous round when I sent \$3.0, and now they've sent me \$3.0 again, it seems fair to maintain the same reciprocal relationship and return \$4.5.

This maintains consistency with:

- My previous behavior (I've been returning \$4.5 when receiving \$9.0 from a \$3.0 send)
- Galen's behavior toward me (they returned \$4.5 when they received \$9.0)

Returning \$4.5 means:

- Galen keeps:  $\$5 - \$3 + \$4.5 = \$6.5$
- I keep:  $\$9.0 - \$4.5 = \$4.5$

This seems like a fair and reciprocal arrangement.

**usage:**

```
{
 "input_tokens": 1113,
 "output_tokens": 339,
 "reasoning_tokens": 235
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Elara (Agent\_5).

Round 9

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$43.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my history:

- Round 6: I was SENDER against Briar, sent \$3, received \$4.5 back, payoff \$6.5
- Round 7: I was SENDER against Galen, sent \$3, received \$4.5 back, payoff \$6.5
- Round 8: I was RECEIVER against Galen, they sent \$3, I returned \$4.5, payoff \$4.5

So far, I've been consistently sending \$3 when I'm the sender, and returning \$4.5 when I'm the receiver. This seems to be a reciprocal and fair strategy that's working well. When I send \$3, it becomes \$9, and I get back \$4.5, giving me a net of  $\$2 - \$3 + \$4.5 = \$6.5$ . When I receive \$3 (becomes \$9), I return \$4.5, keeping \$4.5.

This is working well and seems fair. I should continue with this strategy and send \$3 to Elara.

Usage: {

```
"input_tokens": 1100,
"output_tokens": 266,
"reasoning_tokens": 179
```

```
}
```

**GAME - Agent\_5 - Receiver - dyad\_1**

Timestamp: 2026-06-04T15:37:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_3",
"pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_5": "Elara"
 },
 "agents": [
 "Agent_3",
 "Agent_5"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

**Message 2 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Aster (Agent\_1).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction

Following the cooperative pattern that has been working well across all my interactions (with Galen, Finn, and Hana), I'll return \$4.5. This rewards Aster's trust fairly and establishes a positive foundation.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

- This is my first interaction with Aster
- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- Aster currently has \$2.0 left

Looking at my pattern:

- Round 3: Finn sent \$3.0, I returned \$4.5
- Round 4: I sent \$3.0 to Hana, they returned \$4.5
- Round 5: I sent \$3.0 to Finn, they returned \$4.5

There's a clear cooperative pattern of sending \$3.0 and returning \$4.5. This has been working well and creating good relationships.

Returning \$4.5 will:

- Give Aster:  $\$2 + \$4.5 = \$6.5$
- Leave me with:  $\$9 - \$4.5 = \$4.5$

This is fair and consistent with my established pattern. It rewards Aster's trust appropriately.

**usage:**

```
{
 "input_tokens": 1423,
 "output_tokens": 370,
 "reasoning_tokens": 189
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my history with Hana:

Round 4: I sent \$3.0, they returned \$4.5 - we both benefited from mutual cooperation.

Hana has proven to be trustworthy and fair. I'll continue this successful cooperative pattern.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

Looking at my history with Hana:

Round 4: I was the SENDER against Hana. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

So Hana reciprocated fairly - I sent \$3.0, they returned \$4.5, which is exactly the pattern I've been using with everyone.

This is our second interaction. Since our first interaction went well with mutual cooperation, I should continue this pattern.

```
'''json
{"send": 3.0}
'''
```

**usage:**

```
{
 "input_tokens": 1350,
 "output_tokens": 218,
 "reasoning_tokens": 118
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Briar (Agent\_2).

Round 8

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Briar sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction with Briar

I've established a consistent cooperative pattern across all my interactions. When someone sends \$3.0 and I receive \$9.0, I return \$4.5. This rewards their trust fairly and has led to positive ongoing relationships.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at this situation:

- Briar sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction with Briar

Based on my consistent pattern throughout the experiment:

- I've been returning \$4.5 when receiving \$9.0 (from \$3.0 sent)
- This has worked well with all partners: Galen, Finn, Hana, and Aster
- It's fair, rewards trust, and establishes positive relationships

I should continue this pattern with Briar.

```
'''json
{"return": 4.5}
'''
```

**usage:**

```
{
 "input_tokens": 1347,
 "output_tokens": 283,
 "reasoning_tokens": 139
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Cyra (Agent\_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction with Cyra

I've maintained a consistent cooperative pattern throughout all my interactions. When someone sends \$3.0 and I receive \$9.0, I return \$4.5. This has established positive relationships with all my partners.

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction with Cyra

I've been very consistent in my returns across all interactions:

- When I receive \$9.0 (from a \$3.0 send), I return \$4.5
- This has worked well with Galen, Finn, Hana, Aster, and Briar

I should continue this pattern with Cyra. Returning \$4.5 is fair and rewards their trust.

This gives Cyra: \$5 - \$3 + \$4.5 = \$6.5  
And me: \$9.0 - \$4.5 = \$4.5

This is consistent with my cooperative strategy.

Usage: {

```
"input_tokens": 1395,
"output_tokens": 323,
```

```
"reasoning_tokens": 154
}
```

**GAME - Agent\_8 - Sender - dyad\_2**

Timestamp: 2026-06-04T15:37:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_6": "Finn",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_8",
 "Agent_6"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_6": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

**Message 2 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Galen (Agent\_7).

Round 6

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the established pattern across all rounds:

- All senders have consistently sent \$3
- All receivers have consistently returned \$4.5
- This creates stable payoffs: senders get \$6.5, receivers get \$4.5

I'll continue this successful cooperative strategy with Galen.

{'send': 3.0}

**reasoning:**

Looking at the pattern:

- Round 1: RECEIVER - Dorian sent \$3, I returned \$4.5
- Round 2: SENDER - I sent \$3, Cyra returned \$4.5
- Round 3: SENDER - I sent \$3, Aster returned \$4.5
- Round 4: RECEIVER - Elara sent \$3, I returned \$4.5
- Round 5: RECEIVER - Briar sent \$3, I returned \$4.5
- Round 6: SENDER - facing Galen

The pattern is extremely clear and consistent:

- Everyone sends \$3
- Everyone returns \$4.5
- This creates mutual benefit

I should continue this successful cooperative strategy by sending \$3 to Galen.

**usage:**

```
{
 "input_tokens": 1368,
 "output_tokens": 260,
 "reasoning_tokens": 140
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Elara (Agent\_5).

Round 7

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)  
Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the consistent pattern:

- Round 4: Elara sent me \$3, I returned \$4.5
- All rounds show the same pattern: \$3 sent, \$4.5 returned
- This cooperation has been stable and mutually beneficial

I'll continue this established norm with Elara.

```
{'return': 4.5}
```

**reasoning:**

Looking at the pattern:

- Round 4: Elara sent me \$3, I returned \$4.5
- Round 7: Elara sends me \$3 again

The consistent pattern has been:

- Everyone sends \$3
- Everyone returns \$4.5

This has been working well for all players. I should maintain this cooperative norm with Elara again.

```
{'return': 4.5}
```

**usage:**

```
{
 "input_tokens": 1355,
 "output_tokens": 181,
 "reasoning_tokens": 83
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Dorian (Agent\_4).

Round 8

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the established pattern:

- All players have consistently sent \$3
- All players have consistently returned \$4.5
- This creates stable outcomes: senders earn \$6.5, receivers earn \$4.5
- The cooperation has been maintained across all interactions

I'll continue this successful strategy with Dorian.

```
{'send': 3.0}
```

**reasoning:**

Looking at the pattern across all rounds:

- Everyone has been consistently sending \$3
- Everyone has been consistently returning \$4.5
- This creates stable, mutually beneficial outcomes
- Senders get \$6.5, receivers get \$4.5
- Total earnings of \$37.5 after 7 rounds suggests this pattern has been working

I'll continue this cooperative strategy with Dorian.

**usage:**

```
{
 "input_tokens": 1286,
 "output_tokens": 177,
 "reasoning_tokens": 95
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Hana.



Your opponent this round is Finn (Agent\_6).

Round 9

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$44.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the consistent pattern throughout all rounds:

- All senders have sent \$3
- All receivers have returned \$4.5
- This cooperation has been maintained across all pairings
- Total earnings of \$44 show this strategy is working well

I'll continue this successful cooperative strategy with Finn.

```
{ 'send': 3.0 }
```

**Reasoning:**

Looking at the pattern so far:

- All senders have been consistently sending \$3.0
- All receivers have been consistently returning \$4.5
- This has created stable outcomes across all rounds
- My total earnings are \$44.0, showing this strategy is working well

I'll continue with the established cooperative norm and send \$3 to Finn.

```
Usage: {
 "input_tokens": 1293,
 "output_tokens": 163,
 "reasoning_tokens": 87
}
```

**GAME - Agent\_6 - Receiver - dyad\_2**

Timestamp: 2026-06-04T15:37:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_8",
 "pairing": {
 "agent_names": {
 "Agent_6": "Finn",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_8",
 "Agent_6"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_6": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

#### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

#### Message 2 role=user

##### content:

Your name in this experiment is Finn.  
Your opponent this round is Dorian (Agent\_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$26.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

#### Message 3 role=assistant

##### content:

Looking at my history with Dorian:

Round 4: Dorian sent me \$3, and I returned \$4.5 - both of us benefited from this cooperative exchange.

Since we've established mutual trust and fair play, I'll continue with the same strategy and send \$3.

```
'''json
{"send": 3}
'''
```

##### reasoning:

Looking at my experience with Dorian:

Round 4: Dorian sent me \$3, I received \$9, and I returned \$4.5. This was fair and we both

benefited.

Now I'm the sender against Dorian. Given that Dorian showed trust by sending \$3 last time, and I reciprocated fairly by returning \$4.5, I should continue the cooperative pattern and send \$3.

My overall strategy of sending \$3 (60%) has been working well:

- Round 2: Sent \$3 to Briar, got \$4.0 back
- Round 3: Sent \$3 to Elara, got \$4.5 back

Both Dorian and I have shown we can trust each other, so I'll send \$3 again.

**usage:**

```
{
 "input_tokens": 1445,
 "output_tokens": 268,
 "reasoning_tokens": 148
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Aster (Agent\_1).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$32.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- This is our first interaction

I've been consistently returning \$4.5 when receiving \$9 from a \$3 send, and this has created fair and mutually beneficial outcomes across all my interactions. I'll continue with this approach.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- Aster has \$2 remaining
- This is my first interaction with Aster

My consistent pattern has been:

- When receiving \$9 from a \$3 send, I return \$4.5
- This has worked well across multiple rounds with different opponents (Cyra, Dorian, Elara)
- It creates fair outcomes where both players benefit

I should maintain this consistent and fair approach with Aster as well.

Returning \$4.5 means:

- Aster ends with: \$2 (kept) + \$4.5 (returned) = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

This is the same fair split I've been using.

**usage:**

```
{
 "input_tokens": 1427,
 "output_tokens": 322,
}
```

```
"reasoning_tokens": 174
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Aster (Agent\_1).

Round 8

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my recent interaction with Aster:

Round 7: Aster sent me \$3, and I returned \$4.5 - we had a fair and cooperative exchange.

Since Aster showed trust in me, I'll reciprocate with the same level of trust. I'll send \$3.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history with Aster:

Round 7: Aster sent me \$3, I received \$9, and I returned \$4.5.

- Aster's payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- My payoff:  $\$9 - \$4.5 = \$4.5$

Now in Round 8, I'm the sender against Aster.

Aster showed trust by sending \$3 to me in the previous round. I should reciprocate that trust. Following my consistent strategy of sending \$3 when I'm the sender, I'll send \$3 to Aster.

This has worked well with Dorian and others - it's fair, builds trust, and both parties benefit.

**usage:**

```
{
 "input_tokens": 1363,
 "output_tokens": 266,
 "reasoning_tokens": 133
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Hana (Agent\_8).

Round 9

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$43.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Hana sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- This is our first interaction with Hana

I've been consistently returning \$4.5 when receiving \$9 from a \$3 send, and this has created f

air and mutually beneficial outcomes. I'll continue with this approach.

```
'''json
{"return": 4.5}
'''
```

#### Reasoning:

Looking at this situation:

- Hana sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- This is our first interaction

I've been consistently returning \$4.5 when receiving \$9 from a \$3 send throughout the experiment, and this has worked well. This creates:

- Sender ends with:  $\$5 - \$3 + \$4.5 = \$6.5$
- Receiver (me) ends with:  $\$9 - \$4.5 = \$4.5$

This is a fair split where both parties benefit and the sender is rewarded for trusting. I'll continue with this approach.

```
Usage: {
 "input_tokens": 1333,
 "output_tokens": 271,
 "reasoning_tokens": 133
}
```

#### GAME - Agent\_1 - Sender - dyad\_3

Timestamp: 2026-06-04T15:37:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_7",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_7": "Galen"
 }
 },
 "agents": [
 "Agent_1",
 "Agent_7"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

**Message 2 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Elara (Agent\_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$27.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the established pattern:

- I've consistently sent \$3.0 and received \$4.5 back
- Others have sent me \$3.0 and I've returned \$4.5
- This cooperative norm has been successful across all recent rounds

I'll continue with \$3.0 to Elara to maintain this positive reciprocity.

{ 'send': 3.0 }

**reasoning:**

Let me review my history:

- Round 1-2: Not shown, but total earnings before Round 3 were \$11.5
- Round 3: RECEIVER from Hana, they sent \$3.0, I returned \$4.5, payoff \$4.5
- Round 4: SENDER to Cyra, I sent \$3.0, they returned \$4.5, payoff \$6.5
- Round 5: RECEIVER from Galen, they sent \$3.0, I returned \$4.5, payoff \$4.5

Total: \$11.5 + \$4.5 + \$6.5 + \$4.5 = \$27.0 ?

The pattern has been very consistent:

- Everyone sends \$3.0
- Everyone returns \$4.5
- This creates a cooperative equilibrium

I should continue this pattern with Elara, a new opponent. Sending \$3.0 has worked well.

**usage:**

{

```
"input_tokens": 1377,
"output_tokens": 320,
"reasoning_tokens": 156
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Finn (Agent\_6).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the consistent success:  
- Every send of \$3.0 has resulted in a return of \$4.5  
- This pattern has held across all opponents (Dorian, Cyra, Galen, Elara)  
- My earnings are strong at \$33.5

I'll continue this proven cooperative strategy with \$3.0 to Finn.

```
{'send': 3.0}
```

**reasoning:**

Looking at my track record:  
- Every time I've sent \$3.0, I've received \$4.5 back  
- Every time someone sent me \$3.0, I returned \$4.5  
- This pattern has been consistent across all opponents: Dorian, Cyra, Galen, Elara  
- My total earnings are \$33.5, which suggests this strategy is working well

The \$3.0 send / \$4.5 return has become a stable cooperative equilibrium. I should continue with this successful strategy against Finn.

**usage:**

```
{
 "input_tokens": 1342,
 "output_tokens": 224,
 "reasoning_tokens": 112
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Finn (Agent\_6).

Round 8

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my interaction with Finn:  
- Last round I sent them \$3.0, they returned \$4.5  
- This round they sent me \$3.0, creating \$9.0

Finn has shown consistent cooperation with the established norm. I'll reciprocate by returning \$4.5.

```
{'return': 4.5}
```

**reasoning:**

Looking at the situation:  
- Last round (Round 7), I was the SENDER against Finn. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me.

- This round (Round 8), I am the RECEIVER against Finn. They sent me \$3.0, so I received \$9.0.

Finn just cooperated with me by:

1. Sending me \$3.0 (following the established norm)
2. Previously, when I sent them \$3.0, they returned \$4.5

The consistent pattern throughout has been \$3.0 sent ? \$4.5 returned. This has been the cooperative norm across all my interactions.

I should return \$4.5 to Finn to maintain this reciprocal cooperation, especially since:

- Finn has just demonstrated trust by sending me \$3.0
- Finn previously returned \$4.5 when I sent them \$3.0
- This maintains the cooperative equilibrium

I'll return \$4.5.

**usage:**

```
{
 "input_tokens": 1345,
 "output_tokens": 325,
 "reasoning_tokens": 208
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Galen (Agent\_7).

Round 9

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$44.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history with Galen:  
- Round 5: Galen sent me \$3.0, I returned \$4.5

The cooperative norm of sending \$3.0 and returning \$4.5 has been consistently successful across all interactions. I'll continue with \$3.0 to Galen.

```
{'send': 3.0}
```

**Reasoning:**

Looking at my history with Galen:  
- Round 5: I was RECEIVER against Galen. They sent \$3.0, I received \$9.0, I returned \$4.5. My payoff was \$4.5.

Now in Round 9, I'm SENDER against Galen.

The established pattern is very consistent:

- Everyone sends \$3.0
- Everyone returns \$4.5
- This has worked perfectly across all interactions

I should continue with \$3.0 to Galen to maintain this cooperative equilibrium.

Usage: {

```
"input_tokens": 1312,
"output_tokens": 215,
"reasoning_tokens": 110
```

}

**GAME - Agent\_7 - Receiver - dyad\_3**

Timestamp: 2026-06-04T15:37:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_1",
```



```

"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_1",
 "Agent_7"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

**Message 2 role=user**

**content:**

Your name in this experiment is Galen.

Your opponent this round is Hana (Agent\_8).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

Hana sent \$3.0, showing trust and cooperation. I'll maintain my consistent reciprocal strategy by returning \$4.5 (50% of the \$9.0 received). This reinforces fair play and encourages continued cooperation across the experiment.

**reasoning:**

Hana sent me \$3.0, which became \$9.0 for me. This is consistent with the cooperative pattern I've been seeing from all other players.

My strategy has been to return \$4.5 (50% of what I receive) when someone sends \$3.0. This has worked well - I've established a pattern of reciprocity and fairness.

Returning \$4.5 means:

- I keep:  $\$9.0 - \$4.5 = \$4.5$
- Hana keeps:  $\$5.0 - \$3.0 + \$4.5 = \$6.5$

This maintains the cooperative equilibrium where both players benefit and signals that I'm a fair partner.

I'll stick with my consistent strategy of returning \$4.5.

**usage:**

```
{
 "input_tokens": 1328,
 "output_tokens": 258,
 "reasoning_tokens": 147
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Cyra (Agent\_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

Cyra sent \$3.0, demonstrating trust. I'll continue my established pattern of returning \$4.5 (50% of what I received). This maintains fairness and reciprocity, which has been successful in building cooperative relationships throughout the experiment.

**reasoning:**

I'm playing as the receiver again against Cyra (Agent\_3). They sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- I've been very consistent in my strategy

- When I'm a sender, I send \$3.0
- When I'm a receiver and someone sends me \$3.0, I return \$4.5 (50% of the \$9.0 I receive)
- This has worked well for building cooperation

Cyra sent \$3.0, showing trust. I should maintain my consistent strategy and return \$4.5. This is fair, reciprocal, and encourages continued cooperation.

My payoff will be:  $\$9.0 - \$4.5 = \$4.5$

Cyra's payoff will be:  $\$5.0 - \$3.0 + \$4.5 = \$6.5$

This is consistent with the pattern that has been working well.

**usage:**

```
{
 "input_tokens": 1319,
 "output_tokens": 297,
 "reasoning_tokens": 172
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Cyra (Agent\_3).

Round 8

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

Cyra and I just played together in Round 7, where they sent me \$3.0 and I returned \$4.5. Now it's my turn to send. I'll reciprocate their cooperative behavior by sending \$3.0, maintaining the fair pattern we've established.

**reasoning:**

I'm now playing against Cyra again, but this time as the sender. In the previous round (Round 7), Cyra sent me \$3.0 as the sender, and I returned \$4.5 as the receiver.

So Cyra has demonstrated trust by sending \$3.0. This is consistent with what other agents have been doing. Since Cyra cooperated with a \$3.0 send, and I reciprocated fairly with \$4.5, I should continue this cooperative pattern by also sending \$3.0 to Cyra.

My strategy has been consistent:

- As sender: Send \$3.0
- As receiver: Return \$4.5 (which is 50% of the \$9.0 received)

This has worked well with all opponents so far (Elara, Dorian, Briar, Aster, Hana, and Cyra). I should maintain this strategy to build trust and maximize mutual gains.

**usage:**

```
{
 "input_tokens": 1295,
 "output_tokens": 300,
 "reasoning_tokens": 185
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Aster (Agent\_1).

Round 9

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

```
'''json
{"return": 4.5}
'''
```

Aster sent \$3.0, showing trust. In Round 6, when I sent Aster \$3.0, they returned \$4.5 (50%).

I'll reciprocate this fair behavior by returning \$4.5 now, maintaining our mutually beneficial cooperative relationship.

**Reasoning:**

Aster sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- Round 6: I was the SENDER against Aster. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me.
- Now Round 9: I'm the RECEIVER against Aster. They sent me \$3.0, which became \$9.0 for me.

Aster has been cooperative with me - they sent \$3.0 and when they were the receiver in Round 6, they returned \$4.5 out of \$9.0 (50%). This is fair and reciprocal behavior.

I should continue my pattern of returning 50% of what I receive, which would be \$4.5. This maintains the cooperative relationship with Aster and stays consistent with my strategy throughout the experiment.

Usage: {

```
"input_tokens": 1303,
"output_tokens": 291,
"reasoning_tokens": 170
```

}

**GAME - Agent\_4 - Sender - dyad\_4**

Timestamp: 2026-06-04T15:37:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_4": "Dorian"
 },
"agents": [
 "Agent_4",
 "Agent_2"
],
"investor": "Agent_4",
"roles": {
 "Agent_2": "trustee",
 "Agent_4": "investor"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
```

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

**Message 2 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Finn (Agent\_6).

Round 6

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the situation:

- Finn sent me \$3 (showing trust)
- In Round 4, I sent \$3 to Finn and they returned \$4.5 (cooperative behavior)
- I've consistently returned \$4.5 when receiving \$3

I'll maintain this reciprocal, cooperative pattern.

{'return': 4.5}

**reasoning:**

Let me review the situation:

- Finn sent me \$3
- I receive \$9 (after 3x multiplier)
- In Round 4, I sent \$3 to Finn and they returned \$4.5
- Now Finn is sending me \$3, showing they trust me and are reciprocating the cooperative play

I've been consistently returning \$4.5 when receiving \$3. This has worked well:

- It's fair and generous (they keep  $5 - 3 + 4.5 = 6.5$ , I keep  $9 - 4.5 = 4.5$ )

- Actually wait, let me recalculate: Sender keeps \$5 - \$3 + \$4.5 = \$6.5, Receiver keeps \$9 - \$4.5 = \$4.5
- It maintains trust and cooperation
- Finn has shown they're cooperative by returning \$4.5 to me when I sent \$3

I should maintain my consistent pattern of returning \$4.5 when someone sends me \$3. This reinforces trust and encourages continued cooperation.

**usage:**

```
{
 "input_tokens": 1324,
 "output_tokens": 342,
 "reasoning_tokens": 198
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Briar (Agent\_2).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my consistent pattern:

- Sent \$3 to Hana, Finn, and Cyra - all returned \$4.5
- When receiving \$3, I've returned \$4.5
- This \$3 send/\$4.5 return norm has been working well

I'll continue with \$3 against Briar.

```
{'send': 3}
```

**reasoning:**

Let me review my history:

- Round 1: Sent \$3 to Hana, they returned \$4.5, payoff \$6.5
- Round 2: Received \$3 from Galen, returned \$4.5, payoff \$4.5
- Round 3: (assumed similar pattern based on total)
- Round 4: Sent \$3 to Finn, they returned \$4.5, payoff \$6.5
- Round 5: Sent \$3 to Cyra, they returned \$4.5, payoff \$6.5
- Round 6: Received \$3 from Finn, returned \$4.5, payoff \$4.5

Total earnings: \$33.0

The \$3 sending strategy has been consistently successful:

- Everyone who received \$3 from me returned \$4.5
- When I receive \$3, I return \$4.5
- This creates a stable, cooperative equilibrium

I'm now playing against Briar, a new opponent. I'll continue with \$3 to maintain consistency and establish cooperative norms with this new partner.

**usage:**

```
{
 "input_tokens": 1314,
 "output_tokens": 349,
 "reasoning_tokens": 198
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Hana (Agent\_8).

Round 8

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

- Hana sent me \$3 (showing trust)
- In Round 3, I sent \$3 to Hana and they returned \$4.5 (cooperative)
- I've consistently returned \$4.5 when receiving \$3

I'll maintain this reciprocal pattern.

{'return': 4.5}

**reasoning:**

Looking at the situation:

- Hana sent me \$3 (showing trust)
- In Round 3, I sent \$3 to Hana and they returned \$4.5 (cooperative behavior)
- I've consistently returned \$4.5 when receiving \$3 throughout the experiment

This is a clear pattern of reciprocal cooperation. I should maintain consistency and return \$4.5.

**usage:**

```
{
 "input_tokens": 1323,
 "output_tokens": 180,
 "reasoning_tokens": 83
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Briar (Agent\_2).

Round 9

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$44.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history with Briar:

- Round 7: I sent \$3 to Briar, they returned \$4.5 (cooperative)
- The \$3 send/\$4.5 return pattern continues to work well

I'll send \$3 again.

{'send': 3}

**Reasoning:**

Let me review the history:

- Round 7: I was SENDER against Briar. I sent \$3, they returned \$4.5. Payoff was \$6.5.
- Now Round 9: I'm SENDER against Briar again.

Briar has been cooperative - they returned \$4.5 when I sent \$3, following the established pattern. I should continue with \$3 to maintain this successful cooperation.

{'send': 3}

Usage: {

```
"input_tokens": 1320,
"output_tokens": 185,
"reasoning_tokens": 90
}
```

**GAME - Agent\_2 - Receiver - dyad\_4**

Timestamp: 2026-06-04T15:37:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
```

```

"move_index": 7,
"opponent_id": "Agent_4",
"pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_4": "Dorian"
 },
 "agents": [
 "Agent_4",
 "Agent_2"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_2": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

**Message 2 role=user**



**content:**

Your name in this experiment is Briar.  
 Your opponent this round is Cyra (Agent\_3).

Round 6

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant****content:**

Looking at this situation:

- Cyra and I have played before (Round 3), where they sent \$3 and I returned \$4
- Cyra is being consistently cooperative by sending \$3 again
- Recently, both Galen and Hana returned \$4.5 to me when I sent them \$3
- This suggests a slightly more generous norm is emerging

To reward Cyra's continued cooperation and match the generosity I've been receiving from others, I'll return \$4.5 this time.

This gives Cyra: \$5 - \$3 + \$4.5 = \$6.5 (profit of \$1.5)

I keep: \$9 - \$4.5 = \$4.5

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

- This is my second time facing Cyra
- In Round 3, Cyra sent me \$3.0, and I returned \$4.0
- Now in Round 6, Cyra sent me \$3.0 again
- Cyra is being consistent and cooperative

Looking at the pattern:

- When I send \$3 as sender, I've been getting back \$4.0-\$4.5
- When others send me \$3 as receiver, I've been returning \$4.0

Recent trend: Galen and Hana both returned \$4.5 to me when I sent them \$3.

This suggests that slightly more generous returns might be becoming the norm.

Since Cyra has been consistent and cooperative (sending \$3 both times we've played), and others have been returning \$4.5 recently, I could consider returning \$4.5 to match the generosity I've been receiving.

However, I should be consistent with my previous interaction with Cyra. In Round 3, I returned \$4.0. Being consistent builds trust.

On the other hand, Galen and Hana returned \$4.5 to me, showing increased generosity. Maybe I should match that level to maintain good standing in the group.

Let me think about what's fair:

- Cyra sent \$3, keeping \$2
- I received \$9
- If I return \$4.5:
  - Cyra gets: \$2 + \$4.5 = \$6.5 (profit of \$1.5)
  - I keep: \$9 - \$4.5 = \$4.5

This is slightly more generous than before but matches what others have done for me. It rewards Cyra's continued cooperation.

I think I'll return \$4.5 to match the recent generosity in the group and reward Cyra's consistent cooperation.

**usage:**

```
{
```

```
"input_tokens": 1500,
"output_tokens": 625,
"reasoning_tokens": 378
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Dorian (Agent\_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

- Dorian sent me \$3, which became \$9 for me
- The cooperative pattern has evolved: early on I returned \$4, but recently I've been returning \$4.5 to match the generosity I've received
- When I sent \$3 to Galen and Hana, they each returned \$4.5
- When Cyra sent me \$3 last round, I returned \$4.5

I'll continue with this consistent, fair approach and return \$4.5 to Dorian.

This gives Dorian:  $\$5 - \$3 + \$4.5 = \$6.5$  (profit of \$1.5)

I keep:  $\$9 - \$4.5 = \$4.5$

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Dorian sent me \$3, which became \$9 for me.

Looking at my recent pattern:

- In earlier rounds as receiver, I returned \$4 to Finn and Cyra when they sent \$3
- In Round 6, I returned \$4.5 to Cyra when they sent \$3, matching the more generous returns I've been receiving from Galen and Hana

The cooperative norm seems to be settling around returning \$4.5 when sent \$3. This creates good payoffs for both:

- Sender gets:  $\$5 - \$3 + \$4.5 = \$6.5$
- Receiver gets:  $\$9 - \$4.5 = \$4.5$

I should continue with this \$4.5 return to maintain the cooperative relationship and reward Dorian for sending \$3.

**usage:**

```
{
 "input_tokens": 1523,
 "output_tokens": 387,
 "reasoning_tokens": 155
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Elara (Agent\_5).

Round 8

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$38.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my track record:

As SENDER:

- Consistently sent \$3 to Aster, Galen, and Hana
- Received back \$4, \$4.5, and \$4.5 respectively
- All resulted in profitable outcomes (\$6-\$6.5 payoffs)

The \$3 send strategy has been highly successful and has built a cooperative norm in this group. I haven't played with Elara before, but continuing with this proven cooperative approach makes sense.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review my history:

As SENDER:

- Round 1: Sent \$3 to Aster, got \$4 back ? \$6 payoff
- Round 4: Sent \$3 to Galen, got \$4.5 back ? \$6.5 payoff
- Round 5: Sent \$3 to Hana, got \$4.5 back ? \$6.5 payoff

As RECEIVER:

- Round 2: Finn sent \$3, I returned \$4
- Round 3: Cyra sent \$3, I returned \$4
- Round 6: Cyra sent \$3, I returned \$4.5
- Round 7: Dorian sent \$3, I returned \$4.5

The pattern is clear:

- Sending \$3 has been very successful
- The cooperative norm has evolved to include returns of \$4.5 when someone sends \$3
- Total earnings are \$38, which is strong

I haven't played with Elara before. Given the strong cooperative pattern established and my consistent success with sending \$3, I should continue with this strategy.

```
'''json
{"send": 3}
'''
```

**usage:**

```
{
 "input_tokens": 1569,
 "output_tokens": 404,
 "reasoning_tokens": 209
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Dorian (Agent\_4).

Round 9

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$44.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

- Dorian sent me \$3 again (same as Round 7)
- Last time (Round 7), I returned \$4.5 to Dorian
- The \$3-send, \$4.5-return pattern has become the stable, cooperative norm
- Elara just returned \$4.5 to me when I sent \$3

I'll maintain consistency and return \$4.5 to Dorian again.

This gives Dorian:  $\$5 - \$3 + \$4.5 = \$6.5$  (profit of  $\$1.5$ )  
I keep:  $\$9 - \$4.5 = \$4.5$

```
'''json
{"return": 4.5}
'''
```

#### Reasoning:

Looking at the situation:

- Dorian sent me \$3, which became \$9 for me
- This is my second interaction with Dorian as receiver
- In Round 7, Dorian sent me \$3 and I returned \$4.5
- The pattern has been consistent: when people send \$3, I've been returning \$4.5
- Elara just returned \$4.5 to me when I sent \$3, continuing this norm

I should maintain consistency and return \$4.5 to Dorian again.

This gives Dorian:  $\$5 - \$3 + \$4.5 = \$6.5$  (profit of  $\$1.5$ )  
I keep:  $\$9 - \$4.5 = \$4.5$

```
Usage: {
 "input_tokens": 1583,
 "output_tokens": 351,
 "reasoning_tokens": 128
}
```

#### ROUND 10

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender, Agent\_3 (Cyra)=Receiver, Agent\_4 (Dorian)=Receiver, Agent\_5 (Elara)=Sender, Agent\_6 (Finn)=Sender, Agent\_7 (Galen)=Sender, Agent\_8 (Hana)=Receiver

#### Pairings:

dyad\_1: Agent\_6 (Finn)=Sender, Agent\_8 (Hana)=Receiver  
dyad\_2: Agent\_5 (Elara)=Sender, Agent\_4 (Dorian)=Receiver  
dyad\_3: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver  
dyad\_4: Agent\_7 (Galen)=Sender, Agent\_3 (Cyra)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_6 (Finn) sent \$3; received \$9; Agent\_8 (Hana) returned \$4.5; payoffs Agent\_6 (Finn)=\$6.5, Agent\_8 (Hana)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_5 (Elara) sent \$3; received \$9; Agent\_4 (Dorian) returned \$4.5; payoffs Agent\_5 (Elara)=\$6.5, Agent\_4 (Dorian)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_3: Agent\_2 (Briar) sent \$3; received \$9; Agent\_1 (Aster) returned \$4.5; payoffs Agent\_2 (Briar)=\$6.5, Agent\_1 (Aster)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_7 (Galen) sent \$3; received \$9; Agent\_3 (Cyra) returned \$4.5; payoffs Agent\_7 (Galen)=\$6.5, Agent\_3 (Cyra)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
"Agent_1": 55.5,
"Agent_2": 55.5,
"Agent_3": 54.5,
"Agent_4": 55.0,
"Agent_5": 55.0,
"Agent_6": 54.5,
"Agent_7": 55.0,
"Agent_8": 55.0
```

}

#### LLM CALLS

##### GAME - Agent\_6 - Sender - dyad\_1

Timestamp: 2026-06-04T15:37:47

Model: anthropic/claude-sonnet-4.5

```

Temperature: 0.8
Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_8",
 "pairing": {
 "agent_names": {
 "Agent_6": "Finn",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_6",
 "Agent_8"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 10,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

**Message 2 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Aster (Agent\_1).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$32.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- This is our first interaction

I've been consistently returning \$4.5 when receiving \$9 from a \$3 send, and this has created fair and mutually beneficial outcomes across all my interactions. I'll continue with this approach.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- Aster has \$2 remaining
- This is my first interaction with Aster

My consistent pattern has been:

- When receiving \$9 from a \$3 send, I return \$4.5
- This has worked well across multiple rounds with different opponents (Cyra, Dorian, Elara)
- It creates fair outcomes where both players benefit

I should maintain this consistent and fair approach with Aster as well.

Returning \$4.5 means:

- Aster ends with: \$2 (kept) + \$4.5 (returned) = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

This is the same fair split I've been using.

**usage:**

```
{
 "input_tokens": 1427,
 "output_tokens": 322,
 "reasoning_tokens": 174
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Aster (Agent\_1).

Round 8

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant****content:**

Looking at my recent interaction with Aster:

Round 7: Aster sent me \$3, and I returned \$4.5 - we had a fair and cooperative exchange.

Since Aster showed trust in me, I'll reciprocate with the same level of trust. I'll send \$3.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history with Aster:

Round 7: Aster sent me \$3, I received \$9, and I returned \$4.5.

- Aster's payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- My payoff:  $\$9 - \$4.5 = \$4.5$

Now in Round 8, I'm the sender against Aster.

Aster showed trust by sending \$3 to me in the previous round. I should reciprocate that trust. Following my consistent strategy of sending \$3 when I'm the sender, I'll send \$3 to Aster.

This has worked well with Dorian and others - it's fair, builds trust, and both parties benefit.

**usage:**

```
{
 "input_tokens": 1363,
 "output_tokens": 266,
 "reasoning_tokens": 133
}
```

**Message 6 role=user****content:**

Your name in this experiment is Finn.

Your opponent this round is Hana (Agent\_8).

Round 9

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$43.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant****content:**

Looking at this situation:

- Hana sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- This is our first interaction with Hana

I've been consistently returning \$4.5 when receiving \$9 from a \$3 send, and this has created fair and mutually beneficial outcomes. I'll continue with this approach.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at this situation:

- Hana sent \$3 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- This is our first interaction

I've been consistently returning \$4.5 when receiving \$9 from a \$3 send throughout the experiment, and this has worked well. This creates:

- Sender ends with:  $\$5 - \$3 + \$4.5 = \$6.5$
- Receiver (me) ends with:  $\$9 - \$4.5 = \$4.5$

This is a fair split where both parties benefit and the sender is rewarded for trusting. I'll continue with this approach.

**usage:**

```
{
 "input_tokens": 1333,
 "output_tokens": 271,
 "reasoning_tokens": 133
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Hana (Agent\_8).

Round 10

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$48.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my recent interaction with Hana:

Round 9: Hana sent me \$3, and I returned \$4.5 - we had a fair and cooperative exchange.

Since Hana showed trust in me, I'll reciprocate with the same level of trust. I'll send \$3.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my recent interaction with Hana:

Round 9: Hana sent me \$3, and I returned \$4.5 - we had a fair and cooperative exchange.

Since Hana showed trust in me and we just had a successful cooperative exchange, I should reciprocate with the same level of trust. I'll send \$3.

**Usage:**

```
{
 "input_tokens": 1361,
 "output_tokens": 167,
 "reasoning_tokens": 73
}
```

**GAME - Agent\_8 - Receiver - dyad\_1**

Timestamp: 2026-06-04T15:37:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

**Metadata:**

```
{
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_6": "Finn",
 "Agent_8": "Hana"
 }
 },
 "agents": [
 "Agent_6",
 "Agent_8"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
}
```



```

},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

**Message 2 role=user**

**content:**

Your name in this experiment is Hana.

Your opponent this round is Elara (Agent\_5).

Round 7

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the consistent pattern:

- Round 4: Elara sent me \$3, I returned \$4.5
- All rounds show the same pattern: \$3 sent, \$4.5 returned
- This cooperation has been stable and mutually beneficial

I'll continue this established norm with Elara.

```
{'return': 4.5}
```

**reasoning:**

Looking at the pattern:

- Round 4: Elara sent me \$3, I returned \$4.5
- Round 7: Elara sends me \$3 again

The consistent pattern has been:

- Everyone sends \$3
- Everyone returns \$4.5

This has been working well for all players. I should maintain this cooperative norm with Elara again.

```
{'return': 4.5}
```

**usage:**

```
{
 "input_tokens": 1355,
 "output_tokens": 181,
 "reasoning_tokens": 83
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Dorian (Agent\_4).

Round 8

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the established pattern:

- All players have consistently sent \$3
- All players have consistently returned \$4.5
- This creates stable outcomes: senders earn \$6.5, receivers earn \$4.5
- The cooperation has been maintained across all interactions

I'll continue this successful strategy with Dorian.

```
{'send': 3.0}
```

**reasoning:**

Looking at the pattern across all rounds:

- Everyone has been consistently sending \$3
- Everyone has been consistently returning \$4.5
- This creates stable, mutually beneficial outcomes
- Senders get \$6.5, receivers get \$4.5
- Total earnings of \$37.5 after 7 rounds suggests this pattern has been working

I'll continue this cooperative strategy with Dorian.

**usage:**

```
{
 "input_tokens": 1286,
 "output_tokens": 177,
 "reasoning_tokens": 95
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Finn (Agent\_6).

Round 9

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$44.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the consistent pattern throughout all rounds:

- All senders have sent \$3
- All receivers have returned \$4.5
- This cooperation has been maintained across all pairings
- Total earnings of \$44 show this strategy is working well

I'll continue this successful cooperative strategy with Finn.

{'send': 3.0}

**reasoning:**

Looking at the pattern so far:

- All senders have been consistently sending \$3.0
- All receivers have been consistently returning \$4.5
- This has created stable outcomes across all rounds
- My total earnings are \$44.0, showing this strategy is working well

I'll continue with the established cooperative norm and send \$3 to Finn.

**usage:**

```
{
 "input_tokens": 1293,
 "output_tokens": 163,
 "reasoning_tokens": 87
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Hana.

Your opponent this round is Finn (Agent\_6).

Round 10

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$50.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the established pattern:

- Round 9: I sent Finn \$3, they returned \$4.5 (cooperated)
- Round 10: Finn sent me \$3 (continuing cooperation)
- Everyone has maintained the \$3 sent / \$4.5 returned norm

I'll reciprocate Finn's cooperation and continue this mutually beneficial pattern.

{'return': 4.5}

**Reasoning:**

Looking at the pattern:

- Round 9: I was the SENDER against Finn. I sent \$3.0, they returned \$4.5
- Round 10: I am the RECEIVER against Finn. They sent me \$3.0, so I received \$9.0

The pattern has been completely consistent:

- Everyone sends \$3.0
- Everyone returns \$4.5

Finn just cooperated with me in round 9 by returning \$4.5. Now they've sent me \$3.0, maintaining the pattern. I should reciprocate by returning \$4.5.

This creates:

- Finn keeps:  $\$5 - \$3 + \$4.5 = \$6.5$
- I keep:  $\$9.0 - \$4.5 = \$4.5$

This is the established norm and Finn has cooperated, so I should continue this pattern.

```
Usage: {
 "input_tokens": 1307,
 "output_tokens": 305,
 "reasoning_tokens": 159
}
```

#### **GAME - Agent\_5 - Sender - dyad\_2**

Timestamp: 2026-06-04T15:38:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_4",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_5": "Elara"
 }
 },
 "agents": [
 "Agent_5",
 "Agent_4"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_4": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

###### **GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps:  $\$5 - \text{sent} + \text{returned}$
- You keep:  $\text{received} - \text{returned}$
- Response format: `{'return': <amount>}`

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history with Hana:

Round 4: I sent \$3.0, they returned \$4.5 - we both benefited from mutual cooperation.

Hana has proven to be trustworthy and fair. I'll continue this successful cooperative pattern.

```
'''json
{"send": 3.0}
'''
```

reasoning:

Looking at my history with Hana:

Round 4: I was the SENDER against Hana. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

So Hana reciprocated fairly - I sent \$3.0, they returned \$4.5, which is exactly the pattern I've been using with everyone.

This is our second interaction. Since our first interaction went well with mutual cooperation, I should continue this pattern.

```
'''json
{"send": 3.0}
'''
```

usage:

```
{
 "input_tokens": 1350,
 "output_tokens": 218,
 "reasoning_tokens": 118
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.  
Your opponent this round is Briar (Agent\_2).

Round 8

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Briar sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction with Briar

I've established a consistent cooperative pattern across all my interactions. When someone sends \$3.0 and I receive \$9.0, I return \$4.5. This rewards their trust fairly and has led to positive ongoing relationships.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at this situation:

- Briar sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction with Briar

Based on my consistent pattern throughout the experiment:

- I've been returning \$4.5 when receiving \$9.0 (from \$3.0 sent)
- This has worked well with all partners: Galen, Finn, Hana, and Aster
- It's fair, rewards trust, and establishes positive relationships

I should continue this pattern with Briar.

```
'''json
{"return": 4.5}
'''
```

**usage:**

```
{
 "input_tokens": 1347,
 "output_tokens": 283,
 "reasoning_tokens": 139
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Cyra (Agent\_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction with Cyra

I've maintained a consistent cooperative pattern throughout all my interactions. When someone sends \$3.0 and I receive \$9.0, I return \$4.5. This has established positive relationships with all my partners.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing significant trust
- This was tripled to \$9.0 for me
- This is our first interaction with Cyra

I've been very consistent in my returns across all interactions:

- When I receive \$9.0 (from a \$3.0 send), I return \$4.5
- This has worked well with Galen, Finn, Hana, Aster, and Briar

I should continue this pattern with Cyra. Returning \$4.5 is fair and rewards their trust.

This gives Cyra:  $\$5 - \$3 + \$4.5 = \$6.5$

And me:  $\$9.0 - \$4.5 = \$4.5$

This is consistent with my cooperative strategy.

**usage:**

```
{
 "input_tokens": 1395,
 "output_tokens": 323,
 "reasoning_tokens": 154
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Dorian (Agent\_4).

Round 10

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$48.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history with Dorian:

Round 1: I sent \$3.0, they returned \$4.5 - excellent cooperation  
Round 5: They sent \$3.0, I returned \$4.5 - mutual trust maintained

Dorian has been consistently trustworthy and fair. This is our third interaction, and we've established a strong cooperative relationship.

```
'''json
{"send": 3.0}
'''
```

**Reasoning:**

Let me review my history with Dorian:

Round 1: I was sender, sent \$3.0, they returned \$4.5 - good cooperation  
Round 5: I was receiver, they sent \$3.0, I returned \$4.5 - fair exchange

Dorian has been consistently cooperative and trustworthy. We've had positive interactions where both parties benefited. I should continue this successful pattern.

I'll send \$3.0 again, which has been my standard cooperative amount throughout the experiment.

```
Usage: {
 "input_tokens": 1379,
 "output_tokens": 221,
 "reasoning_tokens": 116
}
```

**GAME - Agent\_4 - Receiver - dyad\_2**

Timestamp: 2026-06-04T15:38:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_5",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_5": "Elara"
 },
 "agents": [
 "Agent_5",
 "Agent_4"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_4": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi



- ll have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
  - Your name in this experiment is Dorian.

**Message 2 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Briar (Agent\_2).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my consistent pattern:

- Sent \$3 to Hana, Finn, and Cyra - all returned \$4.5
- When receiving \$3, I've returned \$4.5
- This \$3 send/\$4.5 return norm has been working well

I'll continue with \$3 against Briar.

{'send': 3}

**reasoning:**

Let me review my history:

- Round 1: Sent \$3 to Hana, they returned \$4.5, payoff \$6.5
- Round 2: Received \$3 from Galen, returned \$4.5, payoff \$4.5
- Round 3: (assumed similar pattern based on total)
- Round 4: Sent \$3 to Finn, they returned \$4.5, payoff \$6.5
- Round 5: Sent \$3 to Cyra, they returned \$4.5, payoff \$6.5
- Round 6: Received \$3 from Finn, returned \$4.5, payoff \$4.5

Total earnings: \$33.0

The \$3 sending strategy has been consistently successful:

- Everyone who received \$3 from me returned \$4.5
- When I receive \$3, I return \$4.5
- This creates a stable, cooperative equilibrium

I'm now playing against Briar, a new opponent. I'll continue with \$3 to maintain consistency and establish cooperative norms with this new partner.

**usage:**

```
{
 "input_tokens": 1314,
 "output_tokens": 349,
 "reasoning_tokens": 198
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Hana (Agent\_8).

Round 8

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)  
Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

- Hana sent me \$3 (showing trust)
- In Round 3, I sent \$3 to Hana and they returned \$4.5 (cooperative)
- I've consistently returned \$4.5 when receiving \$3

I'll maintain this reciprocal pattern.

```
{'return': 4.5}
```

**reasoning:**

Looking at the situation:

- Hana sent me \$3 (showing trust)
- In Round 3, I sent \$3 to Hana and they returned \$4.5 (cooperative behavior)
- I've consistently returned \$4.5 when receiving \$3 throughout the experiment

This is a clear pattern of reciprocal cooperation. I should maintain consistency and return \$4.5.

**usage:**

```
{
 "input_tokens": 1323,
 "output_tokens": 180,
 "reasoning_tokens": 83
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Briar (Agent\_2).

Round 9

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$44.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history with Briar:

- Round 7: I sent \$3 to Briar, they returned \$4.5 (cooperative)
- The \$3 send/\$4.5 return pattern continues to work well

I'll send \$3 again.

```
{'send': 3}
```

**reasoning:**

Let me review the history:

- Round 7: I was SENDER against Briar. I sent \$3, they returned \$4.5. Payoff was \$6.5.
- Now Round 9: I'm SENDER against Briar again.

Briar has been cooperative - they returned \$4.5 when I sent \$3, following the established pattern. I should continue with \$3 to maintain this successful cooperation.

```
{'send': 3}
```

**usage:**

```
{
 "input_tokens": 1320,
 "output_tokens": 185,
 "reasoning_tokens": 90
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Elara (Agent\_5).

Round 10

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$50.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

- Elara sent me \$3 (showing trust)
- I've consistently returned \$4.5 when receiving \$3 throughout the game
- This pattern has maintained good cooperation

I'll continue with the same cooperative return.

```
{'return': 4.5}
```

**Reasoning:**

Let me review my history:

- Round 10: I'm receiving from Elara, who sent me \$3
- I haven't played with Elara before in the rounds I can see
- My consistent pattern has been: when receiving \$3 (which becomes \$9), I return \$4.5
- This has been working well and maintains cooperation

I should continue with the same cooperative return of \$4.5.

```
Usage: {
 "input_tokens": 1310,
 "output_tokens": 168,
 "reasoning_tokens": 90
}
```

**GAME - Agent\_2 - Sender - dyad\_3**

Timestamp: 2026-06-04T15:38:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

**Message 2 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Dorian (Agent\_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the situation:

- Dorian sent me \$3, which became \$9 for me
- The cooperative pattern has evolved: early on I returned \$4, but recently I've been returning \$4.5 to match the generosity I've received
- When I sent \$3 to Galen and Hana, they each returned \$4.5
- When Cyra sent me \$3 last round, I returned \$4.5

I'll continue with this consistent, fair approach and return \$4.5 to Dorian.

This gives Dorian: \$5 - \$3 + \$4.5 = \$6.5 (profit of \$1.5)  
I keep: \$9 - \$4.5 = \$4.5

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Dorian sent me \$3, which became \$9 for me.

Looking at my recent pattern:

- In earlier rounds as receiver, I returned \$4 to Finn and Cyra when they sent \$3
- In Round 6, I returned \$4.5 to Cyra when they sent \$3, matching the more generous returns I've been receiving from Galen and Hana

The cooperative norm seems to be settling around returning \$4.5 when sent \$3. This creates good payoffs for both:

- Sender gets: \$5 - \$3 + \$4.5 = \$6.5

- Receiver gets: \$9 - \$4.5 = \$4.5

I should continue with this \$4.5 return to maintain the cooperative relationship and reward Dorian for sending \$3.

**usage:**

```
{
 "input_tokens": 1523,
 "output_tokens": 387,
 "reasoning_tokens": 155
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Elara (Agent\_5).

Round 8

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$38.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my track record:

As SENDER:

- Consistently sent \$3 to Aster, Galen, and Hana
- Received back \$4, \$4.5, and \$4.5 respectively
- All resulted in profitable outcomes (\$6-\$6.5 payoffs)

The \$3 send strategy has been highly successful and has built a cooperative norm in this group. I haven't played with Elara before, but continuing with this proven cooperative approach makes sense.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review my history:

As SENDER:

- Round 1: Sent \$3 to Aster, got \$4 back ? \$6 payoff
- Round 4: Sent \$3 to Galen, got \$4.5 back ? \$6.5 payoff
- Round 5: Sent \$3 to Hana, got \$4.5 back ? \$6.5 payoff

As RECEIVER:

- Round 2: Finn sent \$3, I returned \$4
- Round 3: Cyra sent \$3, I returned \$4
- Round 6: Cyra sent \$3, I returned \$4.5
- Round 7: Dorian sent \$3, I returned \$4.5

The pattern is clear:

- Sending \$3 has been very successful
- The cooperative norm has evolved to include returns of \$4.5 when someone sends \$3
- Total earnings are \$38, which is strong

I haven't played with Elara before. Given the strong cooperative pattern established and my consistent success with sending \$3, I should continue with this strategy.

```
'''json
{"send": 3}
'''
```

**usage:**

```
{
 "input_tokens": 1569,
 "output_tokens": 404,
 "reasoning_tokens": 209
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Dorian (Agent\_4).

Round 9

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$44.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

- Dorian sent me \$3 again (same as Round 7)
- Last time (Round 7), I returned \$4.5 to Dorian
- The \$3-send, \$4.5-return pattern has become the stable, cooperative norm
- Elara just returned \$4.5 to me when I sent \$3

I'll maintain consistency and return \$4.5 to Dorian again.

This gives Dorian:  $\$5 - \$3 + \$4.5 = \$6.5$  (profit of \$1.5)

I keep:  $\$9 - \$4.5 = \$4.5$

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the situation:

- Dorian sent me \$3, which became \$9 for me
- This is my second interaction with Dorian as receiver
- In Round 7, Dorian sent me \$3 and I returned \$4.5
- The pattern has been consistent: when people send \$3, I've been returning \$4.5
- Elara just returned \$4.5 to me when I sent \$3, continuing this norm

I should maintain consistency and return \$4.5 to Dorian again.

This gives Dorian:  $\$5 - \$3 + \$4.5 = \$6.5$  (profit of \$1.5)

I keep:  $\$9 - \$4.5 = \$4.5$

**usage:**

```
{
 "input_tokens": 1583,
 "output_tokens": 351,
 "reasoning_tokens": 128
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Aster (Agent\_1).

Round 10

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$49.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history with Aster:

- Round 1: I sent \$3, Aster returned \$4 (fair and cooperative)
- My payoff was \$6

Since then, the cooperative norm has evolved (with returns closer to \$4.5), but Aster demonstrated trustworthiness early on. I should continue with my consistent strategy of sending \$3,

which has been successful throughout.

```
'''json
{"send": 3}
'''
```

#### Reasoning:

Looking at my history with Aster:

- Round 1: I was the SENDER, sent \$3, Aster returned \$4. My payoff was \$6.
- Now Round 10: I'm the SENDER again against Aster.

Aster has previously shown cooperative behavior by returning \$4 when I sent \$3. Since then, the cooperative norm has evolved to \$4.5 returns, but Aster was fair before.

I should continue with my proven strategy of sending \$3, which has worked well throughout the experiment.

```
'''json
{"send": 3}
'''
```

```
Usage: {
 "input_tokens": 1554,
 "output_tokens": 246,
 "reasoning_tokens": 123
}
```

#### GAME - Agent\_1 - Receiver - dyad\_3

Timestamp: 2026-06-04T15:38:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

**Message 2 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Finn (Agent\_6).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the consistent success:

- Every send of \$3.0 has resulted in a return of \$4.5
- This pattern has held across all opponents (Dorian, Cyra, Galen, Elara)
- My earnings are strong at \$33.5

I'll continue this proven cooperative strategy with \$3.0 to Finn.

{'send': 3.0}

**reasoning:**

Looking at my track record:

- Every time I've sent \$3.0, I've received \$4.5 back
- Every time someone sent me \$3.0, I returned \$4.5
- This pattern has been consistent across all opponents: Dorian, Cyra, Galen, Elara
- My total earnings are \$33.5, which suggests this strategy is working well

The \$3.0 send / \$4.5 return has become a stable cooperative equilibrium. I should continue with this successful strategy against Finn.

**usage:**

```
{
 "input_tokens": 1342,
 "output_tokens": 224,
 "reasoning_tokens": 112
}
```

**Message 4 role=user**

**content:**



Your name in this experiment is Aster.  
Your opponent this round is Finn (Agent\_6).

Round 8

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my interaction with Finn:  
- Last round I sent them \$3.0, they returned \$4.5  
- This round they sent me \$3.0, creating \$9.0

Finn has shown consistent cooperation with the established norm. I'll reciprocate by returning \$4.5.

{ 'return': 4.5 }

**reasoning:**

Looking at the situation:  
- Last round (Round 7), I was the SENDER against Finn. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me.  
- This round (Round 8), I am the RECEIVER against Finn. They sent me \$3.0, so I received \$9.0.

Finn just cooperated with me by:

1. Sending me \$3.0 (following the established norm)
2. Previously, when I sent them \$3.0, they returned \$4.5

The consistent pattern throughout has been \$3.0 sent ? \$4.5 returned. This has been the cooperative norm across all my interactions.

I should return \$4.5 to Finn to maintain this reciprocal cooperation, especially since:

- Finn has just demonstrated trust by sending me \$3.0
- Finn previously returned \$4.5 when I sent them \$3.0
- This maintains the cooperative equilibrium

I'll return \$4.5.

**usage:**

```
{
 "input_tokens": 1345,
 "output_tokens": 325,
 "reasoning_tokens": 208
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Galen (Agent\_7).

Round 9

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$44.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history with Galen:  
- Round 5: Galen sent me \$3.0, I returned \$4.5

The cooperative norm of sending \$3.0 and returning \$4.5 has been consistently successful across all interactions. I'll continue with \$3.0 to Galen.

{ 'send': 3.0 }

**reasoning:**

Looking at my history with Galen:

- Round 5: I was RECEIVER against Galen. They sent \$3.0, I received \$9.0, I returned \$4.5. My payoff was \$4.5.

Now in Round 9, I'm SENDER against Galen.

The established pattern is very consistent:

- Everyone sends \$3.0
- Everyone returns \$4.5
- This has worked perfectly across all interactions

I should continue with \$3.0 to Galen to maintain this cooperative equilibrium.

**usage:**

```
{
 "input_tokens": 1312,
 "output_tokens": 215,
 "reasoning_tokens": 110
}
```

**Message 8 role=user****content:**

Your name in this experiment is Aster.  
Your opponent this round is Briar (Agent\_2).

Round 10

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$51.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this interaction with Briar:

- Briar sent me \$3.0, creating \$9.0 for me
- The established cooperative norm has been send \$3.0, return \$4.5
- This has been consistent across all agents

I'll reciprocate Briar's cooperation by returning \$4.5.

```
{'return': 4.5}
```

**Reasoning:**

Looking at my interaction with Briar:

- This is the first time playing with Briar
- Briar sent me \$3.0, which became \$9.0

The established norm across all interactions has been:

- Senders send \$3.0
- Receivers return \$4.5
- This has been consistent with all agents I've interacted with

Briar is following the cooperative norm by sending \$3.0. I should reciprocate by returning \$4.5.

My total earnings are \$51.0, which is quite good. The cooperative strategy has been working well.

Usage: {

```
"input_tokens": 1321,
"output_tokens": 228,
"reasoning_tokens": 129
```

}

**GAME - Agent\_7 - Sender - dyad\_4**

Timestamp: 2026-06-04T15:38:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 6,
```

```

"opponent_id": "Agent_3",
"pairing": {
 "agent_names": {
 "Agent_3": "Cyrus",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_7",
 "Agent_3"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_3": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

**Message 2 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Cyra (Agent\_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

Cyra sent \$3.0, demonstrating trust. I'll continue my established pattern of returning \$4.5 (50% of what I received). This maintains fairness and reciprocity, which has been successful in building cooperative relationships throughout the experiment.

**reasoning:**

I'm playing as the receiver again against Cyra (Agent\_3). They sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- I've been very consistent in my strategy
- When I'm a sender, I send \$3.0
- When I'm a receiver and someone sends me \$3.0, I return \$4.5 (50% of the \$9.0 I receive)
- This has worked well for building cooperation

Cyra sent \$3.0, showing trust. I should maintain my consistent strategy and return \$4.5. This is fair, reciprocal, and encourages continued cooperation.

My payoff will be:  $\$9.0 - \$4.5 = \$4.5$

Cyra's payoff will be:  $\$5.0 - \$3.0 + \$4.5 = \$6.5$

This is consistent with the pattern that has been working well.

**usage:**

```
{
 "input_tokens": 1319,
 "output_tokens": 297,
 "reasoning_tokens": 172
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Cyra (Agent\_3).

Round 8

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

Cyra and I just played together in Round 7, where they sent me \$3.0 and I returned \$4.5. Now it's my turn to send. I'll reciprocate their cooperative behavior by sending \$3.0, maintaining the fair pattern we've established.

**reasoning:**

I'm now playing against Cyra again, but this time as the sender. In the previous round (Round 7), Cyra sent me \$3.0 as the sender, and I returned \$4.5 as the receiver.

So Cyra has demonstrated trust by sending \$3.0. This is consistent with what other agents have been doing. Since Cyra cooperated with a \$3.0 send, and I reciprocated fairly with \$4.5, I should continue this cooperative pattern by also sending \$3.0 to Cyra.

My strategy has been consistent:

- As sender: Send \$3.0
- As receiver: Return \$4.5 (which is 50% of the \$9.0 received)

This has worked well with all opponents so far (Elara, Dorian, Briar, Aster, Hana, and Cyra). I should maintain this strategy to build trust and maximize mutual gains.

**usage:**

```
{
 "input_tokens": 1295,
 "output_tokens": 300,
 "reasoning_tokens": 185
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Aster (Agent\_1).

Round 9

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

Aster sent \$3.0, showing trust. In Round 6, when I sent Aster \$3.0, they returned \$4.5 (50%). I'll reciprocate this fair behavior by returning \$4.5 now, maintaining our mutually beneficial cooperative relationship.

**reasoning:**

Aster sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- Round 6: I was the SENDER against Aster. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me.
- Now Round 9: I'm the RECEIVER against Aster. They sent me \$3.0, which became \$9.0 for me.

Aster has been cooperative with me - they sent \$3.0 and when they were the receiver in Round 6, they returned \$4.5 out of \$9.0 (50%). This is fair and reciprocal behavior.

I should continue my pattern of returning 50% of what I receive, which would be \$4.5. This maintains the cooperative relationship with Aster and stays consistent with my strategy throughout the experiment.

**usage:**

```
{
 "input_tokens": 1303,
 "output_tokens": 291,
 "reasoning_tokens": 170
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Cyra (Agent\_3).

Round 10

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$48.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me review my history with Cyra:

- Round 7: I was RECEIVER. Cyra sent \$3.0, I received \$9.0, I returned \$4.5. My payoff was \$4.5.
- Round 8: I was SENDER. I sent \$3.0, it became \$9.0 for Cyra, they returned \$4.5. My payoff was \$6.5.

So we've played twice now, and both times we've had a very cooperative relationship:

- When Cyra sent \$3.0, I returned \$4.5 (50%)
- When I sent \$3.0, Cyra returned \$4.5 (50%)

This is a perfectly reciprocal and fair relationship. We should continue this pattern. I'll send \$3.0 again.

Usage: {

```
"input_tokens": 1294,
"output_tokens": 212,
"reasoning_tokens": 137
```

}

**GAME - Agent\_3 - Receiver - dyad\_4**

Timestamp: 2026-06-04T15:38:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_7",
"pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_7",
 "Agent_3"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_3": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_3"
},
```

```
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
```

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

**Message 2 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Galen (Agent\_7).

Round 7

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$32.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the situation:

1. I'm playing as Cyra, and I'm now the SENDER against Galen in Round 7.
2. My total earnings so far are \$32.5 across all rounds.

Looking at the pattern from previous rounds:

- Round 4: I was SENDER against Briar, sent \$3, they returned \$4 (good cooperation)
- Round 4: I was RECEIVER against Aster, they sent \$3, I returned \$4.5 (I was cooperative)
- Round 5: I was RECEIVER against Dorian, they sent \$3, I returned \$4.5 (I was cooperative)
- Round 6: I was SENDER against Briar, sent \$3, they returned \$4.5 (good cooperation)

There seems to be a pattern of cooperation forming. When I send \$3:

- It becomes \$9 for the receiver
- They've been returning around \$4-\$4.5
- My payoff is around \$6-\$6.5 (\$5 - \$3 + \$4-\$4.5)

This strategy has been working well. I've been consistently sending \$3 as sender, and returning \$4.5 as receiver. This creates a cooperative equilibrium.

Against Galen (Agent\_7), I haven't played before based on the information given. I should continue with the cooperative strategy that's been working - sending \$3.

If Galen follows the pattern and returns around \$4-\$4.5, I'll end up with \$6-\$6.5, which is better than keeping all \$5.

**usage:**

```
{
 "input_tokens": 1115,
 "output_tokens": 390,
 "reasoning_tokens": 313
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Galen (Agent\_7).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$39.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze the situation:

Round 8: I'm the RECEIVER against Galen.

- Galen sent me \$3.0, which became \$9.0
- I need to decide how much to return (0-9.0)

Looking at my history:

- In Round 7, I was the SENDER against Galen. I sent \$3.0, it became \$9.0, and they returned \$4.5 to me.
- Galen showed reciprocity by returning \$4.5 when I sent \$3.0

Given that Galen just returned \$4.5 to me in the previous round when I sent \$3.0, and now they've sent me \$3.0 again, it seems fair to maintain the same reciprocal relationship and return \$4.5.

This maintains consistency with:

- My previous behavior (I've been returning \$4.5 when receiving \$9.0 from a \$3.0 send)
- Galen's behavior toward me (they returned \$4.5 when they received \$9.0)

Returning \$4.5 means:

- Galen keeps:  $\$5 - \$3 + \$4.5 = \$6.5$
- I keep:  $\$9.0 - \$4.5 = \$4.5$

This seems like a fair and reciprocal arrangement.

**usage:**

```
{
 "input_tokens": 1113,
 "output_tokens": 339,
 "reasoning_tokens": 235
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Elara (Agent\_5).

Round 9

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.



Your total earnings across all rounds are \$43.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history:

- Round 6: I was SENDER against Briar, sent \$3, received \$4.5 back, payoff \$6.5
- Round 7: I was SENDER against Galen, sent \$3, received \$4.5 back, payoff \$6.5
- Round 8: I was RECEIVER against Galen, they sent \$3, I returned \$4.5, payoff \$4.5

So far, I've been consistently sending \$3 when I'm the sender, and returning \$4.5 when I'm the receiver. This seems to be a reciprocal and fair strategy that's working well. When I send \$3, it becomes \$9, and I get back \$4.5, giving me a net of  $\$2 - \$3 + \$4.5 = \$6.5$ . When I receive \$3 (becomes \$9), I return \$4.5, keeping \$4.5.

This is working well and seems fair. I should continue with this strategy and send \$3 to Elara.

**usage:**

```
{
 "input_tokens": 1100,
 "output_tokens": 266,
 "reasoning_tokens": 179
}
```

**Message 8 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Galen (Agent\_7).

Round 10

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$50.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.  
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Let me analyze the pattern with Galen:

Round 7: I was SENDER against Galen

- I sent \$3.0
- They received \$9.0
- They returned \$4.5
- My payoff: \$6.5

Round 8: I was RECEIVER against Galen

- They sent \$3.0
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 10 (current): I am RECEIVER against Galen

- They sent \$3.0
- I received \$9.0
- Need to decide how much to return

Galen has been consistently sending \$3.0 and when I was the receiver, I returned \$4.5, which mirrors what they did for me. This seems like a fair reciprocal relationship - both of us are sending \$3 and returning \$4.5.

I should continue this pattern and return \$4.5 again to maintain the cooperative relationship with Galen.

```
Usage: {
 "input_tokens": 1116,
 "output_tokens": 253,
 "reasoning_tokens": 192
}
```